

- 32.** Show the arrangement of three resistors of $2\ \Omega$, $3\ \Omega$ and $6\ \Omega$ which are connected in such a way that the total/equivalent resistance of the combination is (a) $1\ \Omega$, and (b) $4\ \Omega$. Justify your answer in each case.

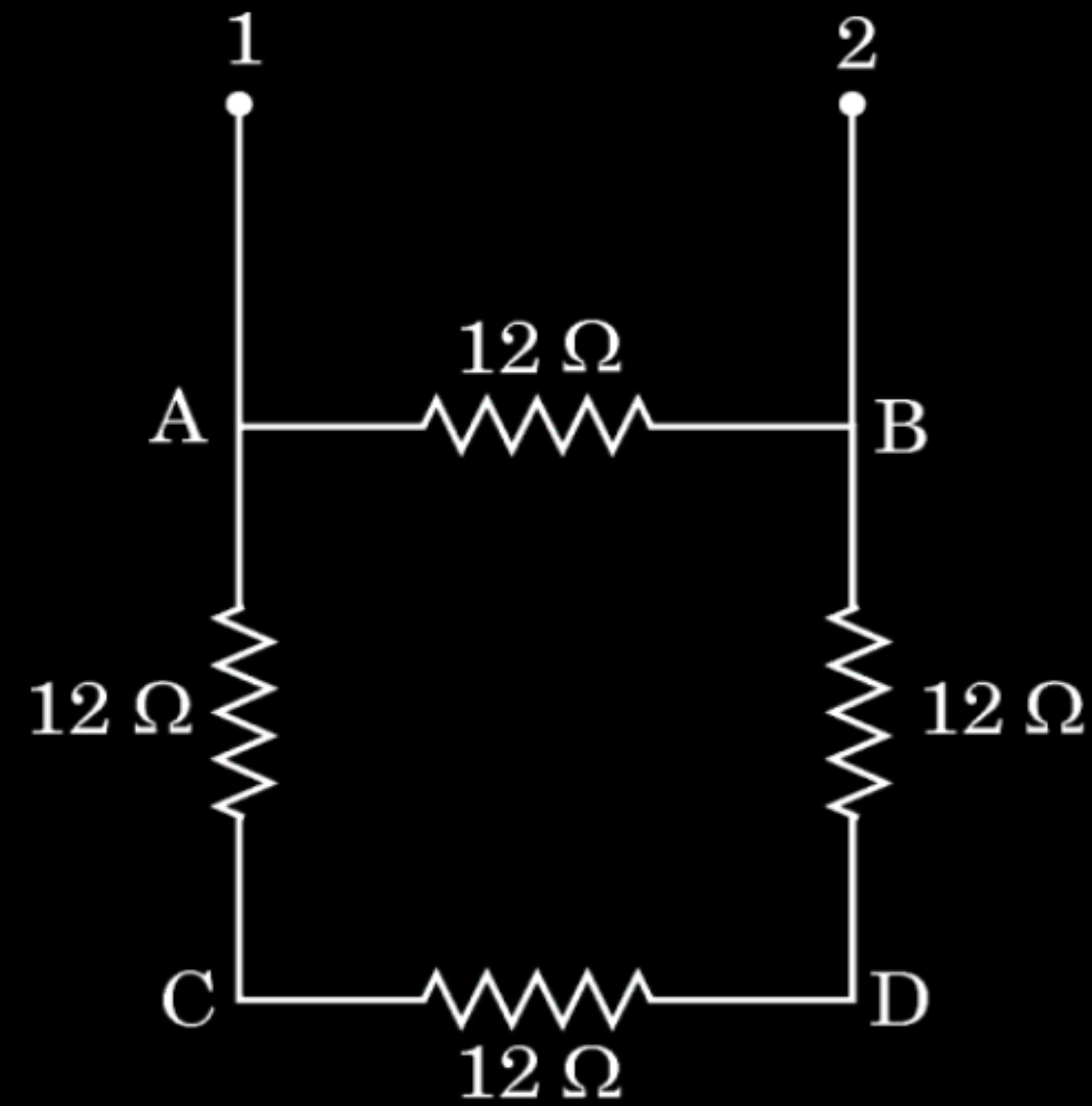
3

CBSE 2025 C

- 36.** (a) (i) If R_s and R_p respectively are the total resistances of the series and parallel combinations of the three resistors of resistances R_1 , R_2 and R_3 , then write the relationships among (I) R_s , R_1 , R_2 and R_3 and (II) R_p , R_1 , R_2 and R_3 .

12. Four identical resistors of $12\ \Omega$ each are connected in series to form a square ABCD as shown in the figure. The resistance of the network between the two points 1 and 2 is :

CBSE 2025



(A) $48\ \Omega$

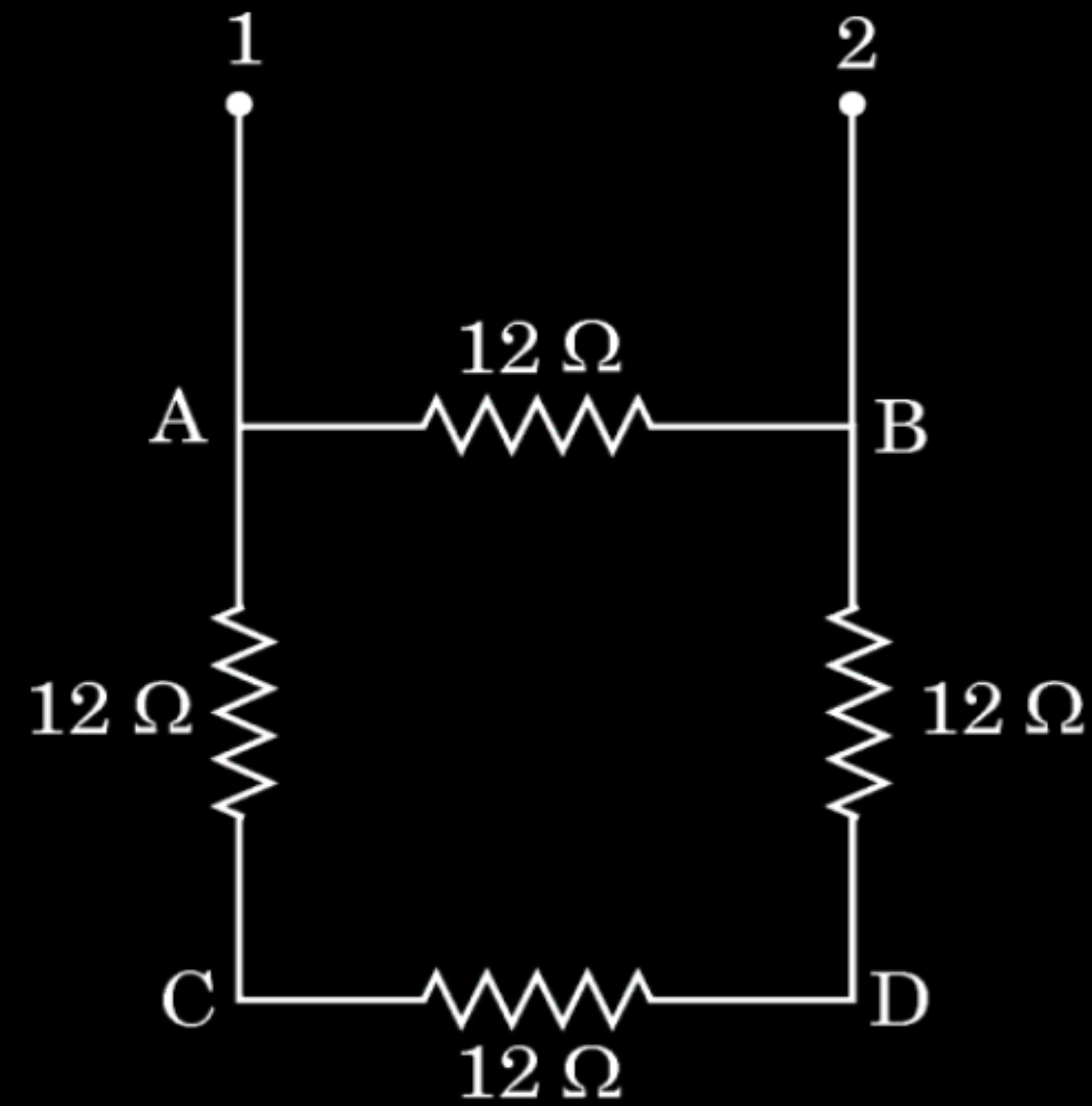
(B) $36\ \Omega$

(C) $9\ \Omega$

(D) $6\ \Omega$

12. Four identical resistors of $12\ \Omega$ each are connected in series to form a square ABCD as shown in the figure. The resistance of the network between the two points 1 and 2 is :

CBSE 2025



(A) $48\ \Omega$

(B) $36\ \Omega$

(C) $9\ \Omega$

(D) $6\ \Omega$

(C) / $9\ \Omega$

32. Three resistors of $2\ \Omega$, $3\ \Omega$ and $6\ \Omega$ are connected in (i) series, and (ii) parallel. Draw the arrangements of the resistors and find the equivalent resistance of each arrangement.

CBSE 2025

32. Three resistors of $2\ \Omega$, $3\ \Omega$ and $6\ \Omega$ are connected in (i) series, and (ii) parallel. Draw the arrangements of the resistors and find the equivalent resistance of each arrangement.

CBSE 2025

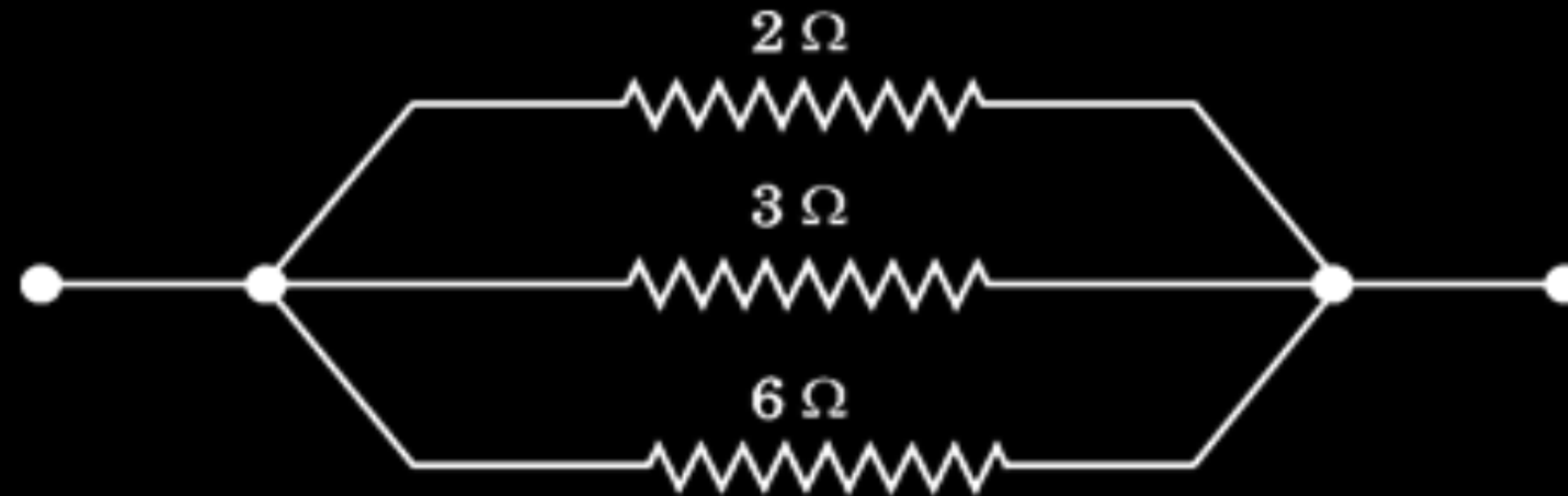
(i)



In series, $R_s = R_1 + R_2 + R_3$

$$= (2 + 3 + 6)\ \Omega = 11\ \Omega$$

(ii)



In parallel, $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

$$= \frac{1}{2} + \frac{1}{3} + \frac{1}{6}$$

$$= \frac{3+2+1}{6}$$

$$R_p = 1.0\ \Omega$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- 25.** (a) A wire of resistance R is cut into three equal parts. If these three parts are then joined in parallel, calculate the total resistance of the combination so formed.

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- 25.** (a) A wire of resistance R is cut into three equal parts. If these three parts are then joined in parallel, calculate the total resistance of the combination so formed.

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2

(a)

$$\text{Resistance of each part} = \frac{R}{3}$$

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R_p} = \frac{1}{R/3} + \frac{1}{R/3} + \frac{1}{R/3}$$

$$\frac{1}{R_p} = \frac{3}{R} + \frac{3}{R} + \frac{3}{R} = \frac{9}{R}$$

$$\Rightarrow R_p = \frac{R}{9}$$

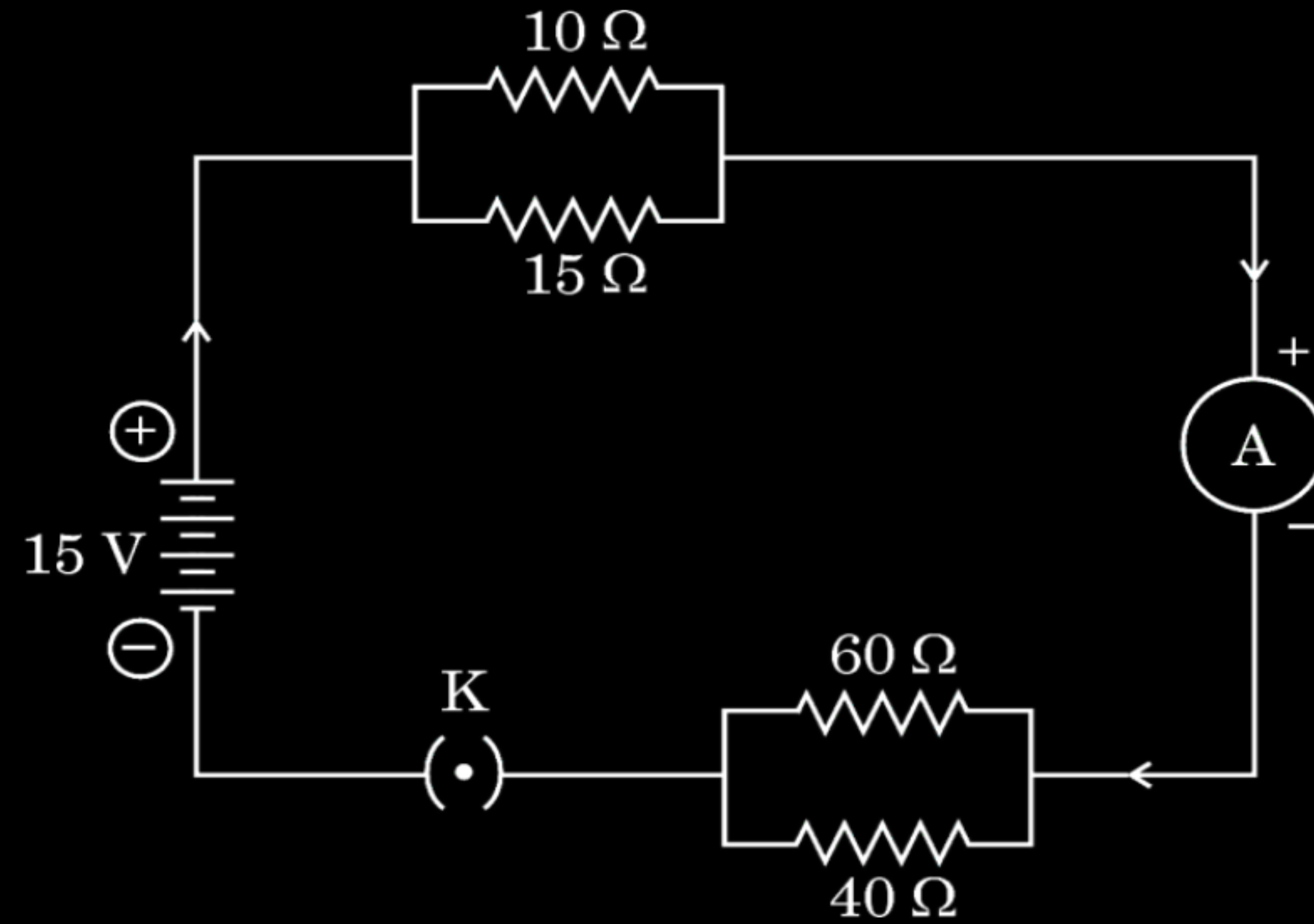
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

32. Consider the following electric circuit :



Calculate the values of the following :

- (a) The total resistance of the circuit
- (b) The total current drawn from the source
- (c) Potential difference across the parallel combination of 10 Ω and 15 Ω resistors

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(a)

$$\frac{1}{R_1} = \frac{1}{10\ \Omega} + \frac{1}{15\ \Omega} = \frac{1}{6\ \Omega} \Rightarrow R_1 = 6\ \Omega$$

$$\frac{1}{R_2} = \frac{1}{60\ \Omega} + \frac{1}{40\ \Omega} = \frac{100}{2400\ \Omega} \Rightarrow R_2 = 24\ \Omega$$

$\therefore R_1$ and R_2 are in series

$$\therefore R_{\text{total}} = R_1 + R_2 = (6 + 24) = 30\ \Omega$$

$$(b) \quad V = IR \quad \Rightarrow \quad I = \frac{V}{R} = \frac{15\ V}{30\ \Omega} = 0.5\ A$$

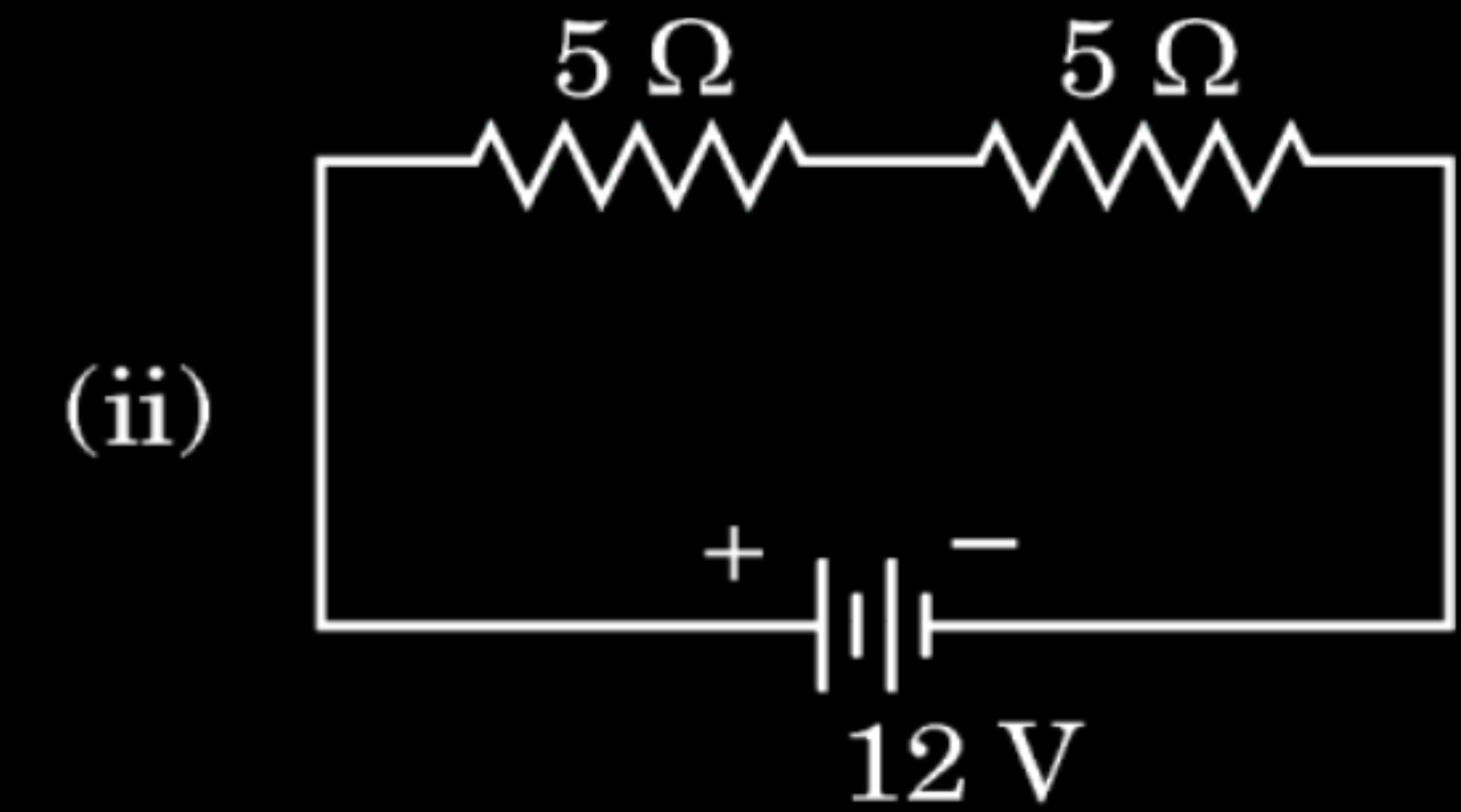
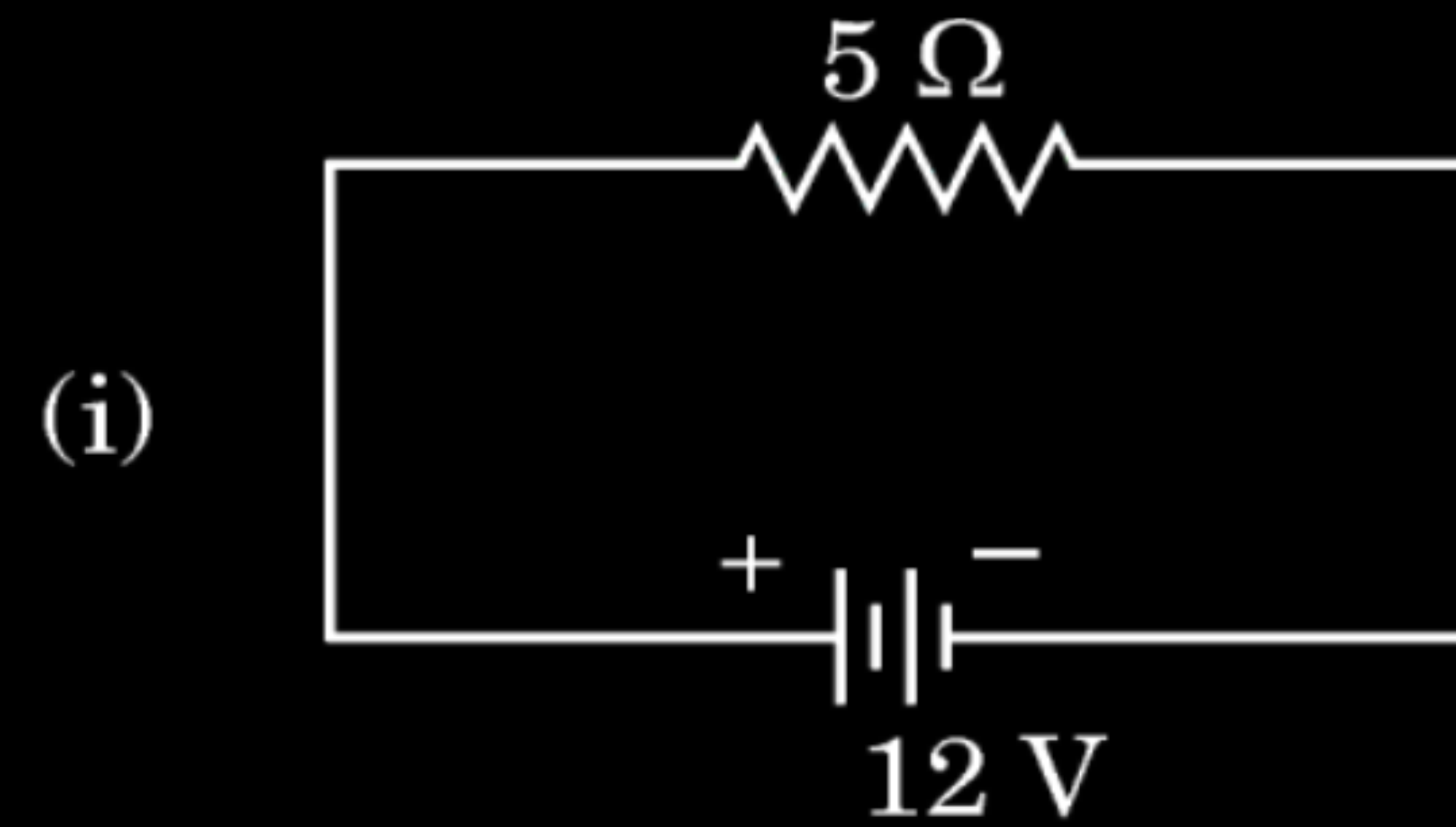
$$(c) \quad V = IR_1 = 0.5\ A \times 6\ \Omega = 3.0\ V$$

1

1

1

22. (a) Consider the following circuits :



In which circuit will the power dissipated in the circuit be
(I) minimum (II) maximum ? Justify your answer.

CBSE 2025

• (a) Power dissipated will be: • (I) minimum in circuit (ii) and • (II) maximum in circuit (iii) • • Justification : Power dissipated = $\frac{V^2}{R}$ • Since voltage is same in three cases, the power dissipation • will be inversely proportional to the resistance of the • circuit. • • (Award marks if explained through calculations)	1/2 1/2 1
---	---

27. (a) Define one volt potential difference between two points in an electric field.

CBSE 2025

(b) Draw a schematic diagram of an electric circuit of a cell of 1.5 V, $5\ \Omega$ and $10\ \Omega$ resistor and a plug key, all connected in series. Calculate the current drawn from the cell when the key is closed.

3

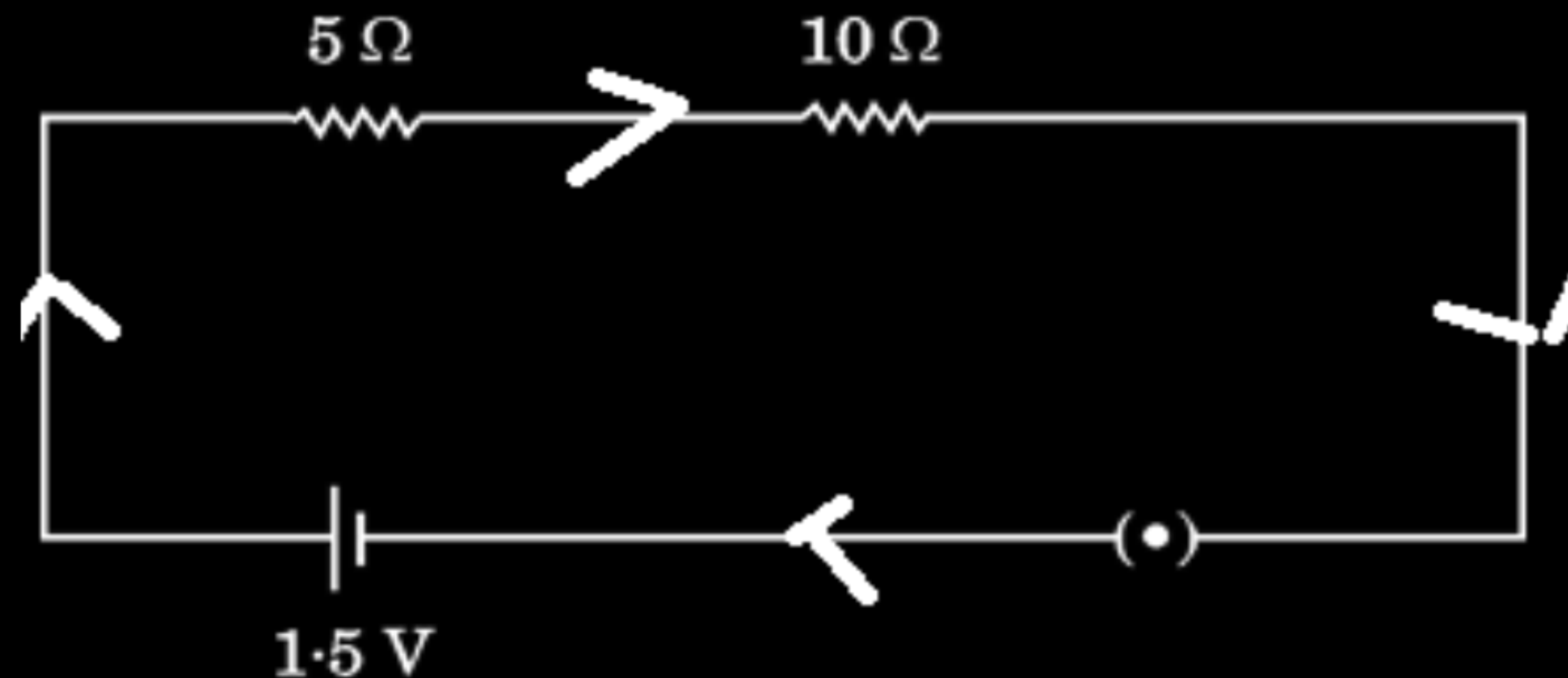
- 27.** (a) Define one volt potential difference between two points in an electric field. CBSE 2025
- (b) Draw a schematic diagram of an electric circuit of a cell of 1.5 V, 5 Ω and 10 Ω resistor and a plug key, all connected in series. Calculate the current drawn from the cell when the key is closed.

3

(a) It is the amount of work done to bring one coulomb charge from one point to another in the field is 1 joule. / $1\text{ V} = \frac{1\text{ J}}{1\text{ C}}$

1

(b)



1

$$\text{Current} = \frac{\text{Potential difference}}{\text{Resistance}} = \frac{1.5\text{ V}}{(5\ \Omega + 10\ \Omega)}$$

$$= \frac{1.5\text{ V}}{15\ \Omega} = 0.1\text{ A}$$

1

12. A piece of wire of resistance 'R' is cut lengthwise into three identical parts. These parts are then connected in parallel. If the equivalent resistance of this combination is R' , then the value of R/R' is :

1

(a) $1/9$

(b) $1/3$

(c) 3

(d) 9

CBSE 2025

12. A piece of wire of resistance 'R' is cut lengthwise into three identical parts. These parts are then connected in parallel. If the equivalent resistance of this combination is R' , then the value of R/R' is :

1

(a) $1/9$

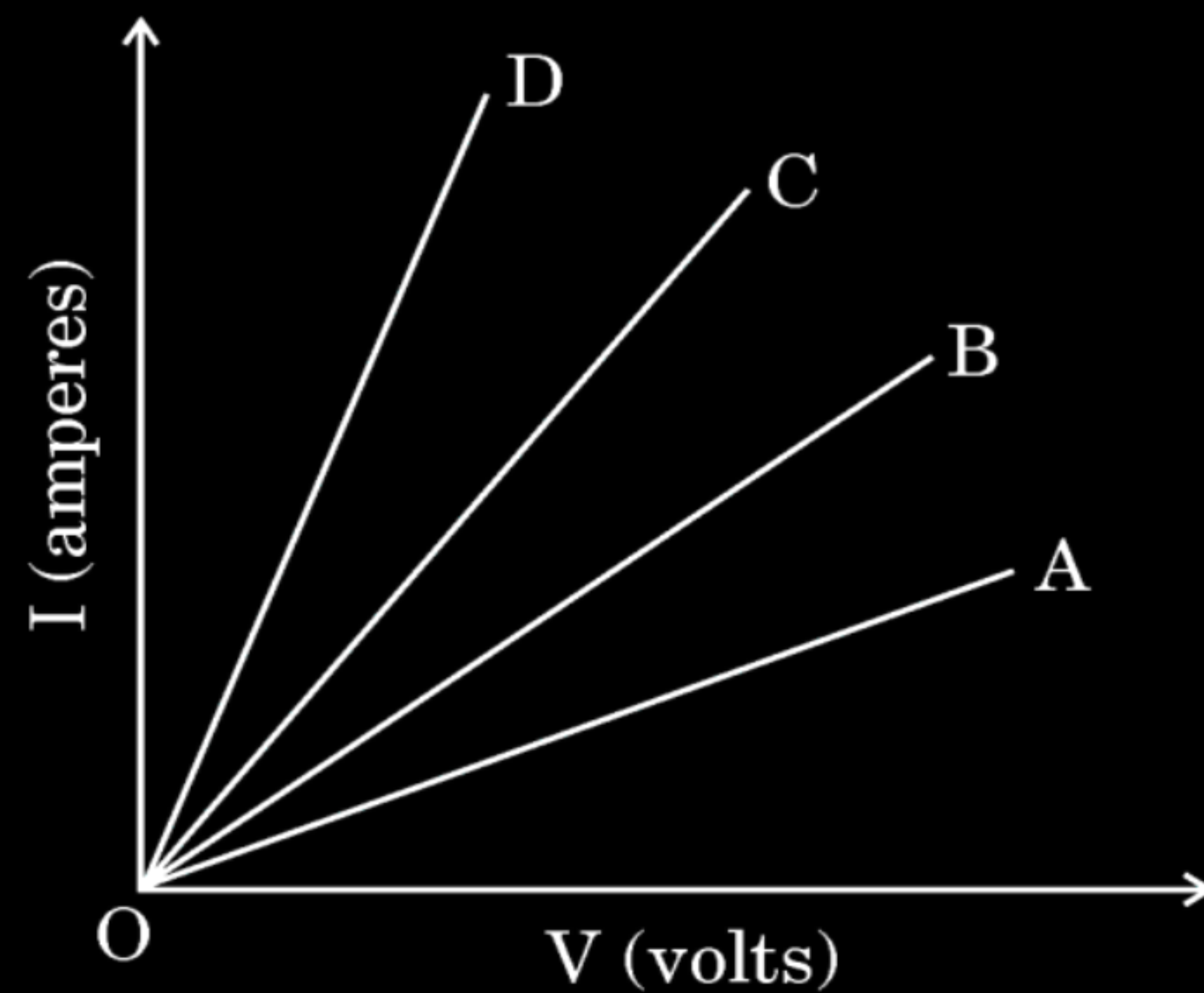
CBSE 2025

(b) $1/3$

(c) 3

(d) 9

(d)/9



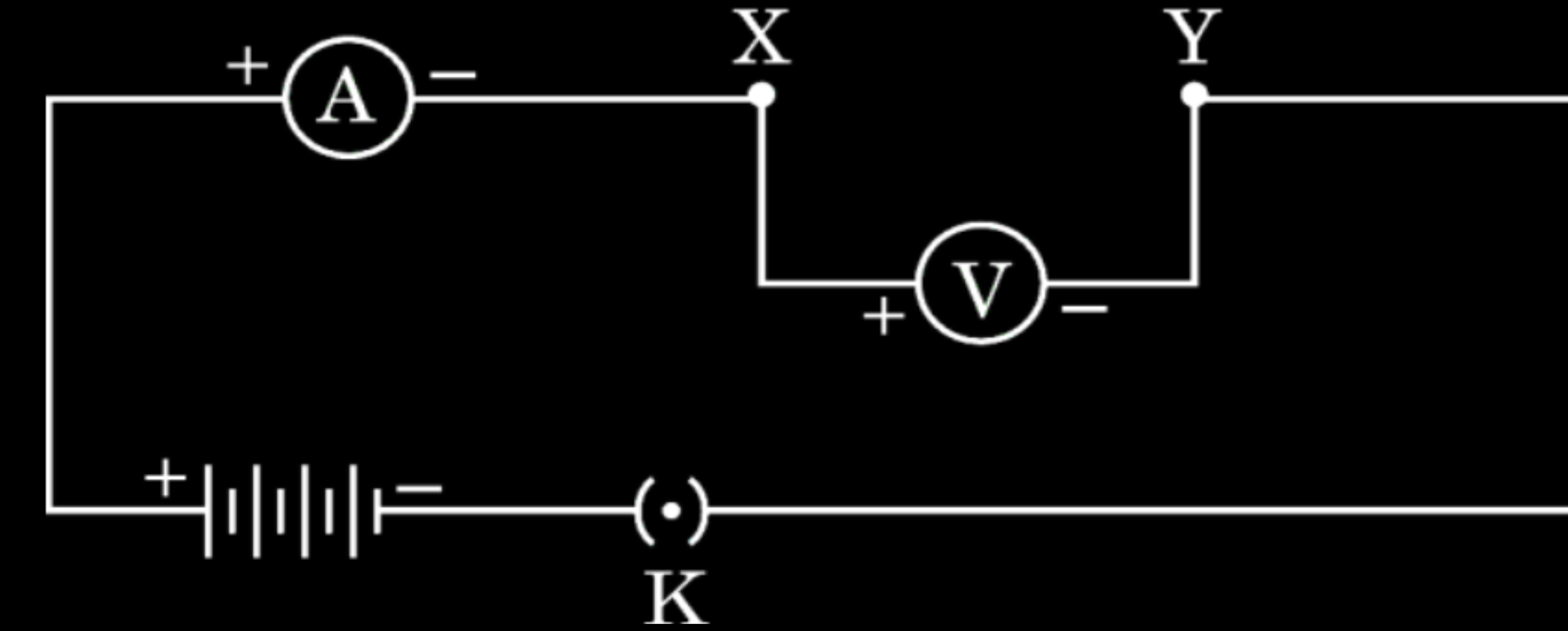
39. As shown in the diagram, an electric circuit consisting of an ammeter, a voltmeter, 4 cells of 1.5 V each, a plug key with a gap XY was set up. Voltmeter and ammeter readings were recorded in the observation table for four arrangements as given below :

Arrangement No. 1 – only resistor R_1 in gap XY

Arrangement No. 2 – only resistor R_2 in gap XY

Arrangement No. 3 – Resistors R_1 and R_2 in series in gap XY

Arrangement No. 4 – Resistors R_1 and R_2 in parallel in gap XY



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Based on the observations, four V – I graphs A, B, C and D as shown in figure were drawn. Study these graphs.

- (a) Which one of the graphs represents the series combination of R_1 and R_2 ? 1
- (b) Which one of these graphs represents the parallel combination of R_1 and R_2 ? 1
- (c) (i) Show an arrangement of three resistors, each of resistance $10\ \Omega$, so that the combination has a resistance of $15\ \Omega$. Give justification for your answer. 2

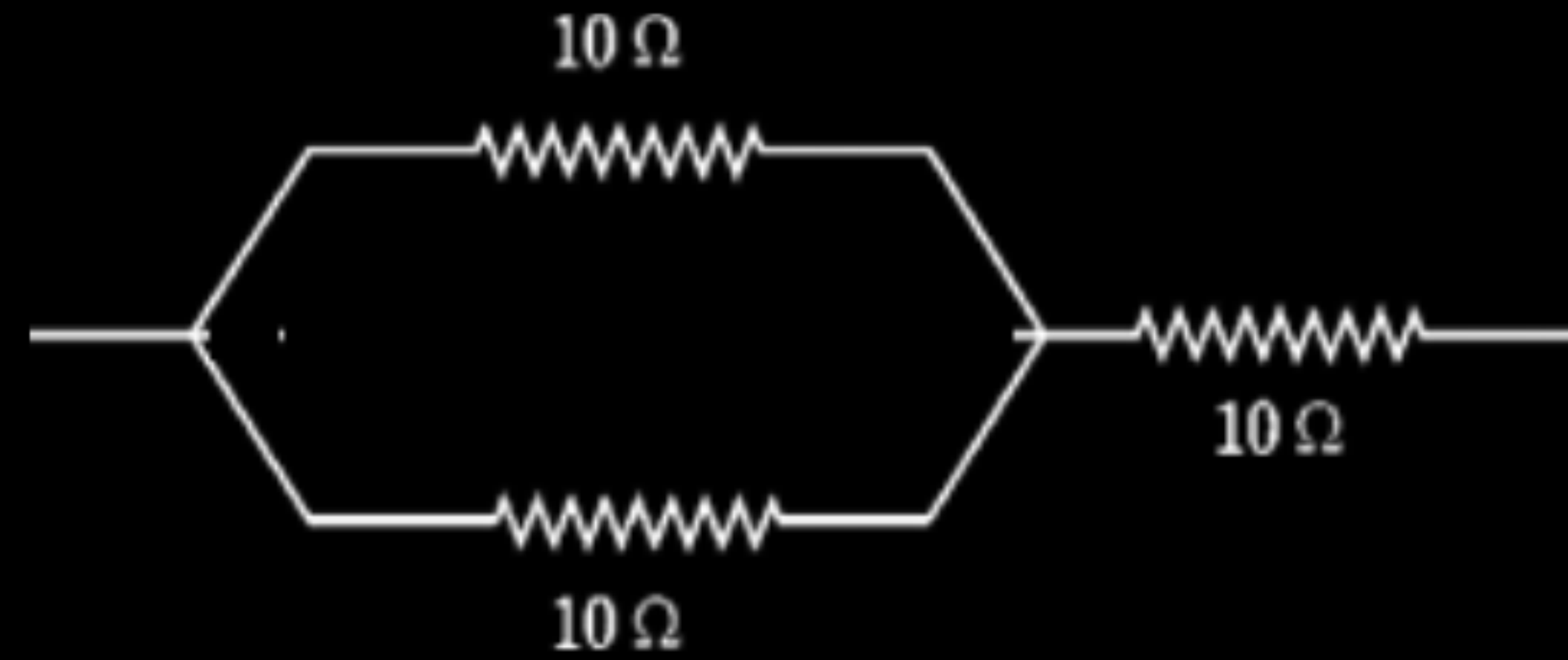
OR

- (c) (ii) A battery of 6 V is connected with a series combination of five resistors of $0.1\ \Omega$, $0.2\ \Omega$, $0.3\ \Omega$, $0.4\ \Omega$ and $0.5\ \Omega$. How much current would flow through the $0.3\ \Omega$ resistor ? Justify your answer. 2

(a) Graph A

(b) Graph D

(c) (i)



$$R = \frac{R_1 R_2}{R_1 + R_2} + R_3$$

$$R = \left(\frac{10 \times 10}{10 + 10} + 10 \right) \Omega = 5\ \Omega + 10\ \Omega = 15\ \Omega$$

OR

(c)

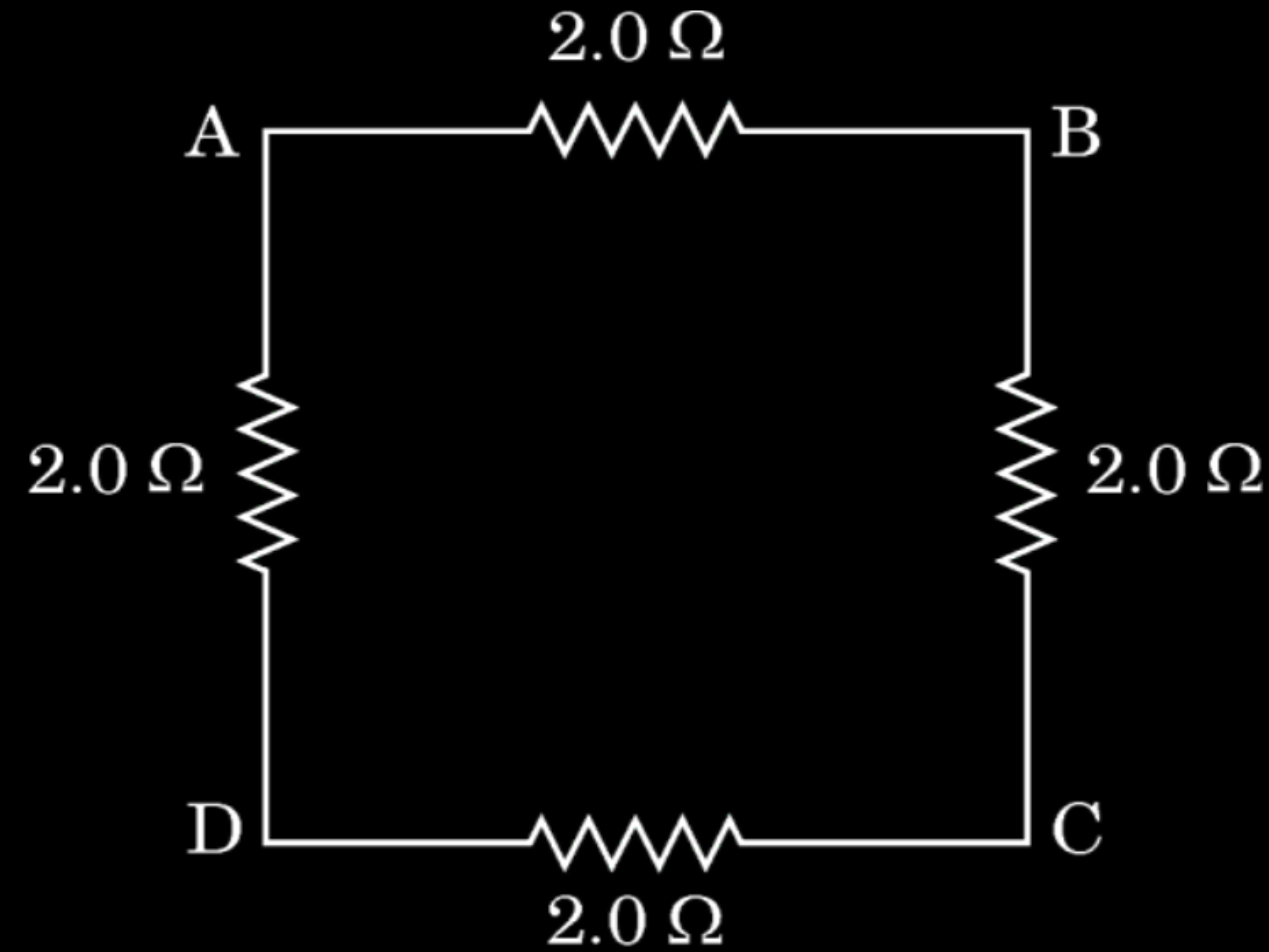
(ii) • $I = \frac{V}{R} = \frac{6\text{ V}}{(0.1 + 0.2 + 0.3 + 0.4 + 0.5)\ \Omega} = \frac{6\text{ V}}{1.5\ \Omega} = 4.0\text{ A}$

• same current flows when resistors are connected in series.

- 26.** Four resistors, each of resistance $2.0\ \Omega$, are joined end to end to form a square ABCD as shown. Using appropriate formula, determine the equivalent resistance of the combination between its two ends A and B.

2

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Between A and B, 3 resistors in series = AD + DC + CB in one branch

Equivalent resistance, $R_s = 2 \Omega + 2 \Omega + 2 \Omega = 6 \Omega$

Now one resistor in arm AD is in parallel combination with the other three.

Two branches:

$$R_1 = 2 \Omega$$

$$R_s = 6 \Omega$$

$$\therefore \frac{1}{R_p} = \frac{1}{6} + \frac{1}{2} = \frac{1+3}{6} = \frac{4}{6}$$

$$R_p = \frac{6}{4} = 1.5 \Omega$$

$$\frac{1}{2}$$

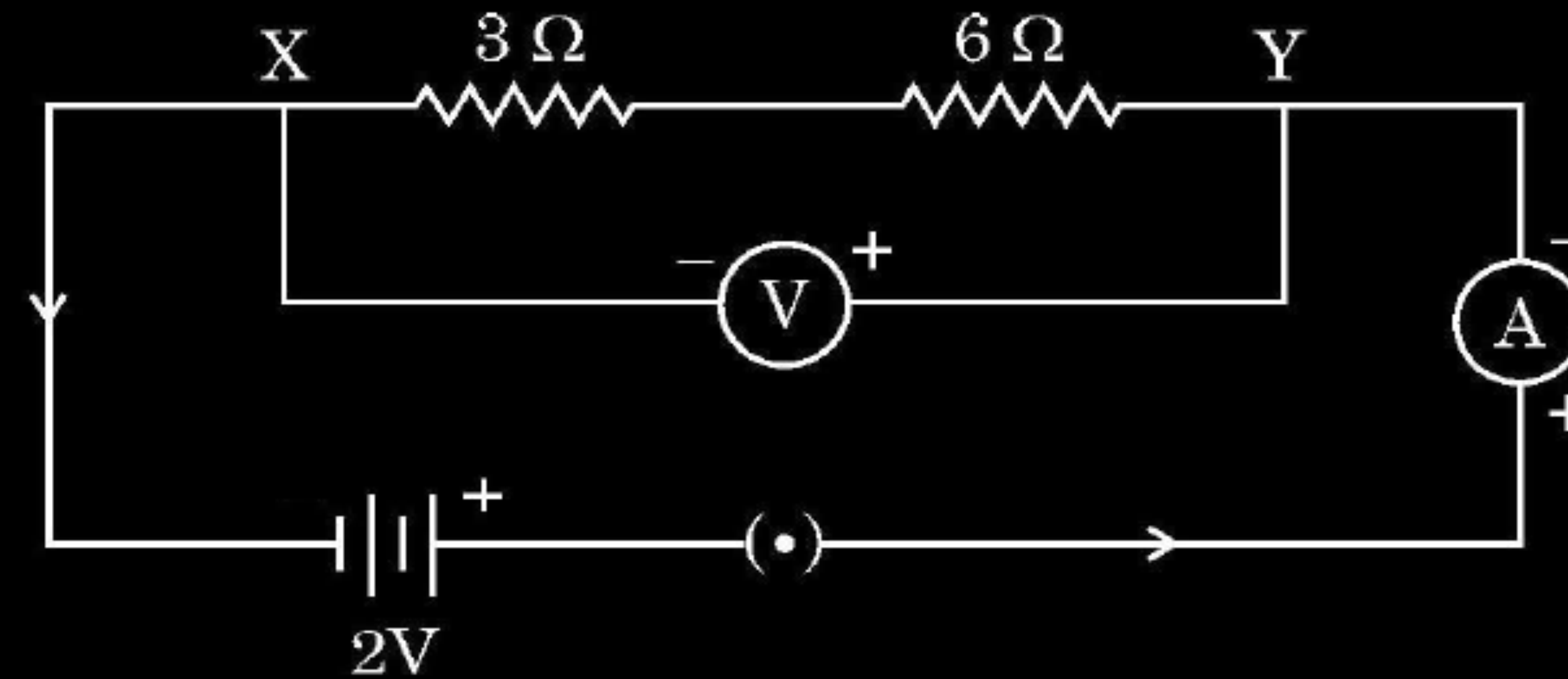
$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

39. Study the circuit shown in which two resistors X and Y of resistances $3\ \Omega$ and $6\ \Omega$ respectively are joined in series with a battery of 2 V .

4

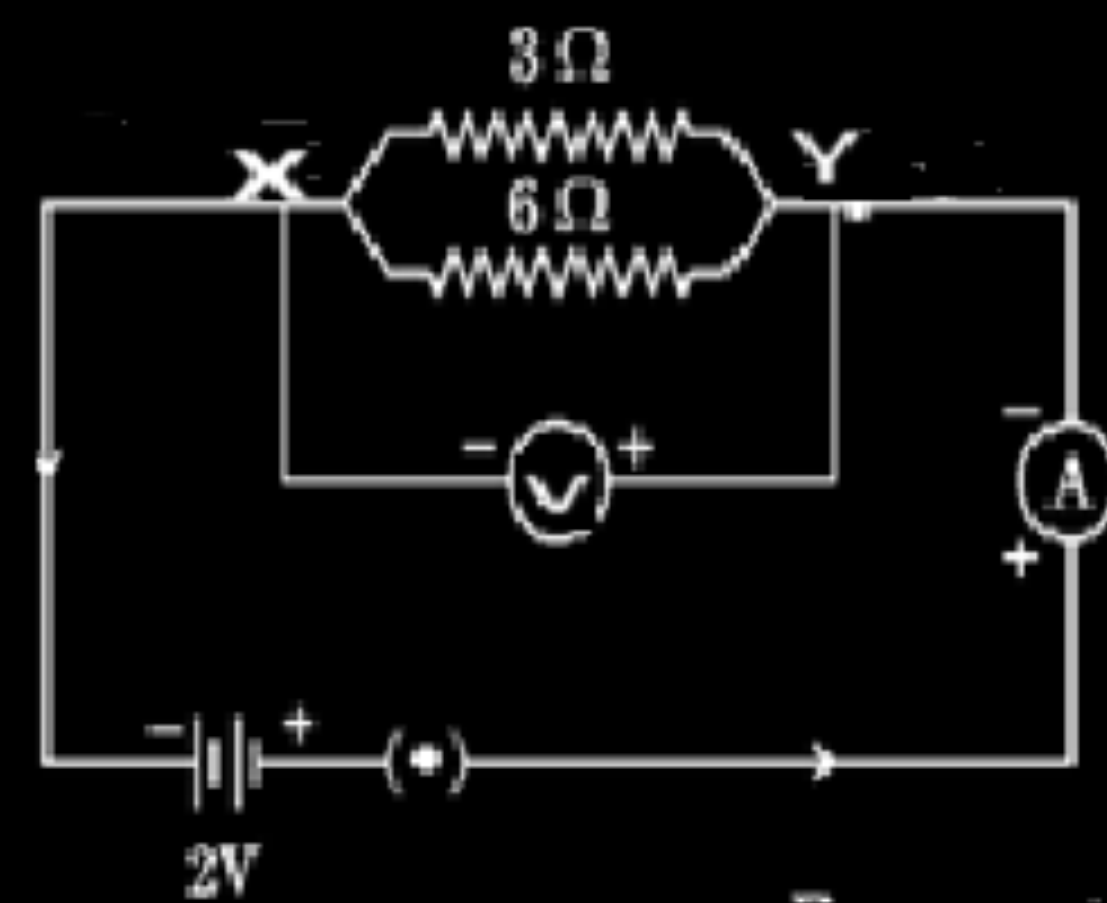


CBSE 2025

- (I) Draw a circuit diagram showing the above two resistors X and Y joined in parallel with same battery and same ammeter and voltmeter.
- (II) In which combination of resistors will the (i) potential difference across X and Y and (ii) current through X and Y, be the same ?
- (III) (a) Find the current drawn from the battery by the series combination of the two resistors (X and Y).

OR

- (III) (b) Determine the equivalent resistance of the parallel combination of the two resistors (X and Y).



(or Any other way)

(II)

- (i) in parallel combination.
- (ii) in series combination.

(III) (a) Resistance $R = 3\ \Omega + 6\ \Omega = 9\ \Omega$

$$V = 2V$$

$$I = \frac{V}{R} = \frac{2\ V}{9\ \Omega} = 0.22\ A$$

OR

$$(b) \quad \frac{1}{R} = \frac{1}{R_1} + \frac{1}{R}$$

$$= \frac{1}{3\ \Omega} + \frac{1}{6\ \Omega}$$

$$= \frac{6 + 3}{18\ \Omega}$$

$$\therefore R = 2\text{ohm}$$

1

$\frac{1}{2}$

$\frac{1}{2}$

1

1

$\frac{1}{2}$

$\frac{1}{2}$

1

22. (A) Determine the maximum and minimum resistance which can be obtained by joining five resistors of $\frac{1}{5} \Omega$ each.

CBSE 2025

22. (A) Determine the maximum and minimum resistance which can be obtained by joining five resistors of $\frac{1}{5} \Omega$ each.

CBSE 2025

(A) The maximum resistance is obtained when resistors are connected in series combination.

$$\text{Thus } R_s = 5 \times \frac{1}{5} \Omega = 1 \Omega$$

The minimum resistance is obtained when resistors are connected in parallel combination.

$$\frac{1}{R_p} = \frac{1}{\frac{1}{5}\Omega} + \frac{1}{\frac{1}{5}\Omega} + \frac{1}{\frac{1}{5}\Omega} + \frac{1}{\frac{1}{5}\Omega} + \frac{1}{\frac{1}{5}\Omega}$$

$$\frac{1}{R_p} = (5 + 5 + 5 + 5 + 5)$$

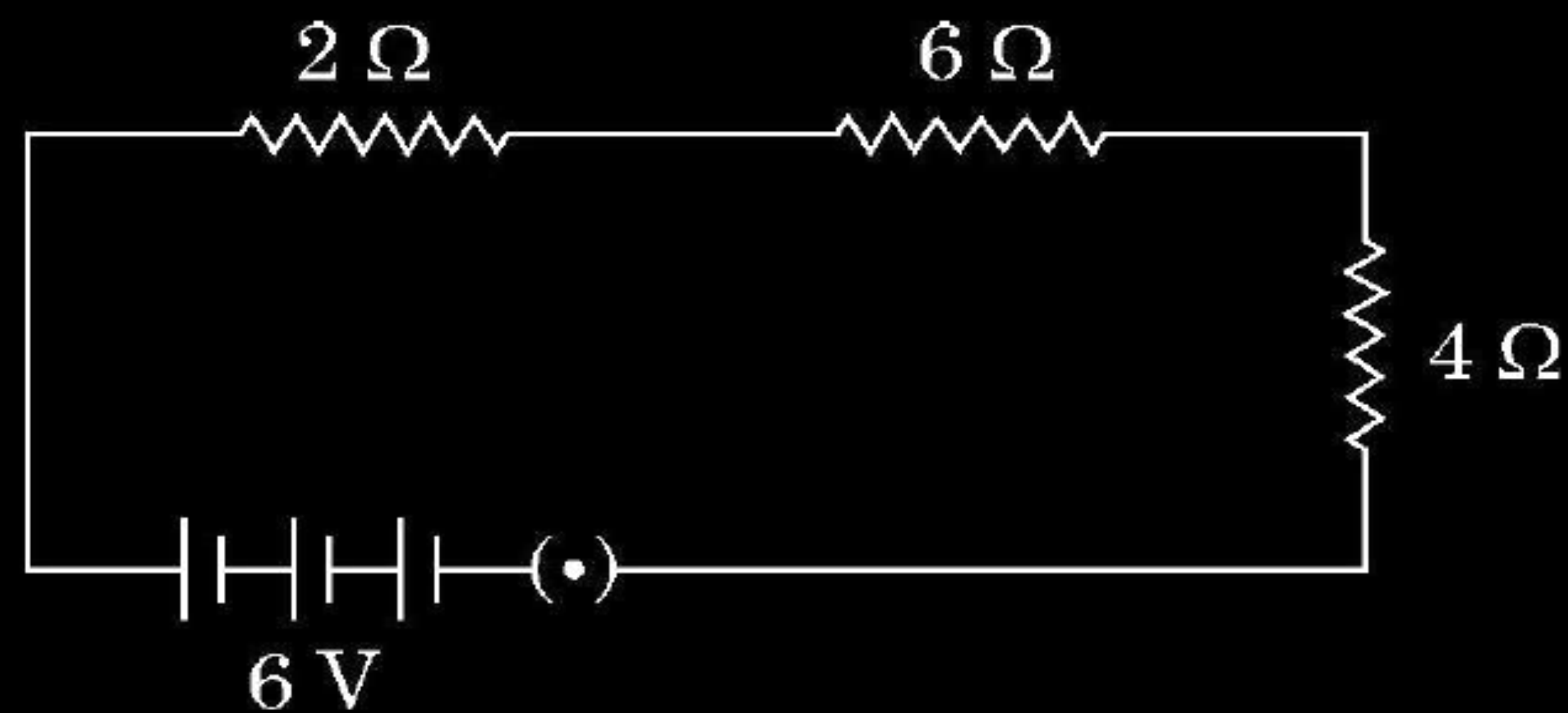
$$\therefore R_p = \frac{1}{25} \Omega$$

1

1

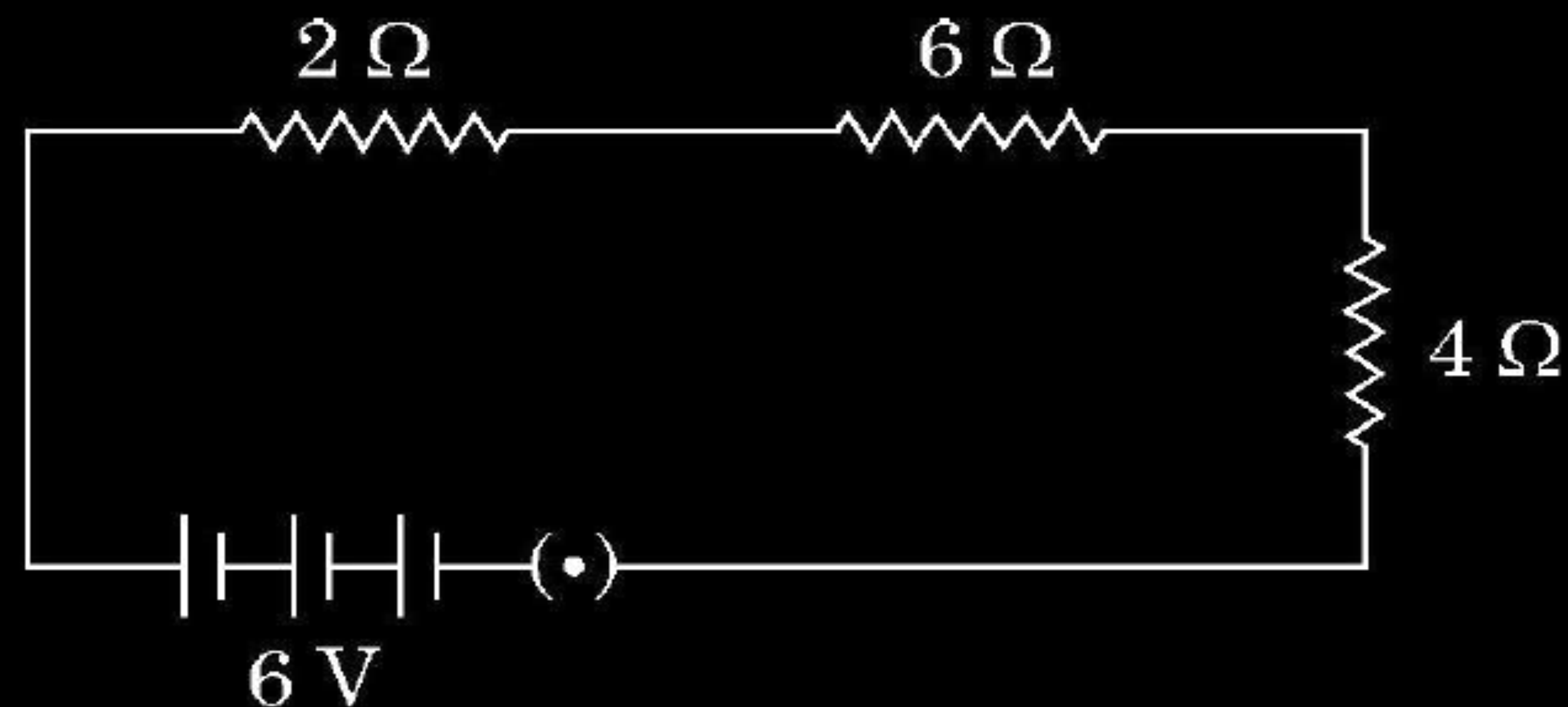
22. (A) Use Ohm's law to determine the potential difference across the $6\ \Omega$ resistor in the following circuit when key is closed :

CBSE 2025



22. (A) Use Ohm's law to determine the potential difference across the $6\ \Omega$ resistor in the following circuit when key is closed :

CBSE 2025



(A) Total resistance in the circuit

$$R = R_1 + R_2 + R_3 = 12\ \Omega$$

Voltage of the battery = 6 V

$$\therefore I = \frac{V}{R} = \frac{6}{12} = 0.5\ \text{A}$$

$$\therefore \text{Potential difference across } 6\ \Omega \text{ resistor} = 0.5\ \text{A} \times 6\ \Omega = 3.0\ \text{V}$$

$\frac{1}{2}$

1

$\frac{1}{2}$

- (b) Determine the total resistance of the parallel combination of three resistances of $2\ \Omega$, $4\ \Omega$ and $6\ \Omega$.

CBSE 2025

- (b) Determine the total resistance of the parallel combination of three resistances of $2\ \Omega$, $4\ \Omega$ and $6\ \Omega$.

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(b)

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

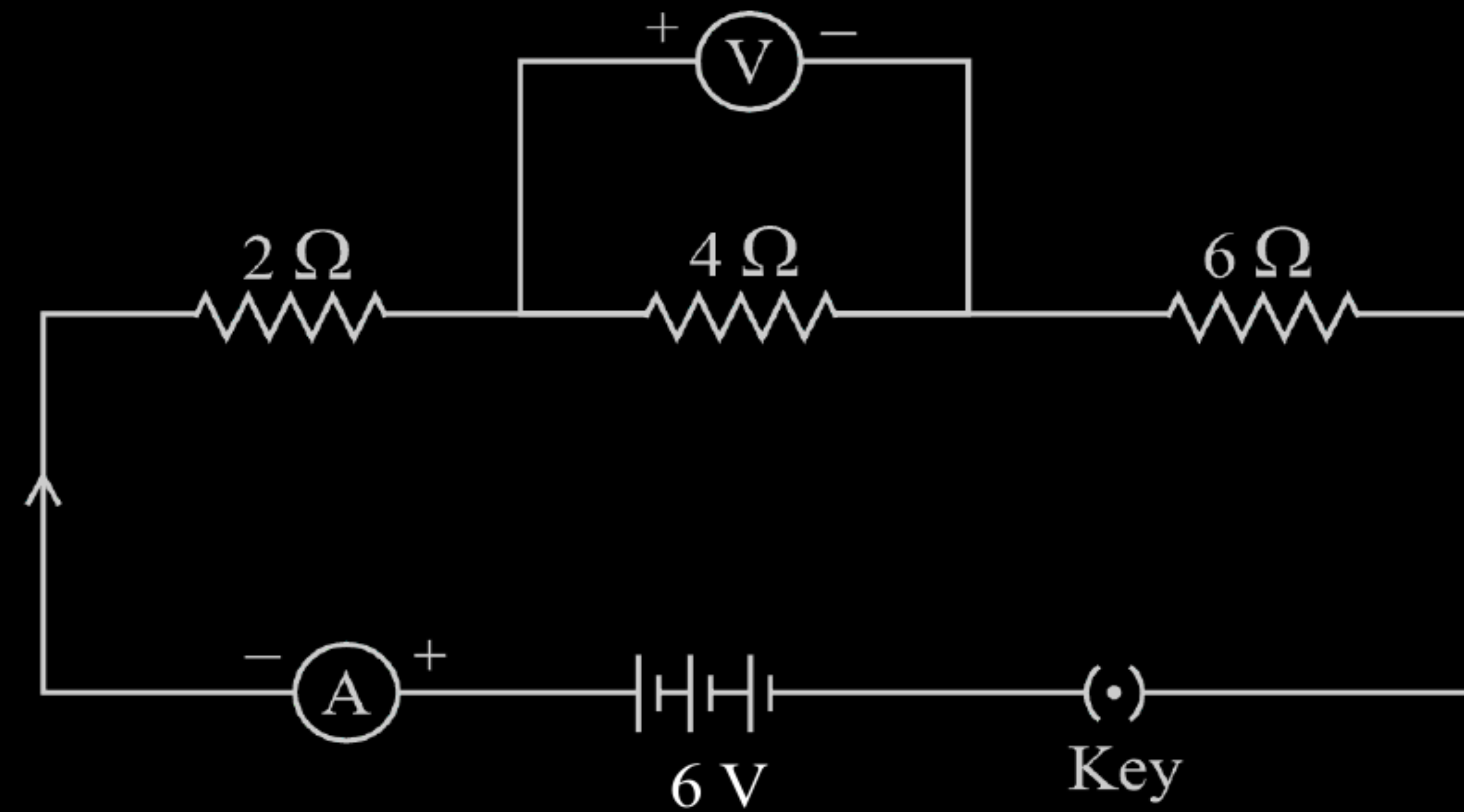
$$\frac{1}{R_p} = \frac{1}{2\ \Omega} + \frac{1}{4\ \Omega} + \frac{1}{6\ \Omega}$$

$$\therefore R_p = \frac{12}{11}\ \Omega = 1.09\ \Omega$$

$\frac{1}{2}$

$\frac{1}{2}$

29. In the given circuit when key is closed, determine the following :

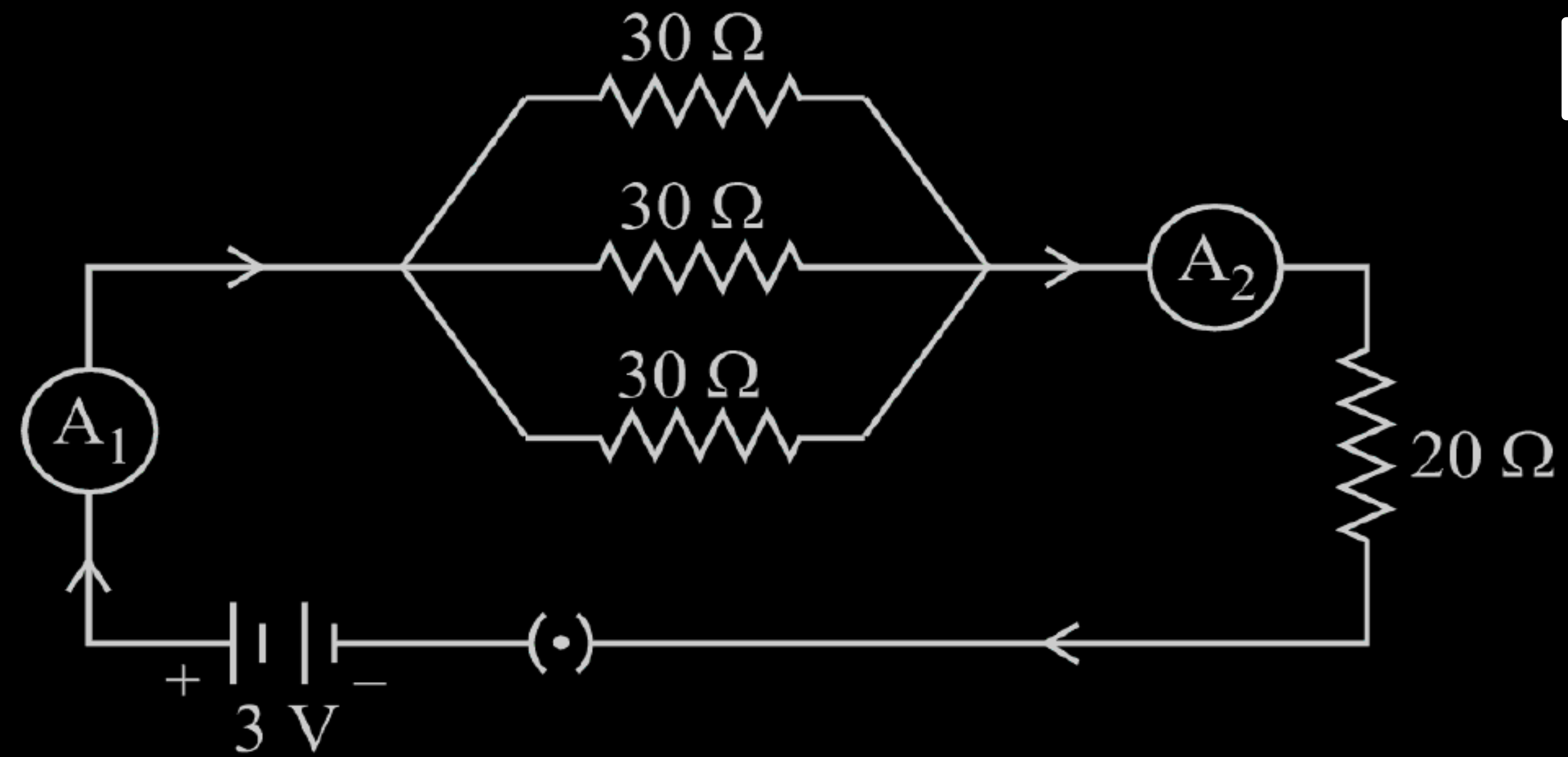


- (i) Total resistance offered by the three resistors
- (ii) Reading of the ammeter
- (iii) Reading of the voltmeter

CBSE 2024 C

(i) $R = 2\ \Omega + 4\ \Omega + 6\ \Omega = 12\ \Omega$	1
(ii) $I = \frac{V}{R} = \frac{6V}{12\Omega} = 0.5\ A$	1
(iii) $V = IR = 0.5A \times 4\Omega = 2V$	1

- (b) (i) You have three resistors of resistance R_1 , R_2 and R_3 . If the equivalent resistance of series and parallel combination of the three resistors are R_s and R_p respectively, write the relationship to determine R_s and R_p in terms of R_1 , R_2 and R_3 .
- (ii) On the basis of the given circuit answer the questions given below :



CBSE 2024 C

- (I) Write the relationship between the readings of the two ammeters A_1 and A_2 . Give reason for your answer.
- (II) What is the equivalent resistance of the parallel combination of three resistors ?
- (III) What is the value of current flowing in the circuit ?

(b)

(i)

$$R_s = R_1 + R_2 + R_3$$

1

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

1

(I)

- Both have same reading.
- As both are connected in series and in series current remains same.

$\frac{1}{2}$

$\frac{1}{2}$

$$(II) \frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$= \frac{1}{30} + \frac{1}{30} + \frac{1}{30}$$

$$= 10 \Omega$$

1

(III)

$$I = \frac{V}{R} = \frac{3V}{30\Omega} = \frac{1}{10} = 0.1A$$

1

3. Two resistors of $1\ \Omega$ and $3\ \Omega$ are connected in parallel in a circuit. If a current of 1 ampere is flowing through the circuit, the current flowing through the $3\ \Omega$ resistor is :

CBSE 2024

- (A) 1.0 ampere
- (B) 0.75 ampere
- (C) 0.33 ampere
- (D) 0.25 ampere

3. Two resistors of $1\ \Omega$ and $3\ \Omega$ are connected in parallel in a circuit. If a current of 1 ampere is flowing through the circuit, the current flowing through the $3\ \Omega$ resistor is : CBSE 2024

- (A) 1.0 ampere
- (B) 0.75 ampere
- (C) 0.33 ampere
- (D) 0.25 ampere

(D)/ 0.25 ampere

4. Two wires of same metal have the same length but their cross-sectional areas are in the ratio of 4 : 1. They are joined in series. The resistance of the thicker wire is $20\ \Omega$. The total resistance of the combination is :

CBSE 2024

- (A) $25\ \Omega$
- (B) $100\ \Omega$
- (C) $16\ \Omega$
- (D) $340\ \Omega$

4. Two wires of same metal have the same length but their cross-sectional areas are in the ratio of 4 : 1. They are joined in series. The resistance of the thicker wire is $20\ \Omega$. The total resistance of the combination is :

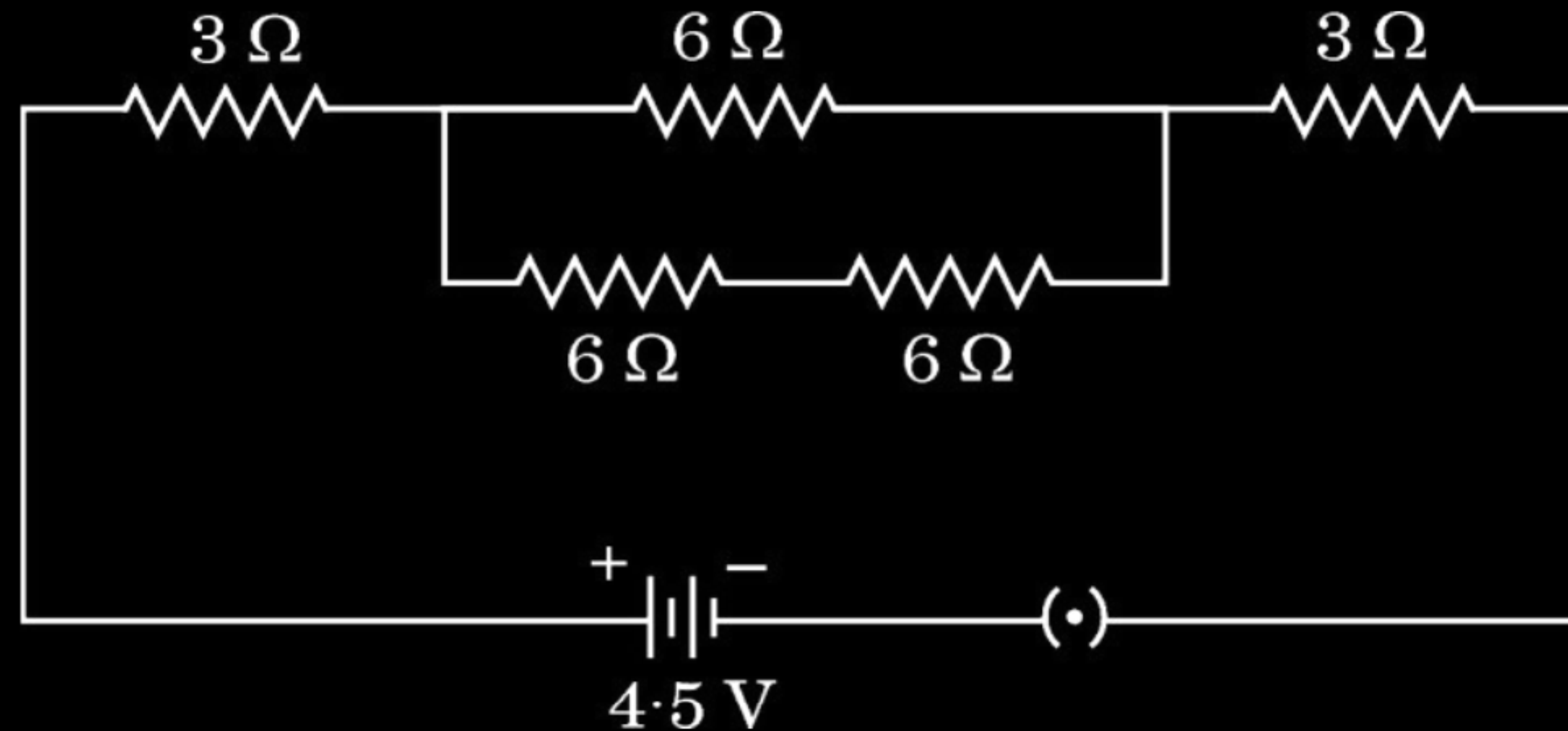
CBSE 2024

- (A) $25\ \Omega$
- (B) $100\ \Omega$
- (C) $16\ \Omega$
- (D) $340\ \Omega$

(B) / $100\ \Omega$

- 36.** (a) (i) Which type of circuits — series or parallel, should be used when you have to operate different electrical gadgets in your house ? List two reasons for your answer.
- (ii) Find the current flowing through the following electric circuit :

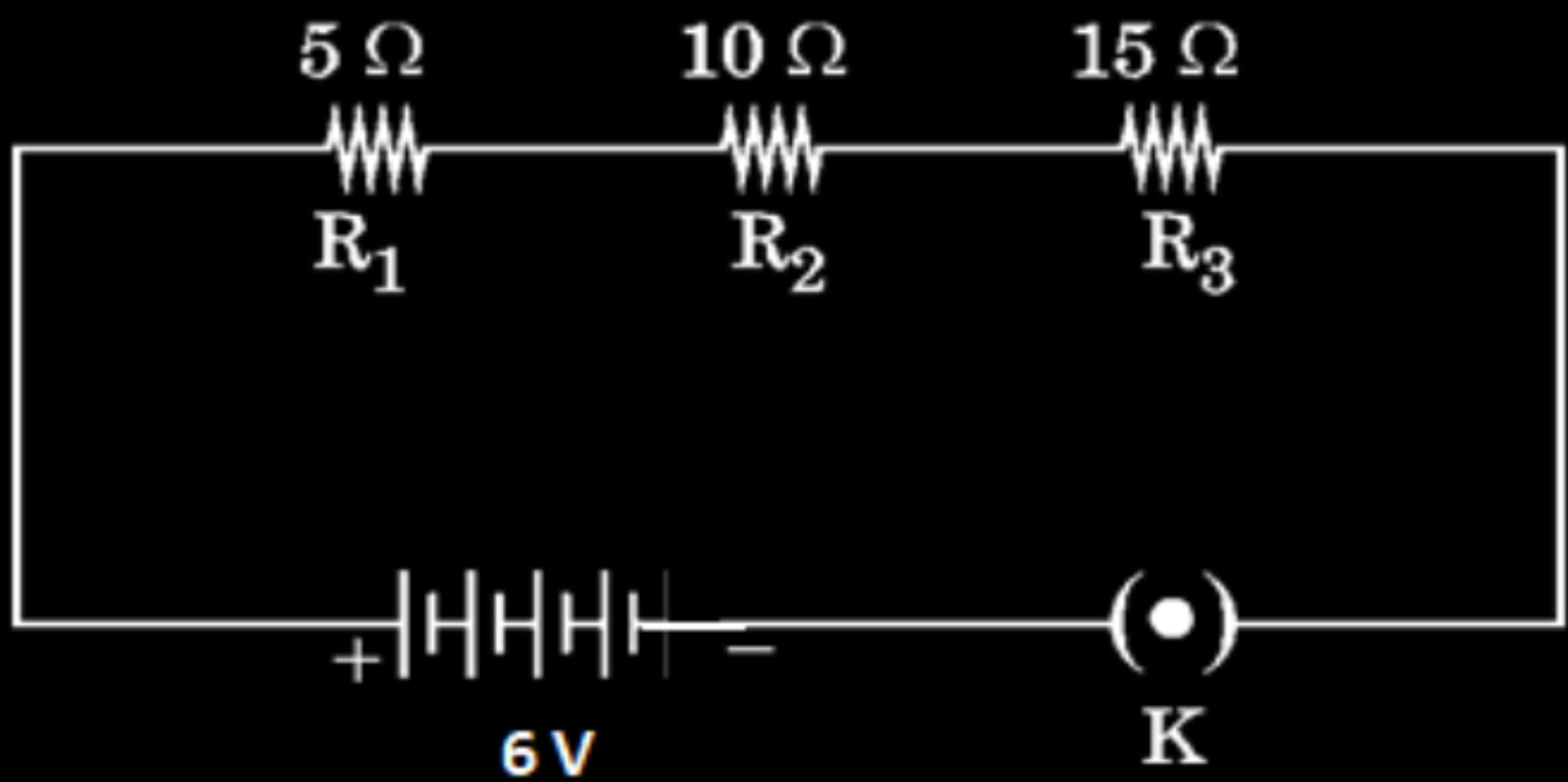
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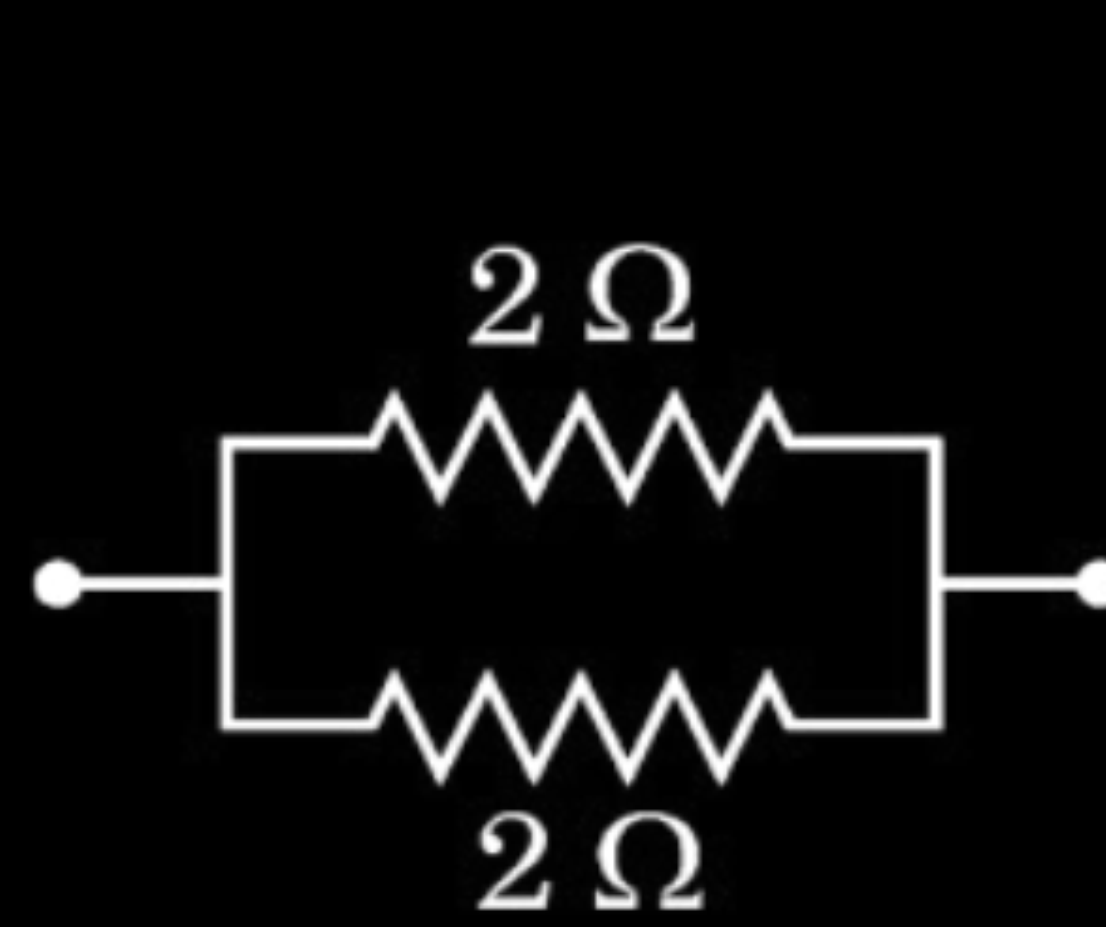
CBSE 2024

(a) (i) Parallel Circuit • Each electrical appliance has its own switch due to which it can be turned ON and OFF separately. • If one electrical appliance stops working, others remain unaffected. • Each appliance has equal potential difference and draws current as per its requirement. • The total resistance in parallel circuit decreases. (any two)	1 $\frac{1}{2} + \frac{1}{2}$
(ii) Combined resistance of the series, $R_1 = 6\ \Omega + 6\ \Omega = 12\ \Omega$ Combined resistance of parallel grouping of $6\ \Omega$ and $R_1 = 12\ \Omega$, resistors is R_2, where $\frac{1}{R_2} = \frac{1}{6} + \frac{1}{12} = \frac{9}{20}$ $R_2 = 4.0\ \Omega$ Total resistance of circuit = $R = 3 + 4 + 3 = 10\ \Omega$ Current flowing = $I = \frac{V}{R}$ $= \frac{4.5\ \text{V}}{10\ \Omega}$ $= 0.45\ \text{A}$	$\frac{1}{2}$ $\frac{1}{2}$ 1 $\frac{1}{2}$ $\frac{1}{2}$

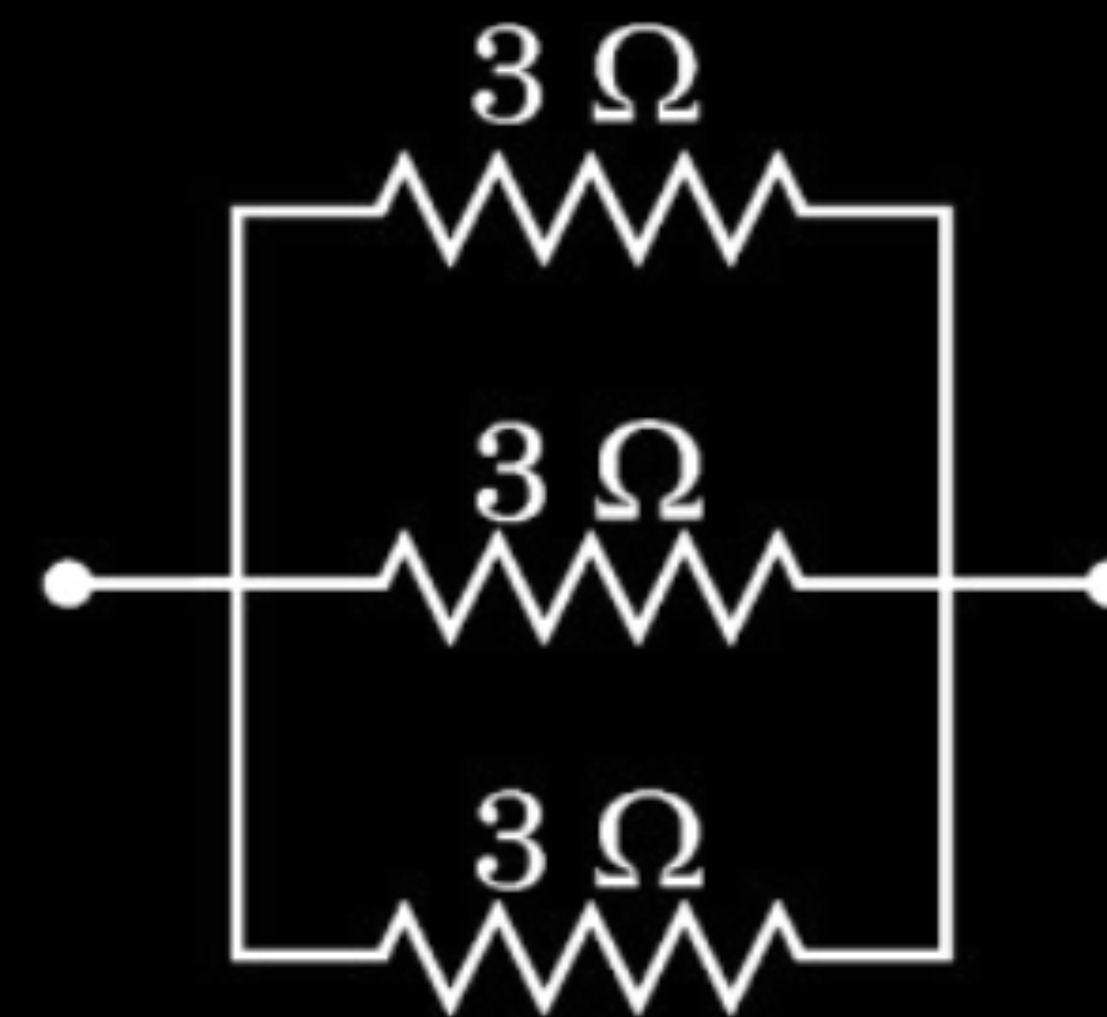
- 35.** (a) (i) The potential difference across the two ends of a circuit component is decreased to one-third of its initial value, while its resistance remains constant. What change will be observed in the current flowing through it ? Name and state the law which helps us to answer this question.
- (ii) Draw a schematic diagram of a circuit consisting of a battery of four 1.5 V cells, a $5\ \Omega$ resistor, a $10\ \Omega$ resistor and a $15\ \Omega$ resistor and a plug key, all connected in series. Now find (I) the electric current passing through the circuit, and (II) potential difference across the $10\ \Omega$ resistor when the plug key is closed.

(a)	(i) • Current becomes one-third of its initial value. • Ohm's Law The potential difference across the ends of a conductor is directly proportional to the current flowing through it, provided its temperature remains the same.	$\frac{1}{2}$ $\frac{1}{2}$
(ii)	 The diagram shows a rectangular circuit loop. At the top, three resistors are connected in series. The first resistor is labeled $5\ \Omega$ and R_1. The second is labeled $10\ \Omega$ and R_2. The third is labeled $15\ \Omega$ and R_3. At the bottom, there is a 6V battery (represented by four cells) and a key labeled K (represented by a circle with a dot). The battery is on the left and the key is on the right.	1 1
(I)	Total Voltage = $V = 4 \times 1.5\text{ V} = 6\text{ V}$ Total resistance, $R(s) = R_1 + R_2 + R_3$ $= 5\ \Omega + 10\ \Omega + 15\ \Omega = 30\ \Omega$ Current, $I = \frac{V}{R} = \frac{6\text{ V}}{30\ \Omega} = 0.2\text{ A}$	1
(II)	$V = IR = 0.2\text{ A} \times 10\ \Omega = 2\text{ V}$	1

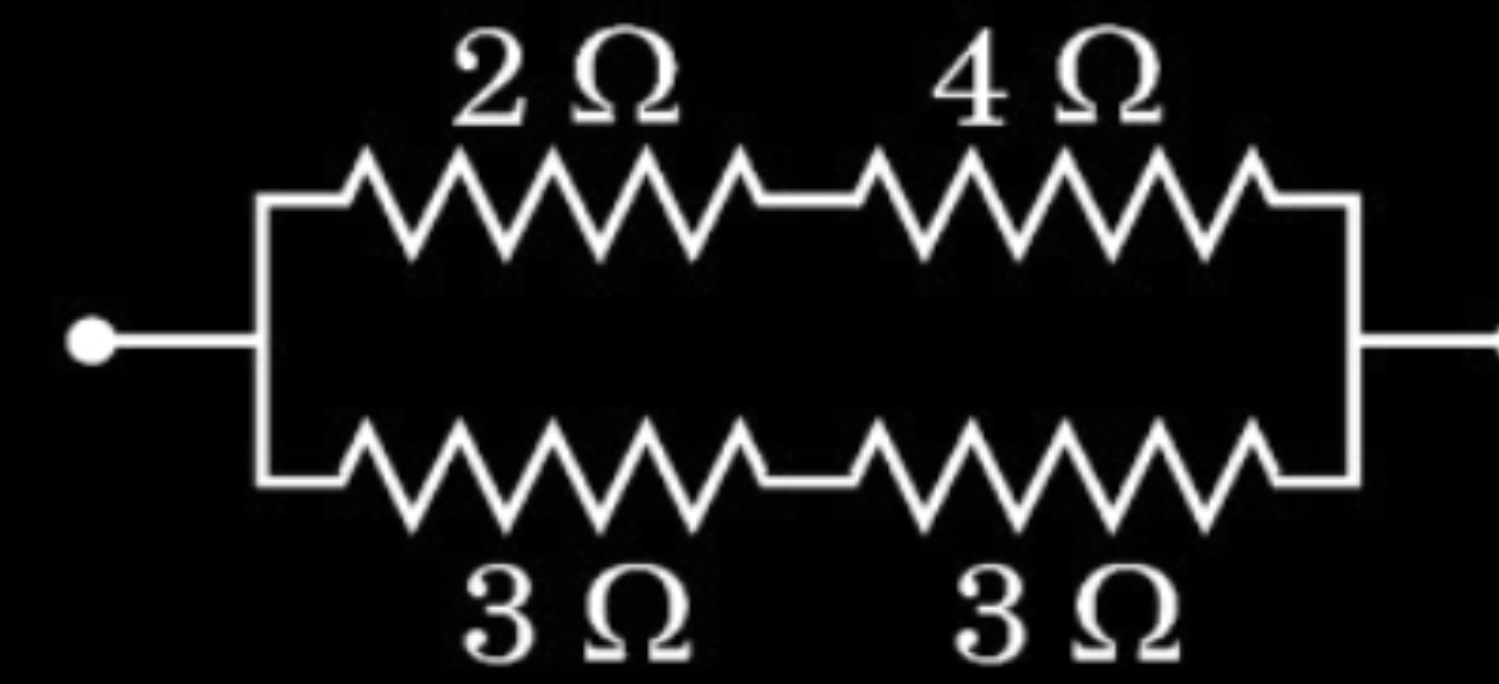
16. Consider the following combinations of resistors :



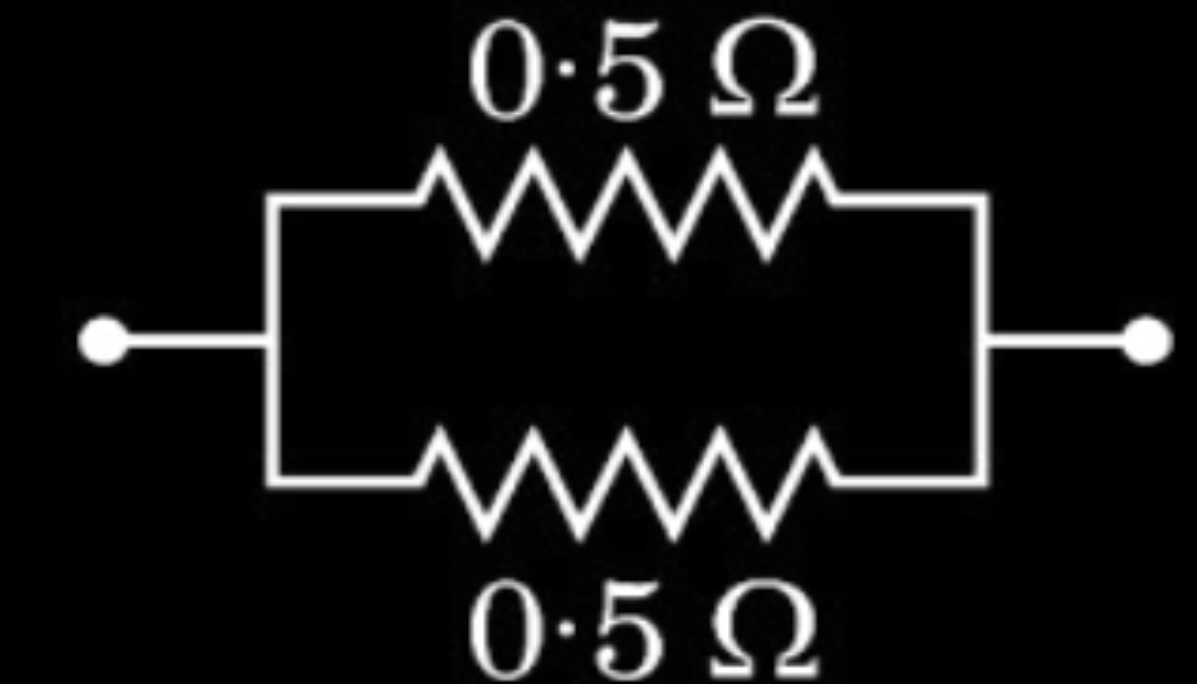
I



II



III



IV

The combinations having equivalent resistance $1\ \Omega$ is/are :

(A) I and IV

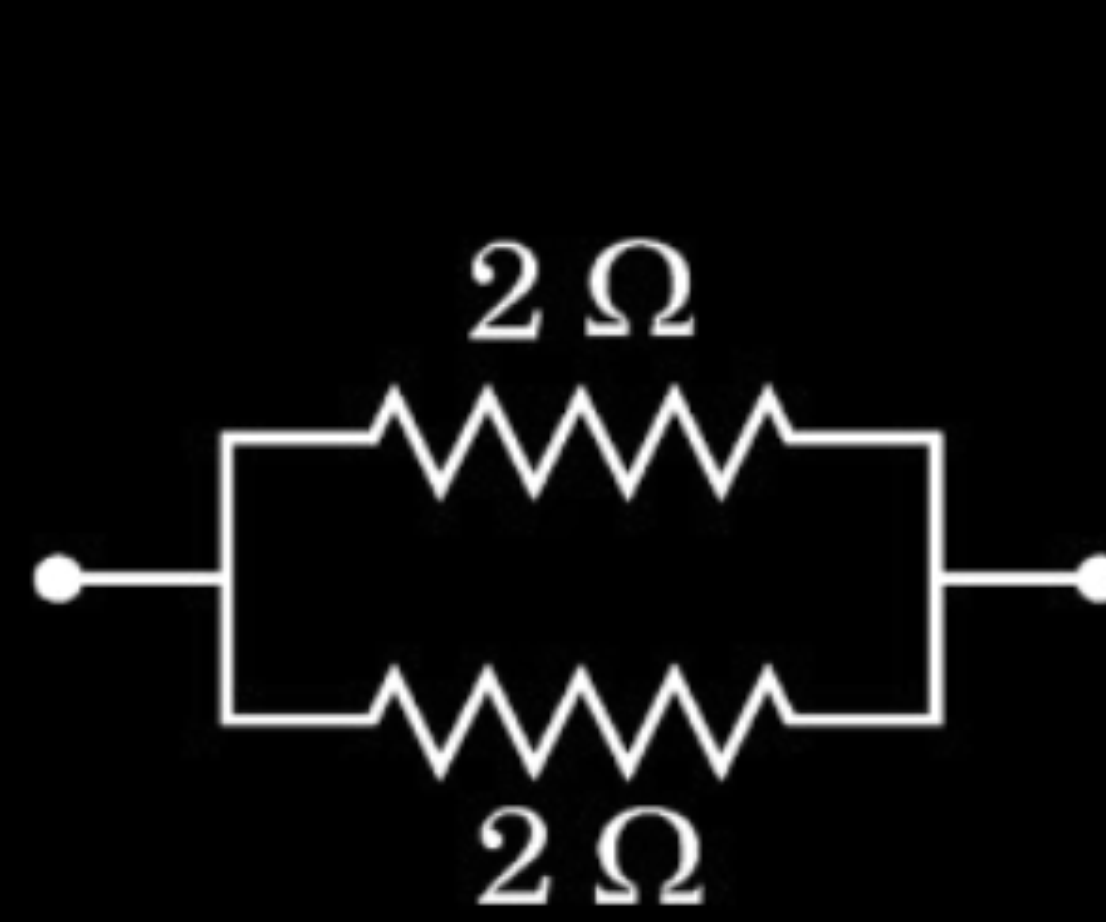
(B) Only IV

(C) I and II

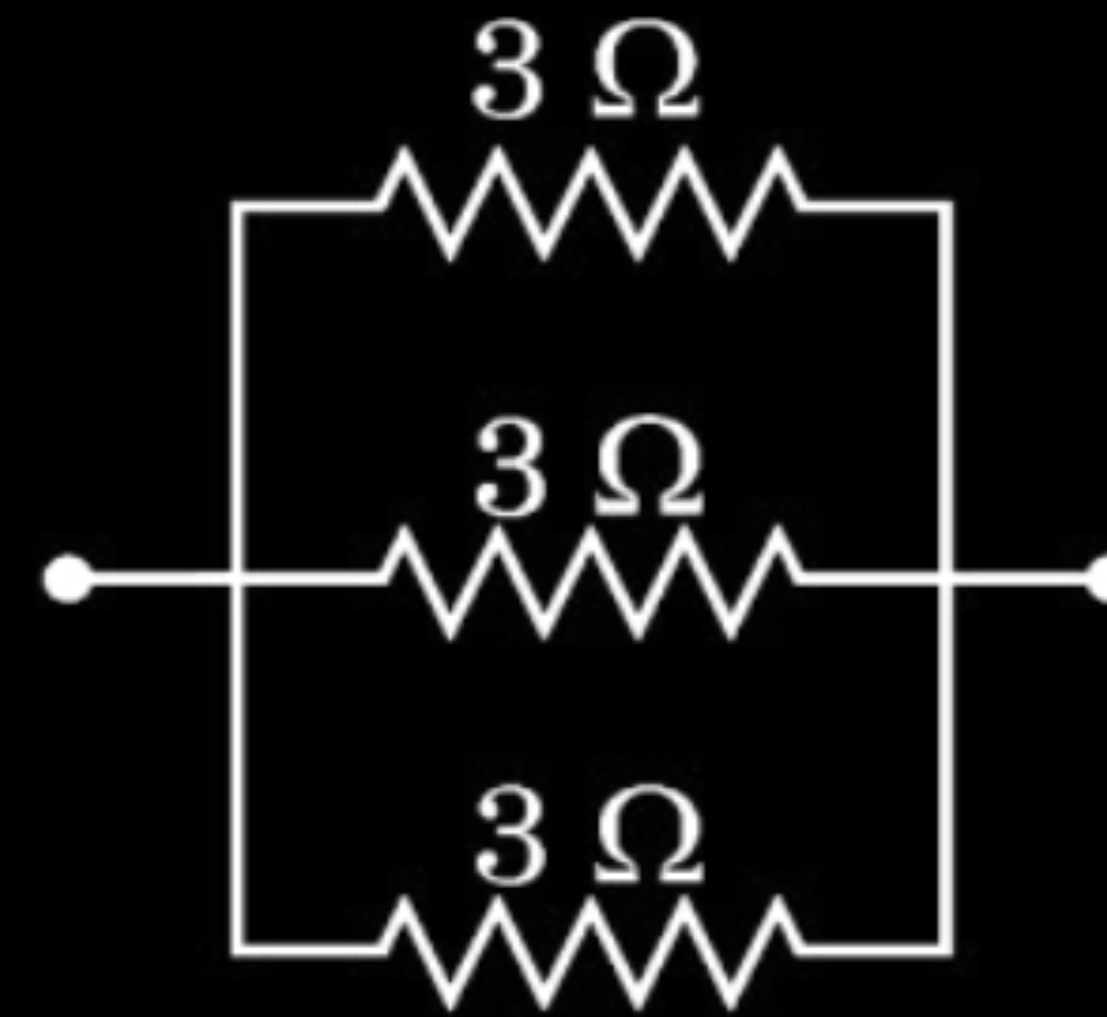
(D) I, II and III

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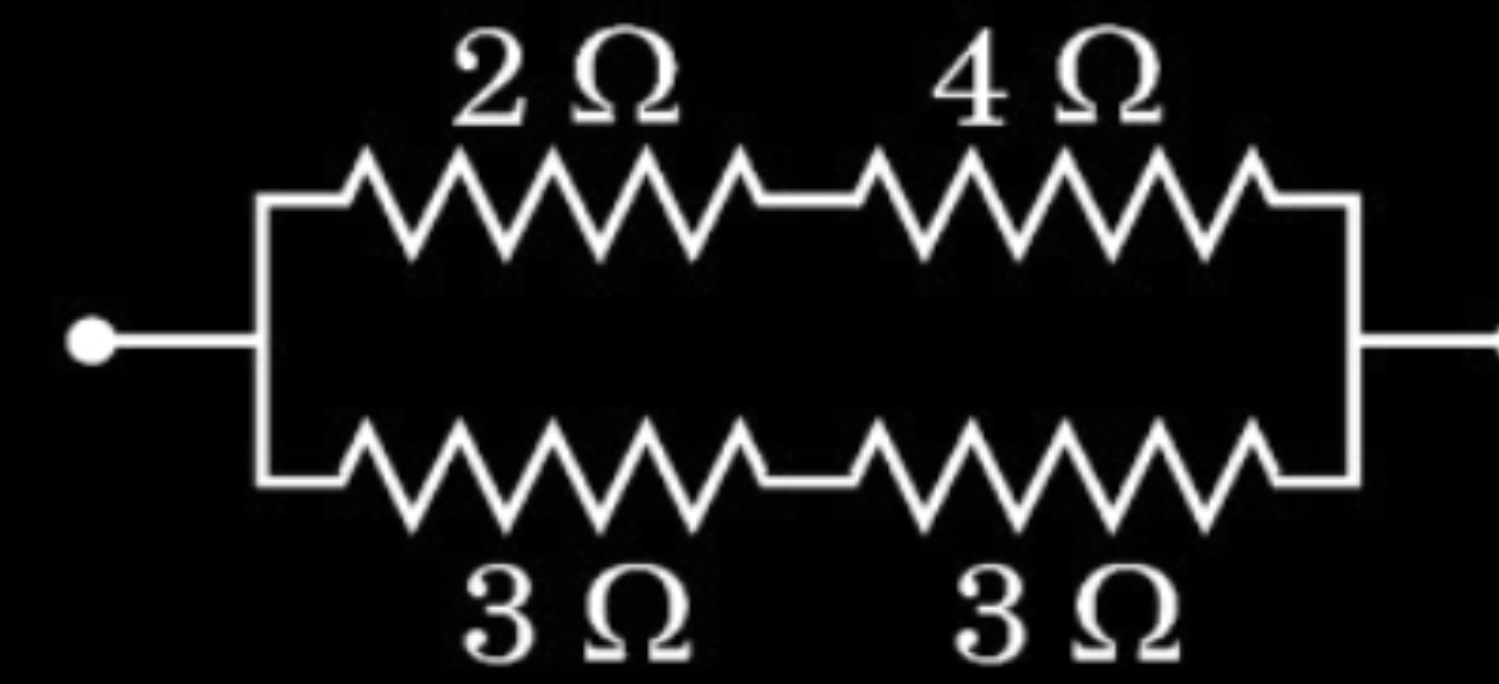
16. Consider the following combinations of resistors :



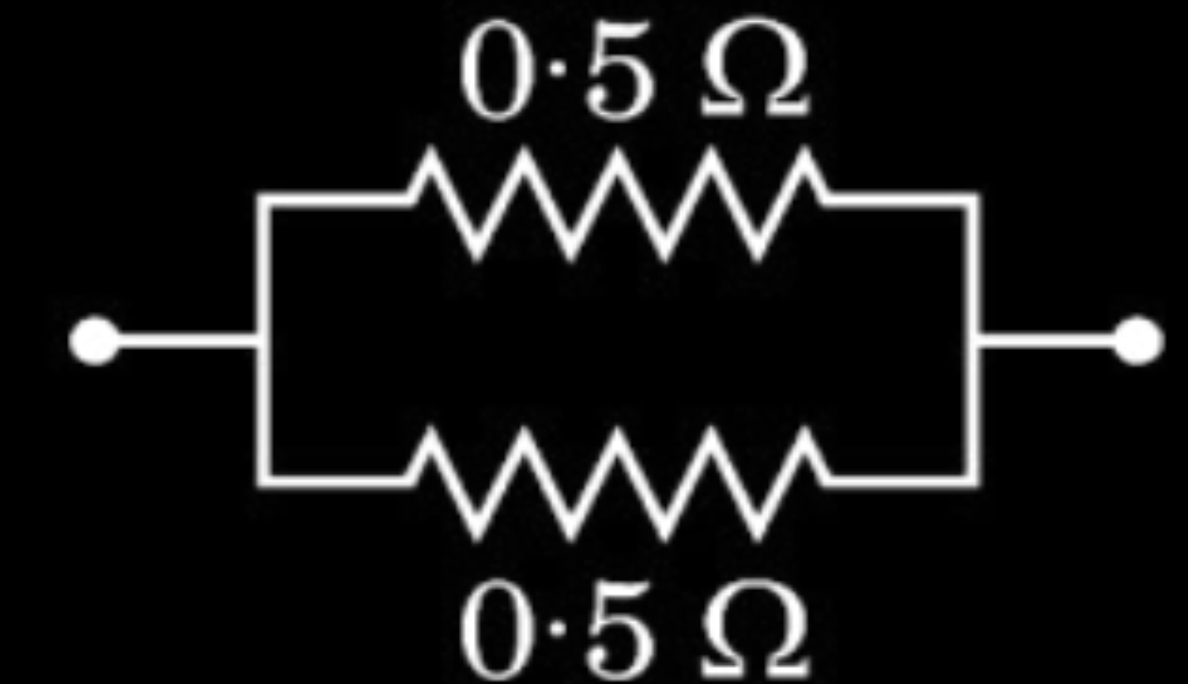
I



II



III



IV

The combinations having equivalent resistance $1\ \Omega$ is/are :

(A) I and IV

(B) Only IV

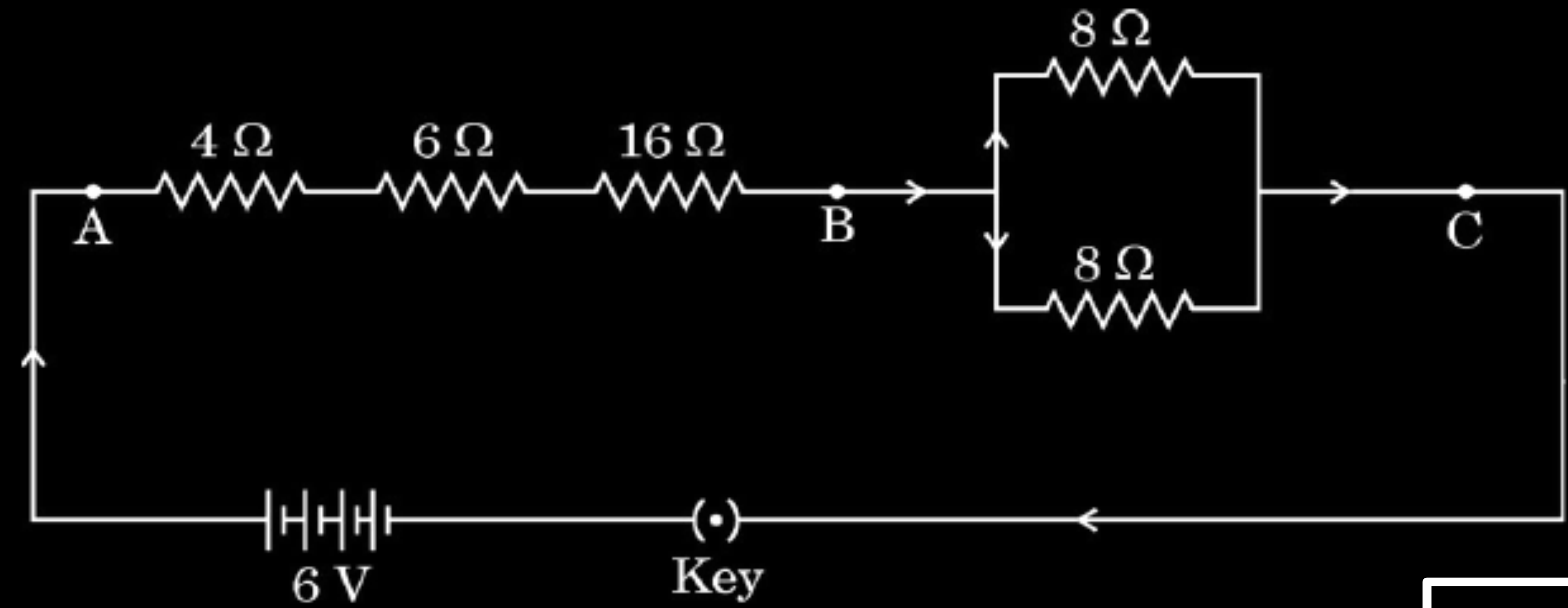
(C) I and II

(D) I, II and III

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(C)/I and II

37. Study the following circuit :



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On the basis of this circuit, answer the following questions :

- (a) Find the value of total resistance between the points A and B. 1
- (b) Find the resistance between the points B and C. 1
- (c) (i) Calculate the current drawn from the battery, when the key is closed. 2

OR

- (c) (ii) In the above circuit, the $16\ \Omega$ resistor or the parallel combination of two resistors of $8\ \Omega$, which one of the two will have more potential difference across its two ends ? Justify your answer. 2

$$(a) \quad R_S = 4 \, \Omega + 6 \, \Omega + 16 \, \Omega = 26 \, \Omega$$

$$(b) \quad \frac{1}{R_P} = \frac{1}{8 \, \Omega} + \frac{1}{8 \, \Omega} = \frac{1}{4} \, \Omega$$

$$R_P = 4 \, \Omega$$

$$(c) (i) \quad \text{Total resistance} = 26 \, \Omega + 4 \, \Omega = 30 \, \Omega$$

$$\text{Potential difference} = V = 6V$$

$$\text{Current } I = \frac{V}{R}$$

$$\frac{6}{30} = \frac{1}{5} \, A \quad \text{or } 0.2 \, A.$$

OR

$$(c)(ii) \quad 16 \, \Omega$$

Justification: According to Ohm's law when same current flows, the potential difference across a higher resistance is always higher./

$$\text{Potential difference across } 16 \, \Omega = V = IR = 0.2 \times 16 = 3.2V$$

$$\text{Potential difference across } 8 \, \Omega = V = IR_{(total)} = 0.2 \times 4 = 0.8V$$

1

1

1

$\frac{1}{2}$

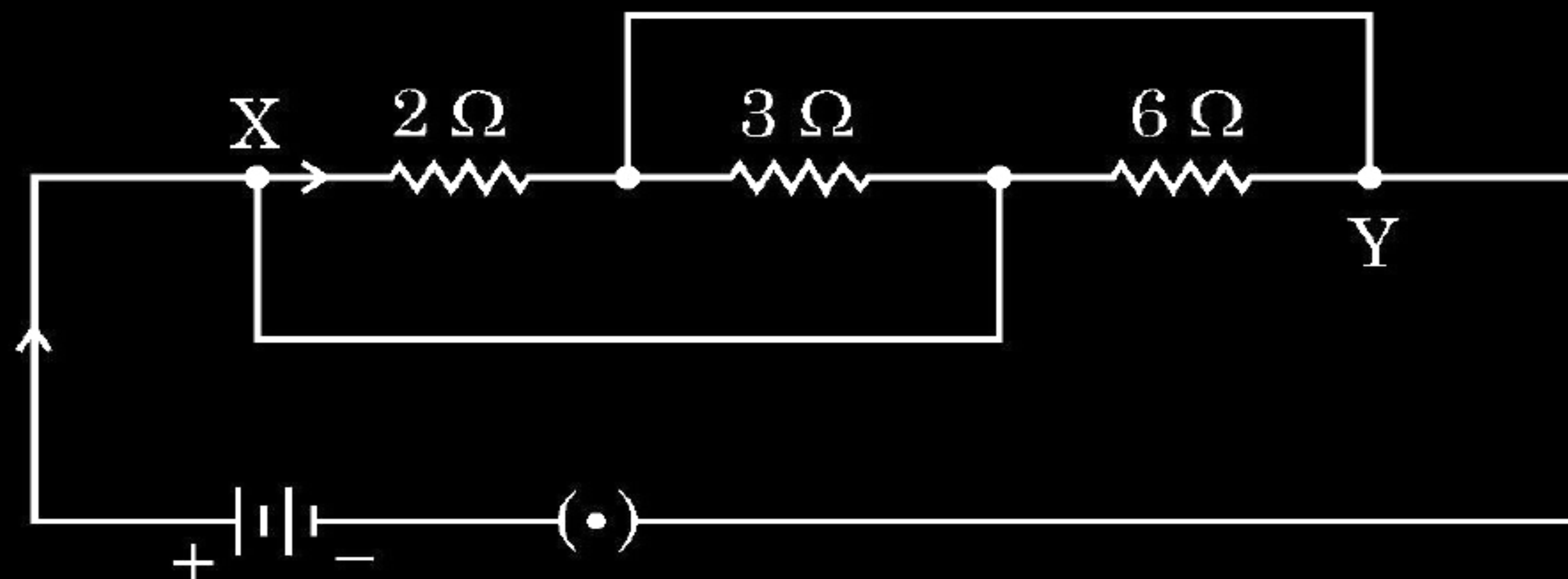
$\frac{1}{2}$

1

1

14. In the given circuit the total resistance between X and Y is :

1



(A) $12\ \Omega$

(B) $4\ \Omega$

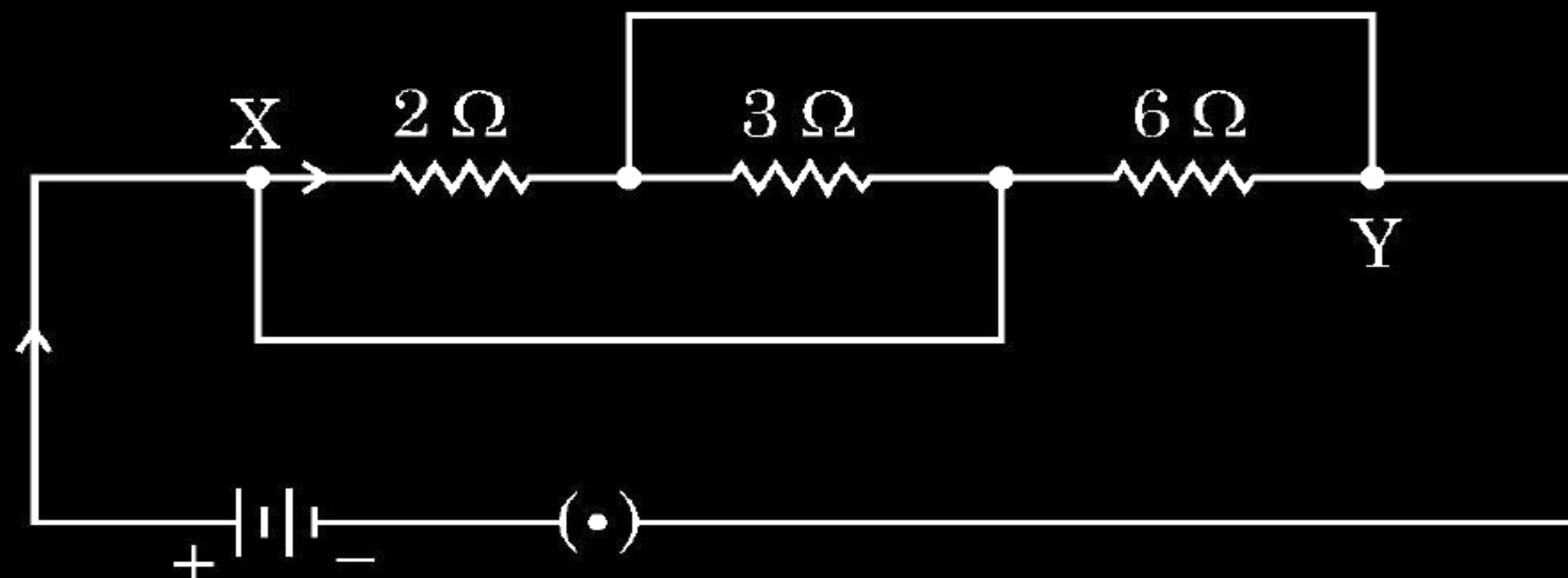
(C) $6\ \Omega$

(D) $1\ \Omega$

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14. In the given circuit the total resistance between X and Y is :

1



(A) $12\ \Omega$

(B) $4\ \Omega$

(C) $6\ \Omega$

(D) $1\ \Omega$

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(D) / $1\ \Omega$

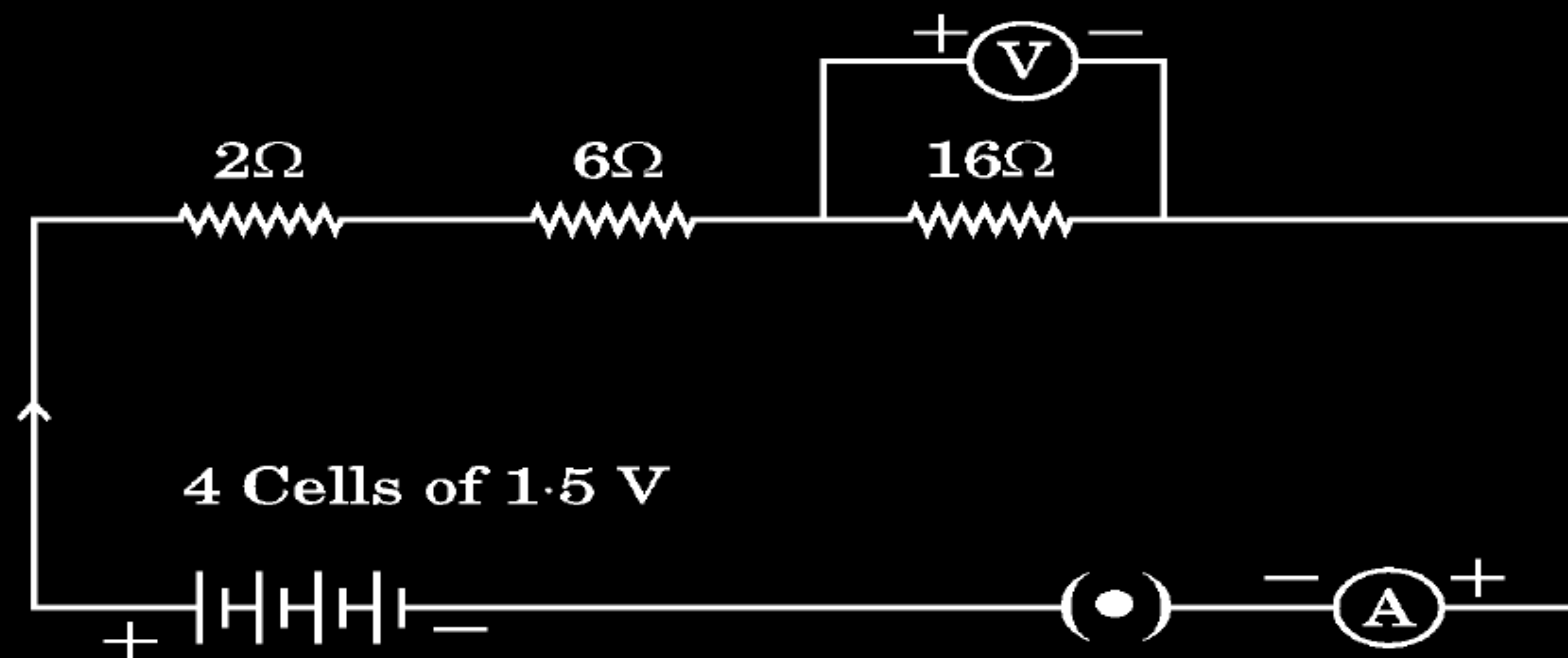
32. Draw a schematic diagram of a circuit consisting of a battery of four dry cells of 1.5 V each, a $2\ \Omega$ resistor, a $6\ \Omega$ resistor, $16\ \Omega$ resistor and a plug key all connected in series. Put an ammeter to measure the current in the circuit and a voltmeter across the $16\ \Omega$ resistor to measure potential difference across its two ends. Use Ohm's law to determine

3

(a) ammeter reading, and

(b) voltmeter reading when key is closed.

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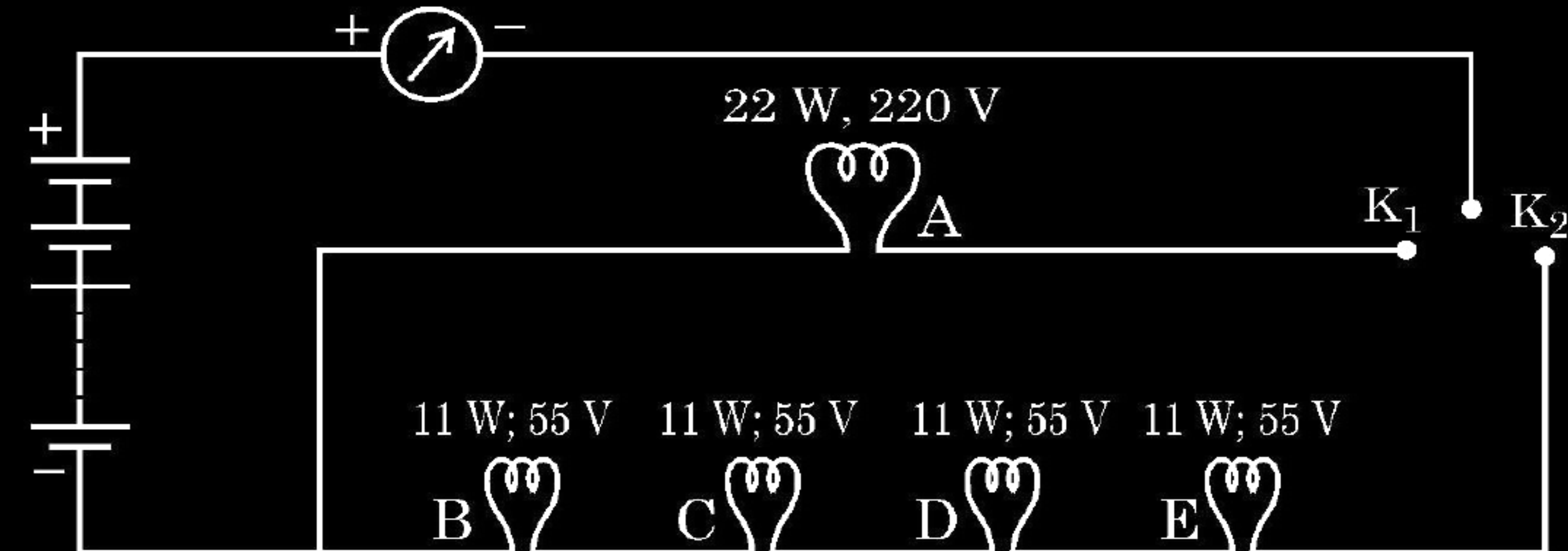
1

(a) Current (I) = $\frac{V}{R} = \frac{4 \times 1.5 \text{ V}}{2\Omega + 6\Omega + 16\Omega} = \frac{6 \text{ V}}{24\Omega} = \frac{1}{4} \text{ A}$

1

(b) Potential Difference across 16 Ω resistor = $\frac{1}{4} \text{ A} \times 16 \Omega = 4 \text{ V}$

1



39. In a domestic circuit five LED bulbs are arranged as shown. The source voltage is 220 V and the power rating of each bulb is marked in the circuit diagram. Based on the following circuit diagram, answer the following questions :

(a) State what happens when

(i) key K_1 is closed.

(ii) key K_2 is closed.

(b) Find the current drawn by the bulb B when it glows.

(c) Calculate

(i) the resistance of bulb B, and

(ii) total resistance of the combination of four bulbs B, C, D and E.

OR

(c) What would happen to the glow of all the bulbs in the circuit when keys K_1 and K_2 both are closed and the bulb C suddenly get fused ?

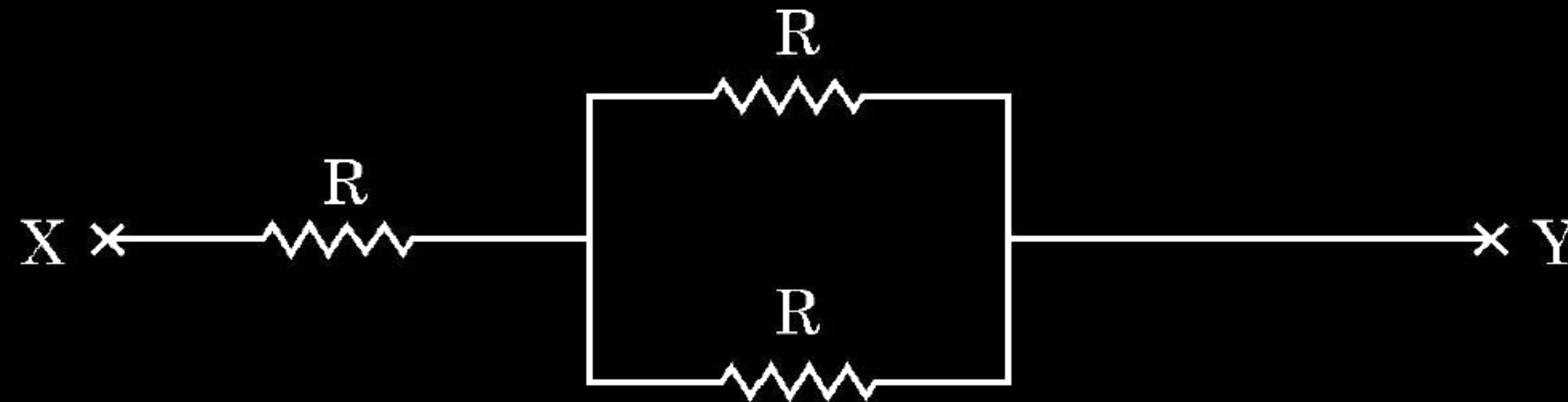
Give reason to justify your answer.

(a) (i) Bulb A glows (ii) Bulbs B, C, D and E glow	$\frac{1}{2}$ $\frac{1}{2}$
(b) $P = V \times I$ $11 = 55 \times I$ $I = \frac{1}{5} = 0.2 \text{ amp}$	$\frac{1}{2}$ $\frac{1}{2}$
(c) (i) Resistance of bulb B, $R = \frac{V}{I} = \frac{55 \text{ V}}{0.2 \text{ A}} = 275 \Omega$ (alternative formula for calculation $R = \frac{V^2}{P}$) (ii) Total resistance of the series combination of four bulbs $= 4 \times 275 = 1100 \Omega$ OR	1 1
(c) - Bulb A will keep glowing with same brightness. - Other bulbs i.e., B, D and E will stop glowing.	$\frac{1}{2}$ $\frac{1}{2}$
Reason : As the bulbs B, D and E are connected in series with fused bulb C, so no current flows through them and thus they will not glow. The bulb A remains unaffected as it is connected in parallel combination.	1

33. (a) State Ohm's law. Write formula for the equivalent resistance R_p of the parallel combination of three resistors of values R_1 , R_2 and R_3 .

(b) Find the resistance of the following network of resistors :

3

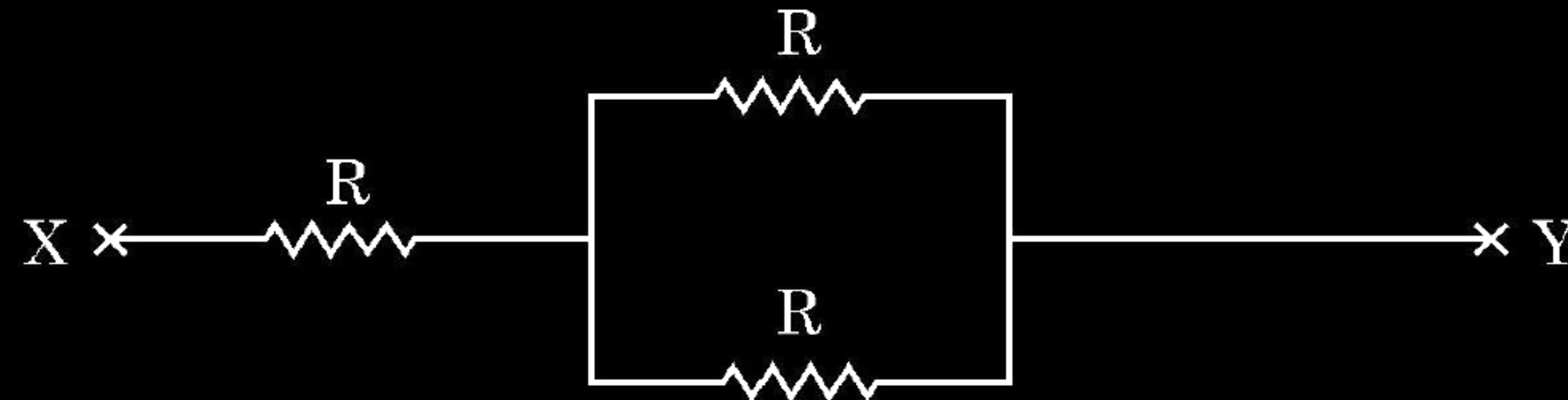


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33. (a) State Ohm's law. Write formula for the equivalent resistance R_p of the parallel combination of three resistors of values R_1 , R_2 and R_3 .

(b) Find the resistance of the following network of resistors :

3



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(a) Ohm's Law – The potential difference, V , across the ends of a given metallic wire in an electric circuit is directly proportional to the current flowing through it, provided its temperature remains the same.

Formula :-
$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

(b)
$$R + \frac{R}{2} = \frac{3R}{2}$$

1

1

1

14. The maximum resistance of a network of five identical resistors of $\frac{1}{5} \Omega$ each can be –

(A) 1Ω

(B) 0.5Ω

(C) 0.25Ω

(D) 0.1Ω

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14. The maximum resistance of a network of five identical resistors of $\frac{1}{5} \Omega$ each can be –

1

(A) 1Ω

(B) 0.5Ω

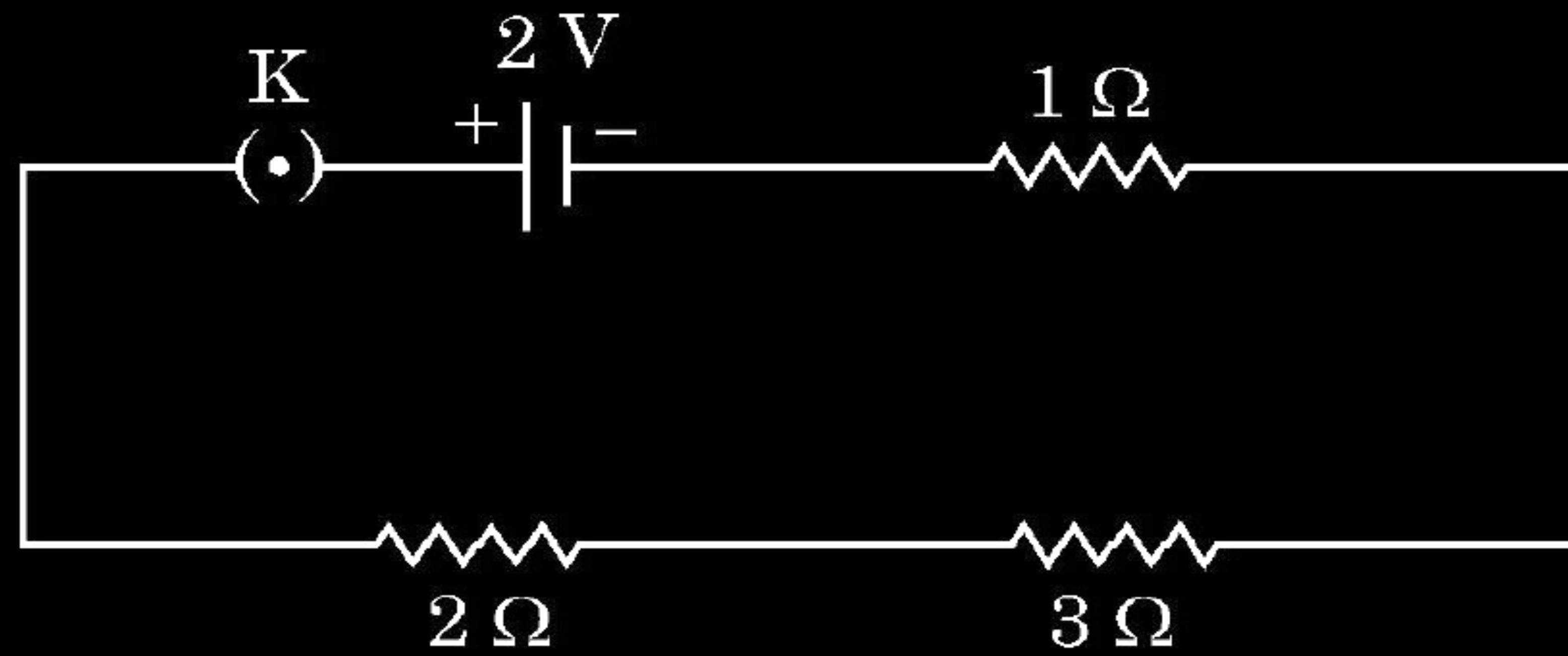
(C) 0.25Ω

(D) 0.1Ω

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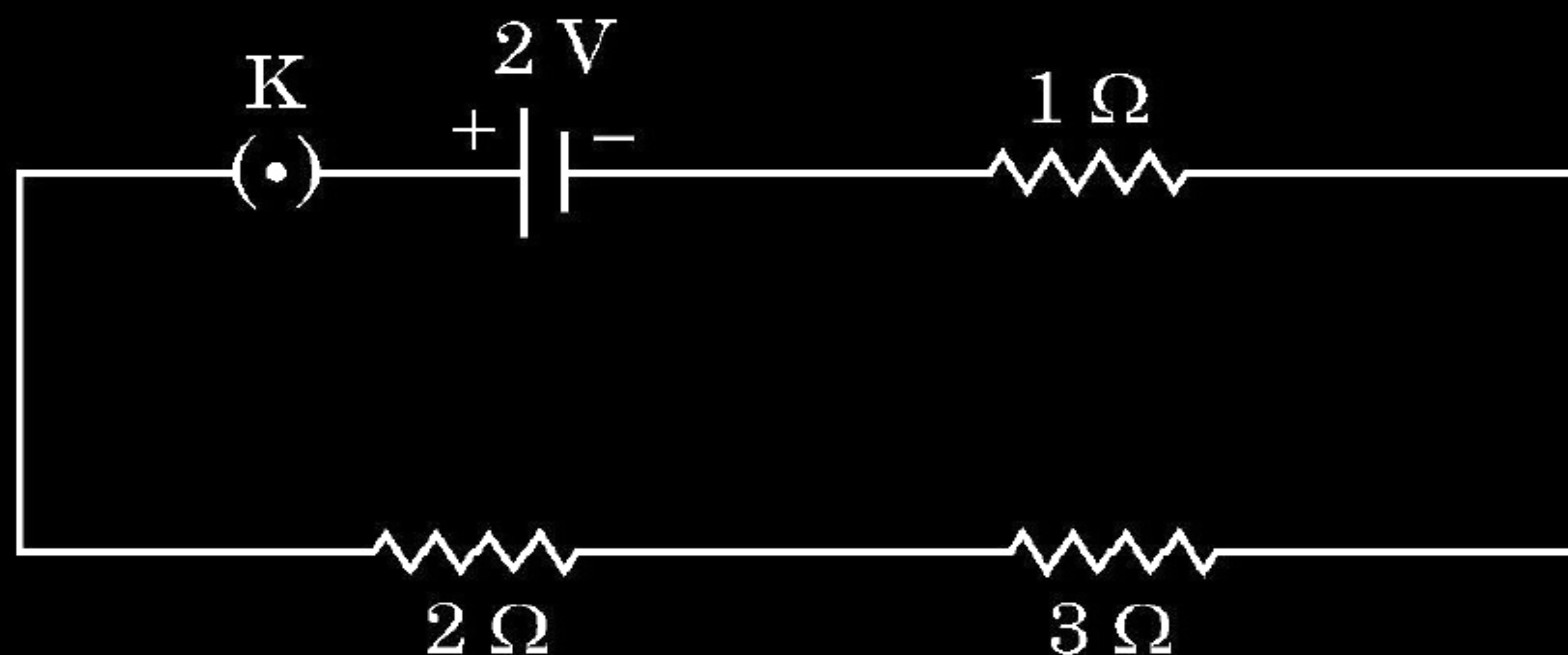
(A) / 1Ω

24. Use Ohm's law to determine the potential difference across the $3\ \Omega$ resistor in the circuit shown in the following diagram when key is closed : **2**



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24. Use Ohm's law to determine the potential difference across the $3\ \Omega$ resistor in the circuit shown in the following diagram when key is closed : 2



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$$R_s = R_1 + R_2 + R_3$$

$$= 1 + 2 + 3 = 6\ \Omega$$

$$I = \frac{V}{R}$$

$$= \frac{2V}{6\Omega} = \frac{1}{3} A$$

$$V = IR$$

$$= \frac{1}{3} A \times 3(\Omega) = 1\ V$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

15. The minimum resistance which can be made using five resistors each of resistance $10\ \Omega$ is :

1

(A) $\frac{1}{50}\ \Omega$

(B) $\frac{1}{5}\ \Omega$

(C) $2\ \Omega$

(D) $1\ \Omega$

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15. The minimum resistance which can be made using five resistors each of resistance $10\ \Omega$ is :

1

(A) $\frac{1}{50}\ \Omega$

(B) $\frac{1}{5}\ \Omega$

(C) $2\ \Omega$

(D) $1\ \Omega$

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(C) / $2\ \Omega$

(B) Draw a schematic diagram of a circuit consisting of a battery of six 2V cells, a $6\ \Omega$ resistor, a $12\ \Omega$ resistor and a $18\ \Omega$ resistor and a plug key all connected in series. Calculate the following (when key is closed) :

5

- (i) Electric current flowing in the circuit.
- (ii) Potential difference across $18\ \Omega$ resistor.
- (iii) Electric power consumed in $18\ \Omega$ resistor.

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(b)

$$\therefore R_s = 24 \Omega + 24 \Omega = 48 \Omega$$

$$\begin{aligned} \text{Power consumed } P_2 &= \frac{V^2}{R_s} \\ &= \frac{6 V \times 6 V}{48 \Omega} \\ &= \frac{3}{4} W \end{aligned}$$

 $\frac{1}{2}$

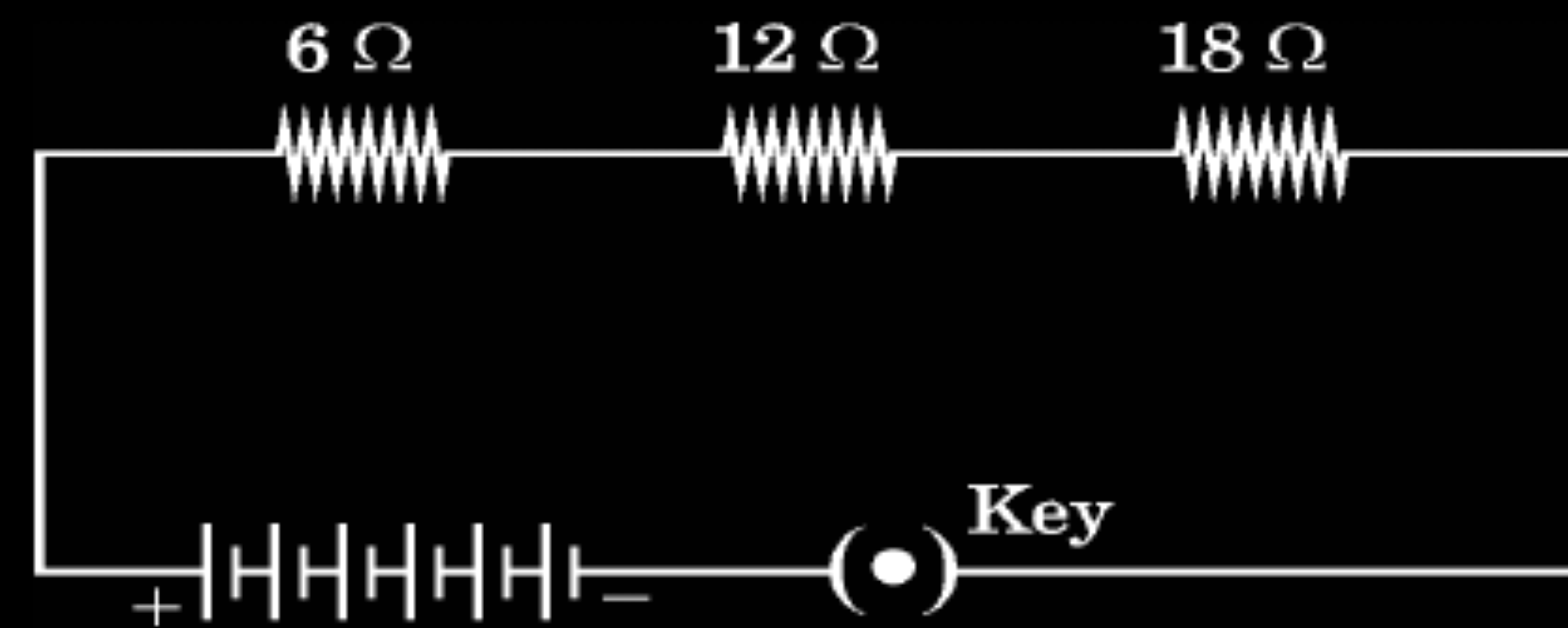
$$\therefore \frac{P_1}{P_2} = \frac{\frac{3}{4}}{\frac{3}{4}} = \frac{4}{1}$$

from P_1 and P_2

$$\Rightarrow P_p : P_s = 4 : 1$$

 $\frac{1}{2}$ **OR**

(B)



12 V (Six cells of 2 V each)

2

$$(i) \text{Current} = \frac{V}{R} = \frac{12}{(6+12+18) \Omega} = \frac{1}{3} A$$

1

$$(ii) \text{Potential difference across } 18 \Omega \text{ resistor} = I \times R = \frac{1}{3} A \times 18 \Omega = 6 V$$

1

$$(iii) \text{Power consumed in } 18 \Omega \text{ resistor} = V \times I = 6 V \times \frac{1}{3} A = 2 W$$

1

22. (A) Show how you would connect three resistors each of resistance $6\ \Omega$, so that the combination has a resistance of $9\ \Omega$. Also justify your answer.

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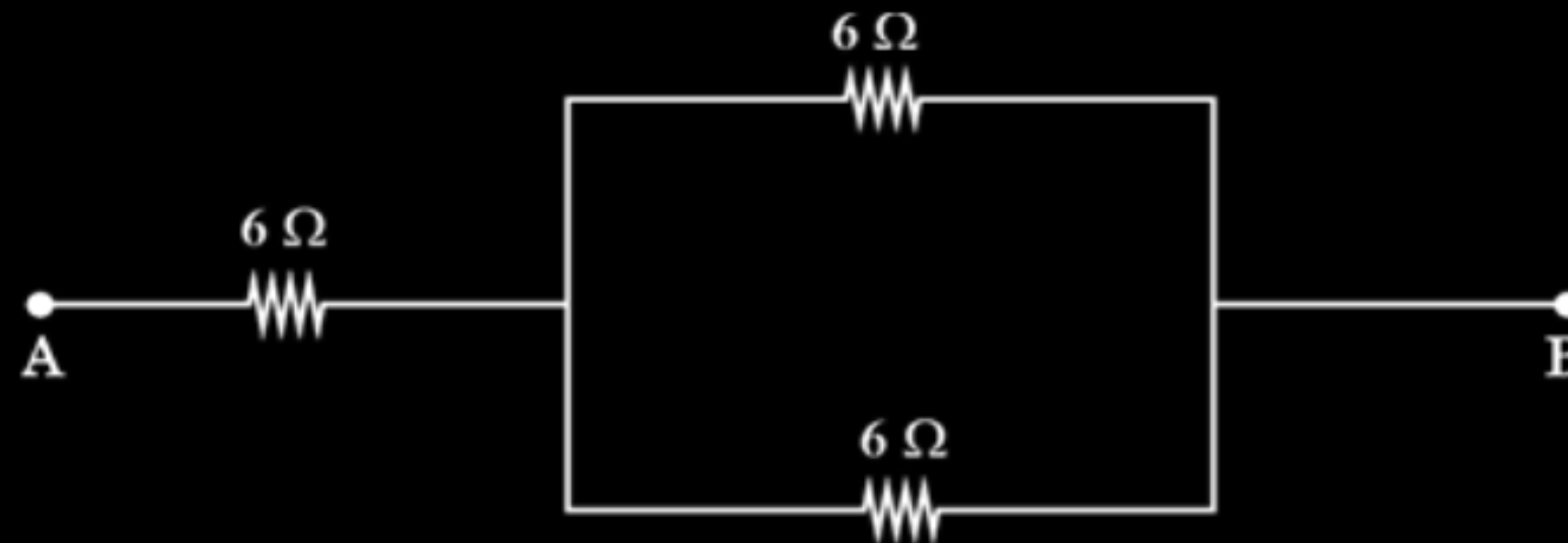
2

22. (A) Show how you would connect three resistors each of resistance $6\ \Omega$, so that the combination has a resistance of $9\ \Omega$. Also justify your answer.

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2

(A) When two $6\ \Omega$ resistances are connected in parallel and the third resistance of $6\ \Omega$ is connected in series combinations to this, then equivalent resistance will be $9\ \Omega$ /



[Award marks for writing the statement or drawing the diagram]

$$\frac{1}{R_P} = \frac{1}{6\ \Omega} + \frac{1}{6\ \Omega}$$

$$\therefore R_P = 3\ \Omega$$
$$R_S = 6 + 3 = 9\ \Omega$$

1

1

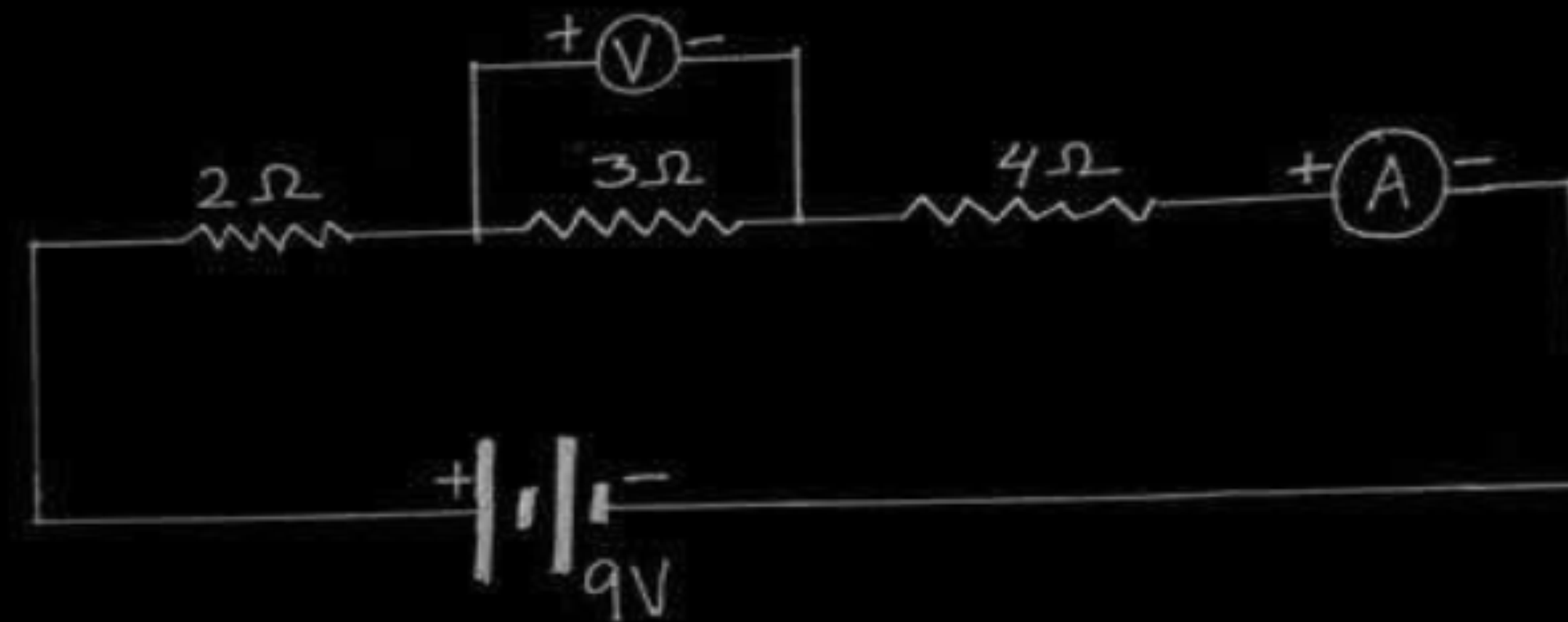
- 36.** (a) You have three resistors R_1 , R_2 and R_3 . Write the formula for the equivalent resistance of the combination, when these resistors are joined (i) in series and (ii) in parallel.
- (b) Three resistors of $2\ \Omega$, $3\ \Omega$ and $4\ \Omega$ are connected in series and the combination is connected to a battery of $9\ \text{V}$. An ammeter and a voltmeter are also connected in the circuit. Draw a circuit diagram to show the connection of the above said devices in a correct order. Determine (i) current shown by the ammeter and (ii) potential difference across the $3\ \Omega$ resistor.

(a) (i) $R_s = R_1 + R_2 + R_3$

(ii) $1/R_p = 1/R_1 + 1/R_2 + 1/R_3$

(b)

•



(i) $R_s = 2\ \Omega + 3\ \Omega + 4\ \Omega = 9\ \Omega$

$$V = 9V$$

$$I = \frac{v}{R_s}$$

$$= \frac{9v}{9\Omega} = 1\ A$$

Ammeter reading = 1.0 A

(ii) Potential Difference across 3 Ω resistor

$$V = I R$$

$$= 1.0\ A \times 3.0\ \Omega$$

$$= 3\ V$$

- (b) (i) Write the formula for finding the value of the equivalent resistance of three resistors of resistance R_1 , R_2 and R_3 , when the resistors are connected in (i) series, and (ii) parallel.
- (ii) List two advantages of connecting electrical devices in parallel with the source of electricity instead of connecting them in series.

- (b) (i) Write the formula for finding the value of the equivalent resistance of three resistors of resistance R_1 , R_2 and R_3 , when the resistors are connected in (i) series, and (ii) parallel.
- (ii) List two advantages of connecting electrical devices in parallel with the source of electricity instead of connecting them in series.

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3

(b) (i) Formula for Equivalent resistance in series : $R_s = R_1 + R_2 + R_3$ Equivalent resistance in parallel : $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$	1 1
(ii) (1) Each device can draw current as per need (2) Separate 'ON/OFF' switch for every device can be put (3) Total resistance of the combination decreases hence larger current can be drawn from the source. (4) If one device fails, other devices keep on working (any two)	2 x ½

24. Three resistors of $6\ \Omega$, $4\ \Omega$ and $4\ \Omega$ are connected together so that the total resistance is $8\ \Omega$. Draw a diagram to show this arrangement and give reason to justify your answer.

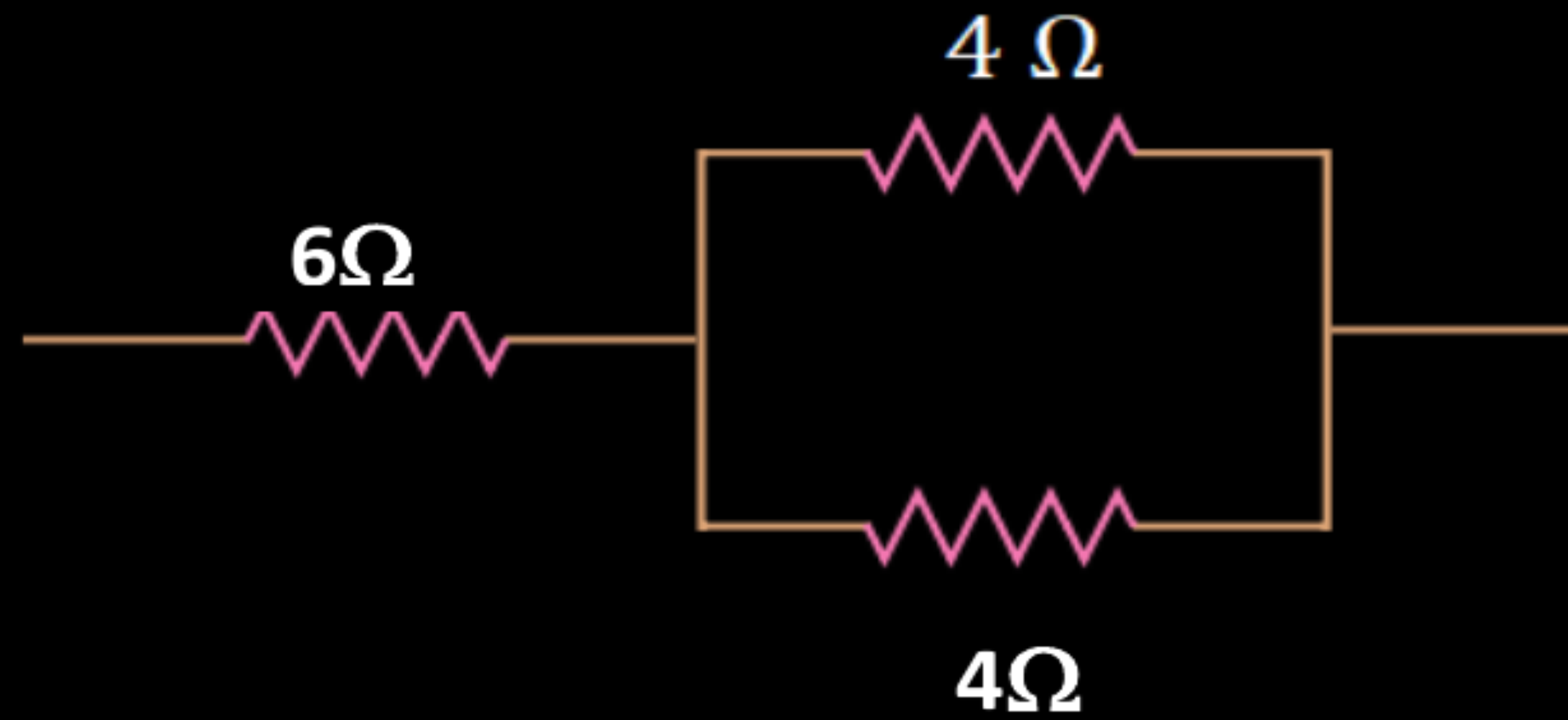
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2

24. Three resistors of $6\ \Omega$, $4\ \Omega$ and $4\ \Omega$ are connected together so that the total resistance is $8\ \Omega$. Draw a diagram to show this arrangement and give reason to justify your answer.

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2



$$\frac{1}{R_p} = \left(\frac{1}{4} + \frac{1}{4} \right) = \frac{2}{4} \Rightarrow R_p = 2\Omega$$

$$R = 6\Omega + 2\Omega$$

1

$\frac{1}{2}$

$\frac{1}{2}$

34. There are three resistors of $10\ \Omega$, $20\ \Omega$ and $30\ \Omega$ joined in parallel in a circuit. The potential difference across the electric circuit is 10 V .

- (a) Draw a circuit diagram for the above case.
- (b) Find the total resistance of the combination of resistors.
- (c) Calculate the electric current drawn from the same source.

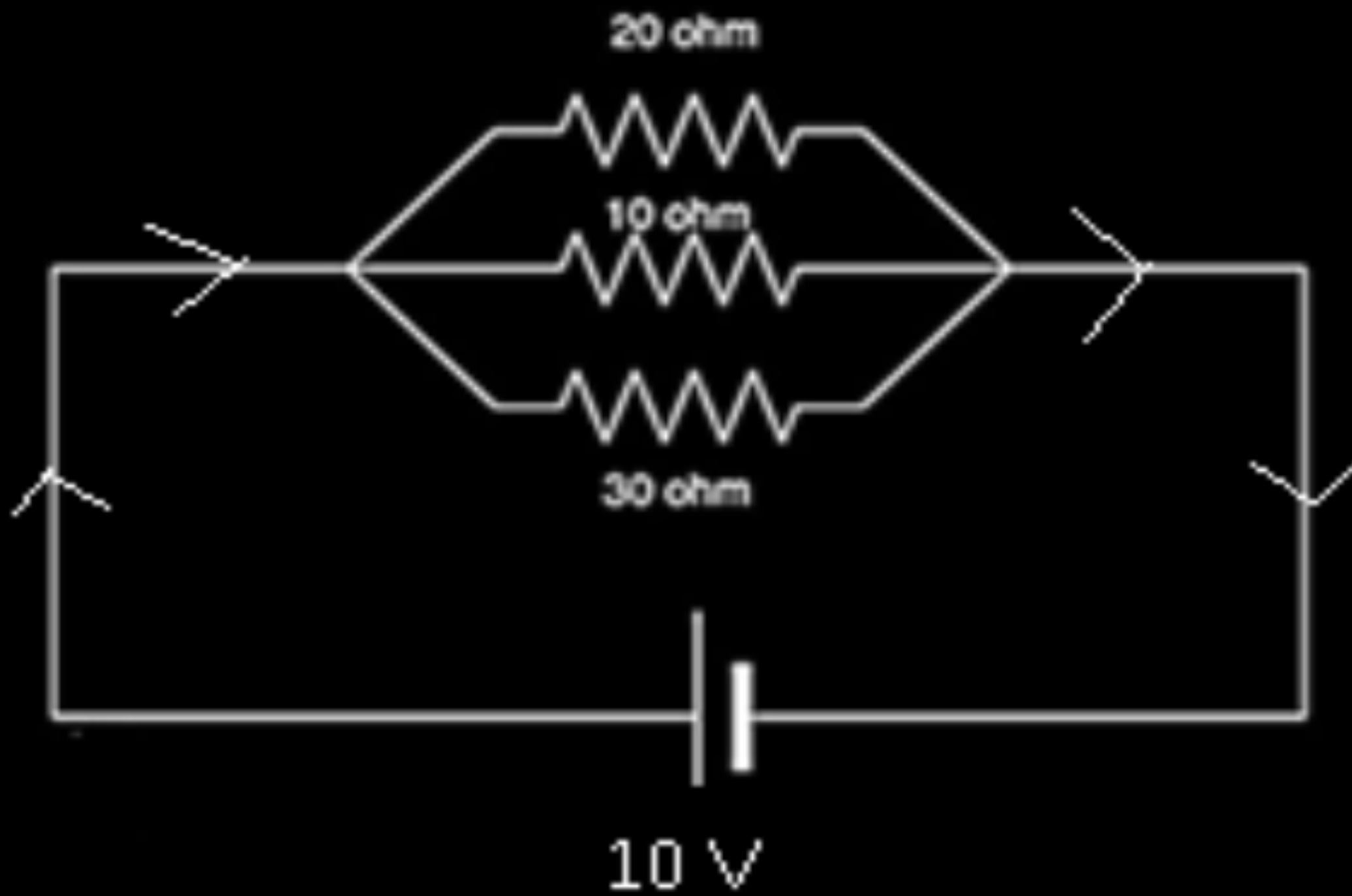
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- 34.** There are three resistors of $10\ \Omega$, $20\ \Omega$ and $30\ \Omega$ joined in parallel in a circuit. The potential difference across the electric circuit is 10 V .

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- (a) Draw a circuit diagram for the above case.
- (b) Find the total resistance of the combination of resistors.
- (c) Calculate the electric current drawn from the same source.

5



2

(b)
$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$= \frac{1}{10} + \frac{1}{20} + \frac{1}{30}$$

$$= \frac{6+3+2}{60}$$

$\frac{1}{2}$

(c)
$$\therefore R = \frac{60}{11}\ \Omega \quad / 5.45\ \Omega$$

$$I = \frac{V}{R}$$

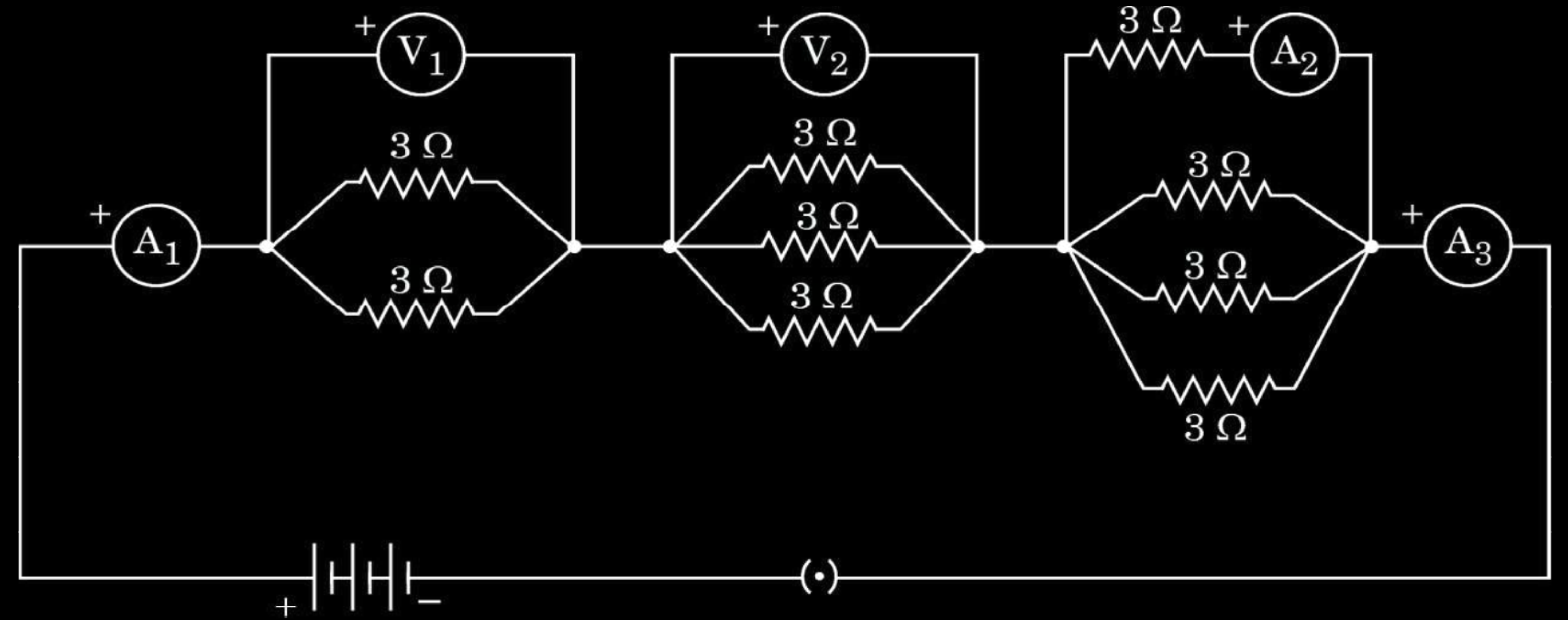
1

$\frac{1}{2}$

$$I = \frac{10\text{ V}}{\frac{60}{11}\ \Omega} = \frac{110}{60}\text{ A} = 1.83\text{ A}$$

1

39. Consider the following electrical circuit diagram in which nine identical resistors of $3\ \Omega$ each are connected as shown. If the reading of the ammeter A_1 is 1 ampere, answer the following questions :



- (a) What is the relationship between the readings of A_1 and A_3 ? Give reasons for your answer. 1
- (b) What is the relationship between the readings of A_2 and A_3 ? 1
- (c) Determine the reading of the voltmeter V_1 . 2

OR

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- (c) Find the total resistance of the circuit. 2

(a) • Both have same reading / $A_1 = A_3$	$\frac{1}{2}$
• Both are connected in series	$\frac{1}{2}$
(b) Reading of $A_2 = \frac{1}{4}A$ as current is equally divided in the four identical resistors . / Reading of $A_2 = \frac{1}{4}$ times Reading of A_3 . / $A_2 = 0.25 A$ / $A_2 < A_3$	1
(c) $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$ / $R_p = \frac{R}{n}$	$\frac{1}{2}$
$\frac{1}{R_p} = \frac{1}{3 \Omega} + \frac{1}{3 \Omega}$	$\frac{1}{2}$
$R_p = \frac{3}{2} \Omega$	$\frac{1}{2}$
$V = I R$	$\frac{1}{2}$
$V_1 = 1A \times \frac{3}{2} \Omega = \frac{3}{2} V = 1.5V$	$\frac{1}{2}$
OR	
$\frac{1}{R_p} = \frac{1}{3 \Omega} + \frac{1}{3 \Omega}$	
$\therefore R_p = \frac{3}{2} \Omega$	$\frac{1}{2}$
$\frac{1}{R_p} = \frac{1}{3 \Omega} + \frac{1}{3 \Omega} + \frac{1}{3 \Omega}$	
$\therefore R_{p2} = 1 \Omega$	$\frac{1}{2}$
$\frac{1}{R_p} = \frac{1}{3 \Omega} + \frac{1}{3 \Omega} + \frac{1}{3 \Omega} + \frac{1}{3 \Omega}$	
$\therefore R_{p3} = \frac{3}{4} \Omega$	$\frac{1}{2}$
$\therefore R = R_{p1} + R_{p2} + R_{p3} = \left(\frac{3}{2} + 1 + \frac{3}{4} \right) \Omega = \frac{13}{4} \Omega / 3.25\Omega$	$\frac{1}{2}$

14. If four identical resistors, of resistance 8 ohm, are first connected in series so as to give an effective resistance R_s , and then connected in parallel so as to give an effective resistance R_p , then the ratio $\frac{R_s}{R_p}$ is

1

(a) 32

(b) 2

(c) 0.5

(d) 16

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14. If four identical resistors, of resistance 8 ohm, are first connected in series so as to give an effective resistance R_s , and then connected in parallel so as to give an effective resistance R_p , then the ratio $\frac{R_s}{R_p}$ is

1

(a) 32

(b) 2

(c) 0.5

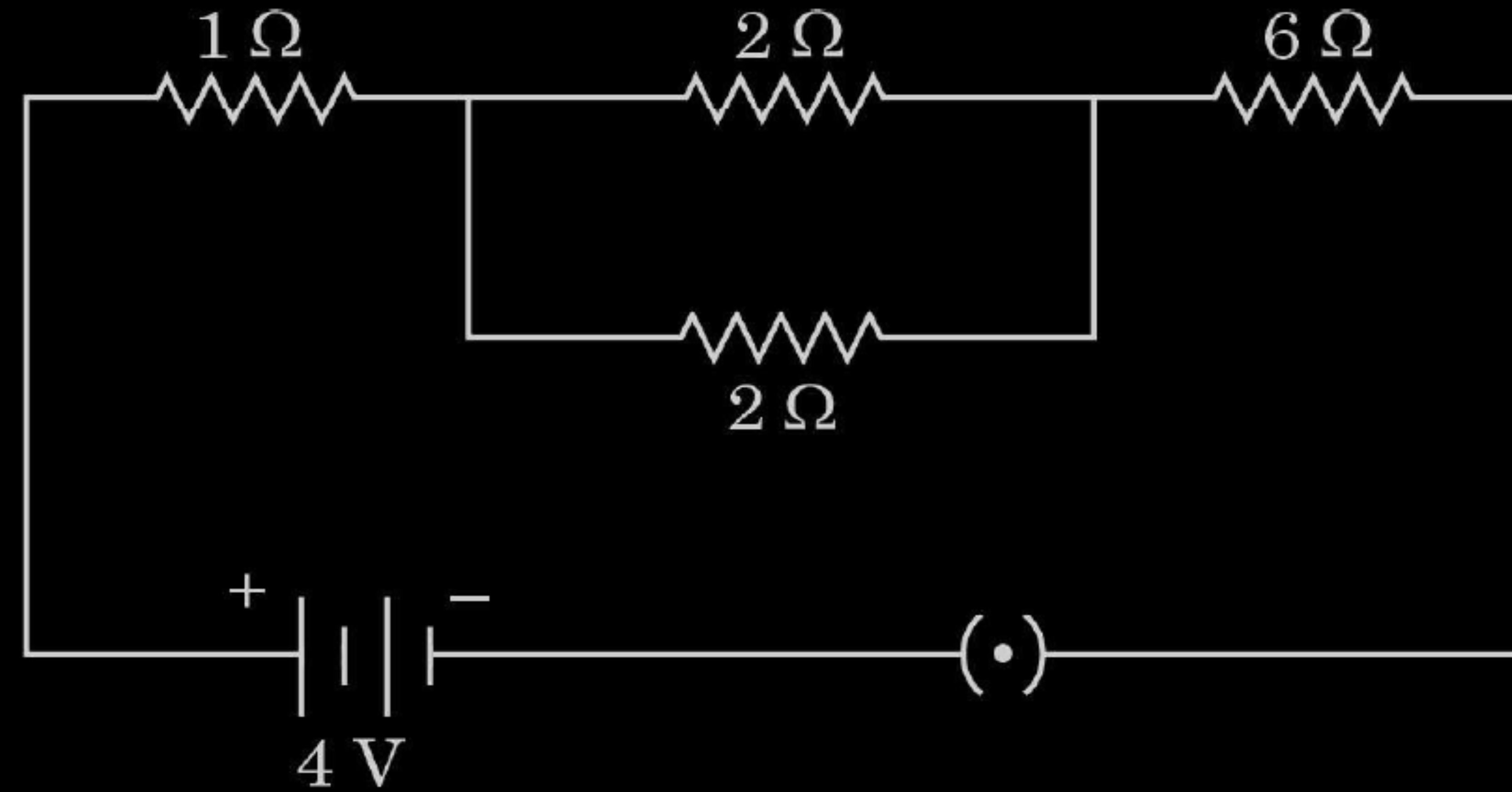
(d) 16

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(d)

10. (a) Find the current flowing through the following electric circuit :

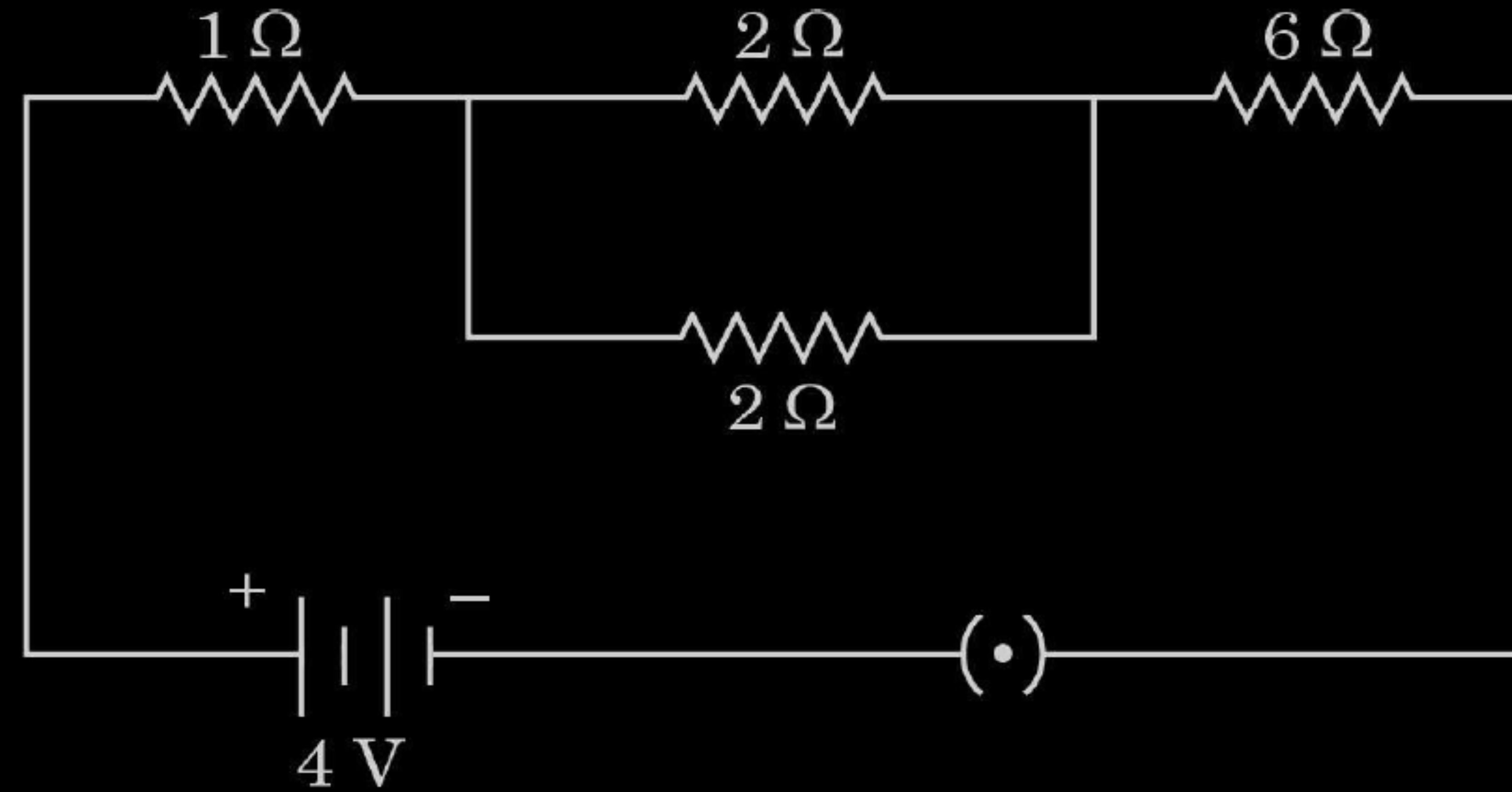
3



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10. (a) Find the current flowing through the following electric circuit :

3



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• Parallel combination,

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_p} = \frac{1}{2} + \frac{1}{2} = \frac{2}{2} \Omega$$

$$R_p = 1 \Omega$$

• Total resistance,

$$R = 1 \Omega + 1 \Omega + 6 \Omega = 8 \Omega$$

• Current,

$$I = \frac{V}{R} = \frac{4 \text{ V}}{8 \Omega} = 0.5 \text{ A}$$

$\frac{1}{2}$

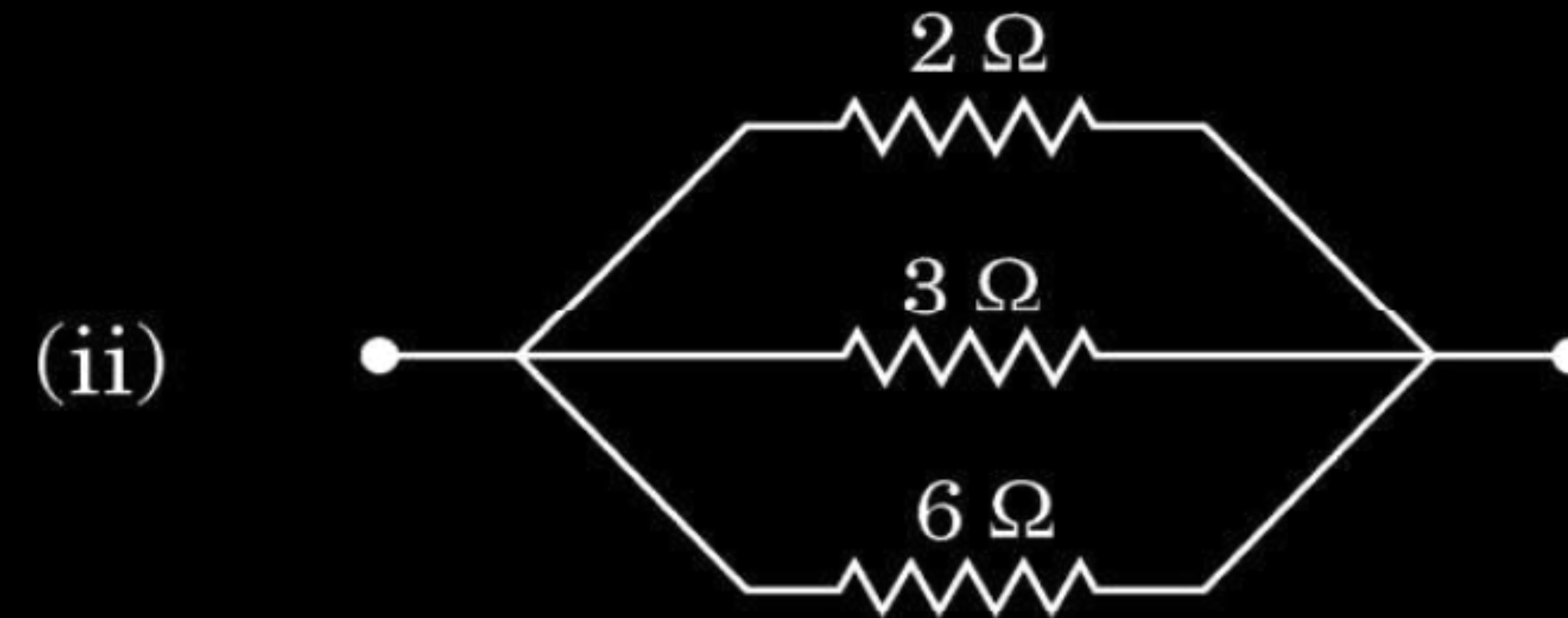
$\frac{1}{2}$

1

1

- (b) Find the equivalent resistance of the following combinations of resistors :

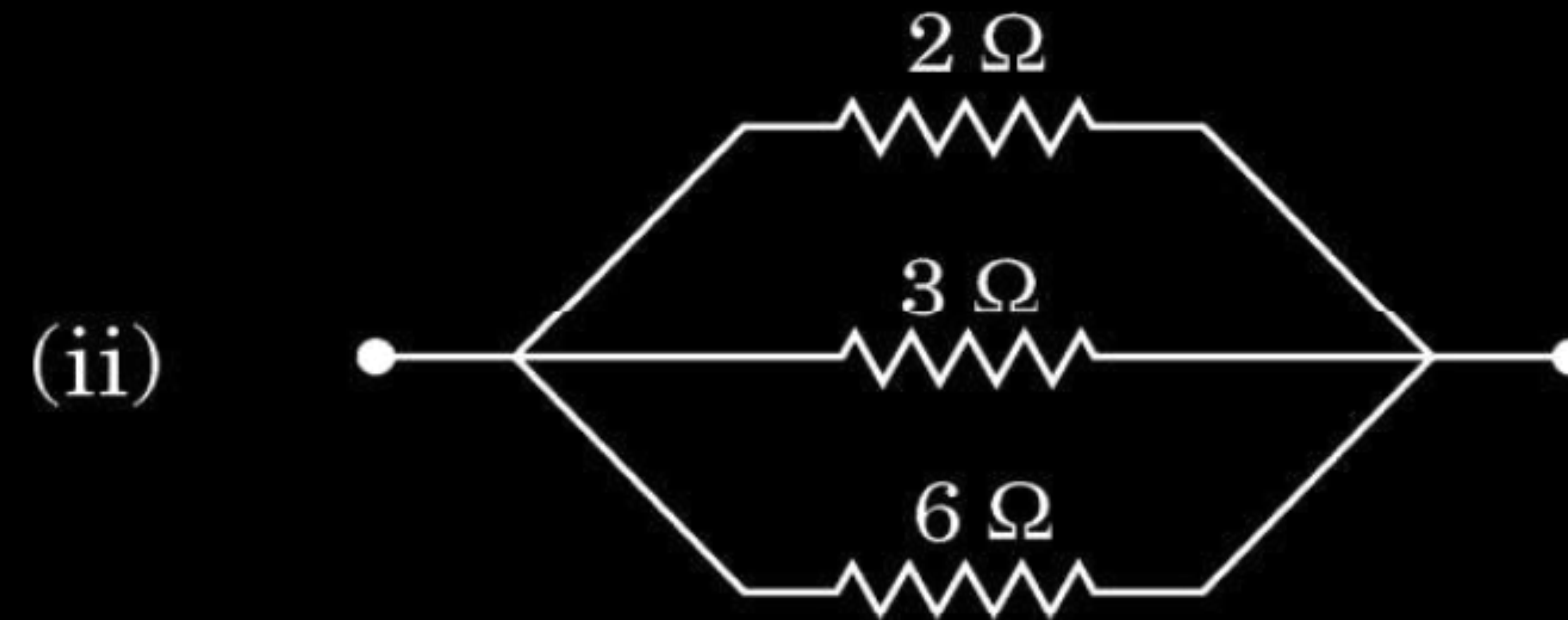
3



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(b) Find the equivalent resistance of the following combinations of resistors :

3



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$$\begin{aligned} \text{(b) (i) } R_s &= R_1 + R_2 + R_3 \\ &= 2 \, \Omega + 3 \, \Omega + 6 \, \Omega = 11 \, \Omega \end{aligned}$$

$$\begin{aligned} \text{(ii) } \frac{1}{R_p} &= \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \\ &= \frac{1}{2 \, \Omega} + \frac{1}{3 \, \Omega} + \frac{1}{6 \, \Omega} = 1 \, \Omega \end{aligned}$$

$\frac{1}{2}$
1

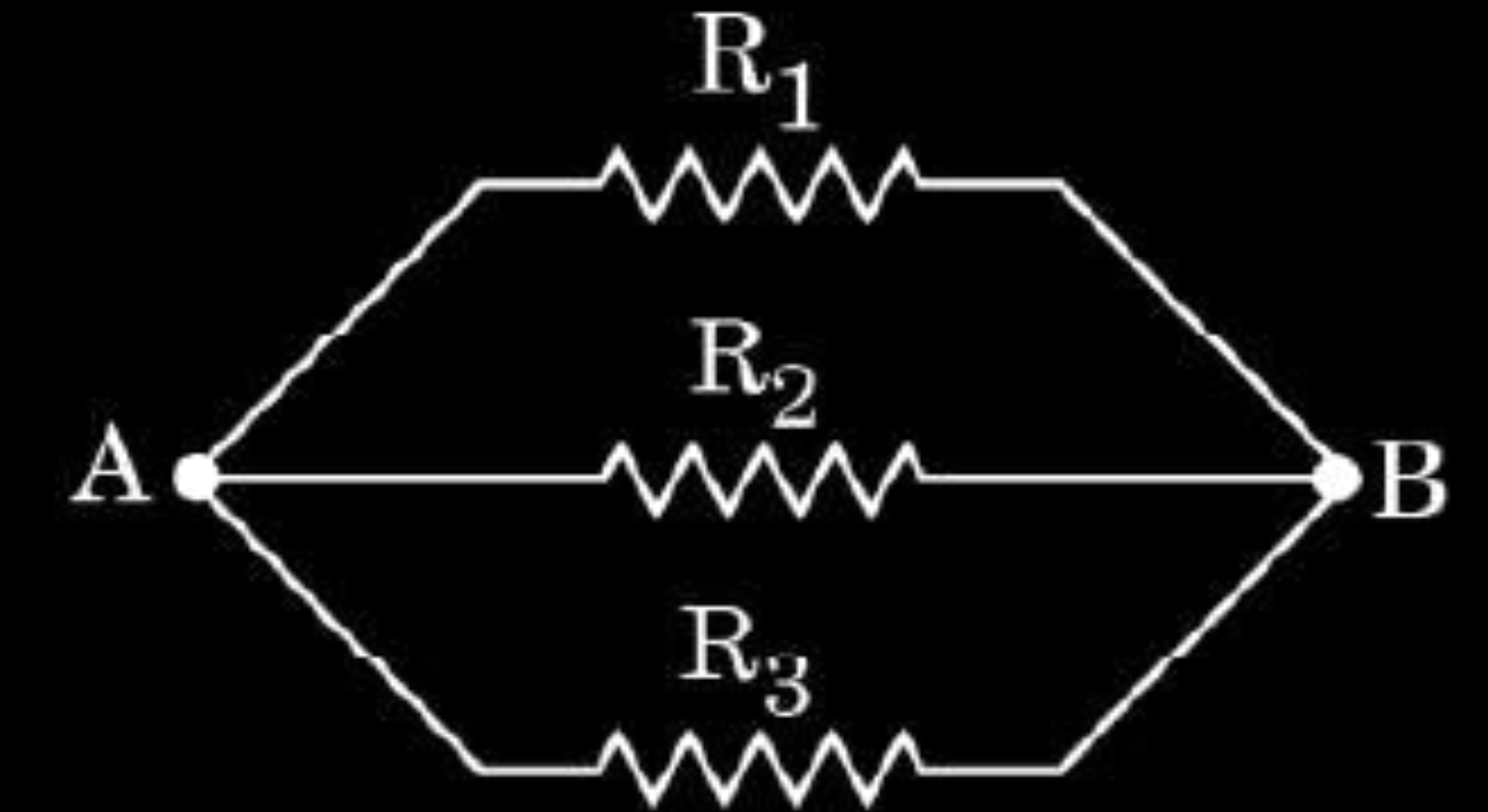
$\frac{1}{2}$

1

- (b) (i) Write the formula for determining the equivalent resistance between A and B of the two combinations (I) and (II) of three resistors R_1 , R_2 and R_3 arranged as follows :



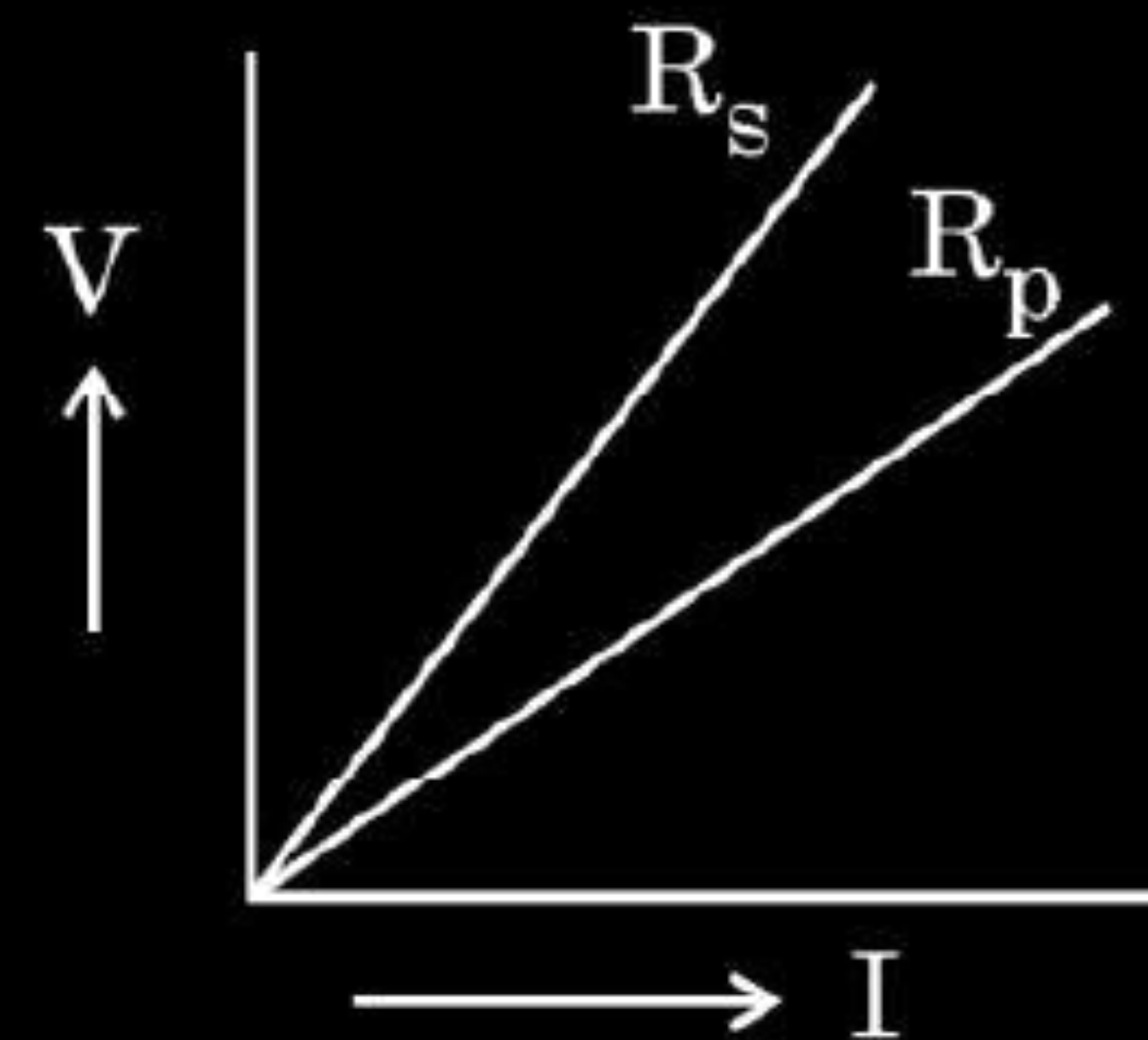
(I)



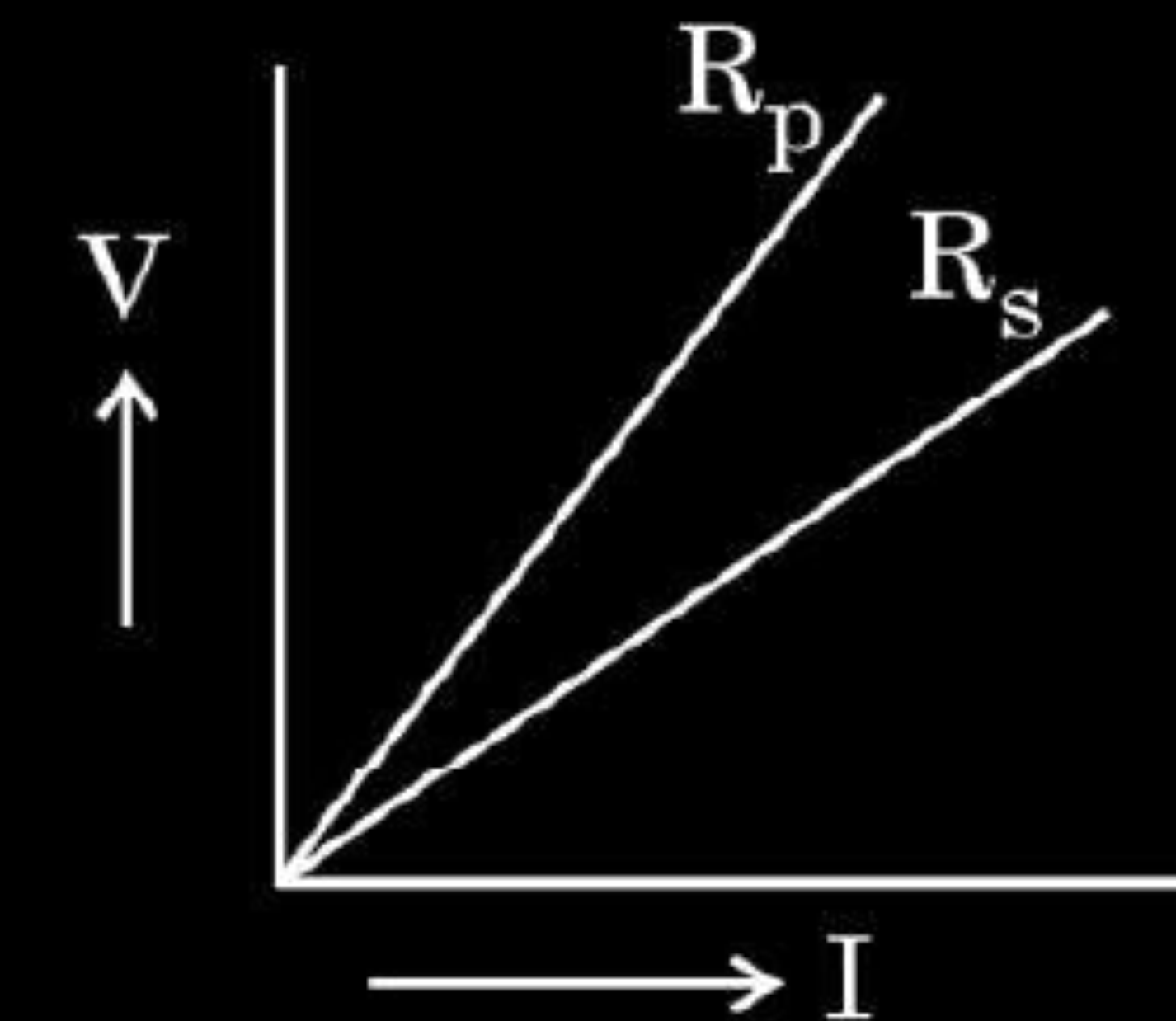
(II)

- (ii) If the equivalent resistance of the arrangements (I) and (II) are R_s and R_p respectively, then which one of the following $V - I$ graphs is correctly labelled ? Justify your answer.

3



(I)



(II)

(b) (i) (I) $R_s = R_1 + R_2 + R_3$	$\frac{1}{2}$
(II) $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$	$\frac{1}{2}$
(ii) Graph (I)	1
Reason : The resistance for the series combination is more than that of parallel combination and it is directly proportional to slope of V-I graph. As in the graph (I), slope of R_s is greater than R_p , hence graph I is correct.	1

- 15.** A student has designed an electric circuit containing a 6 V battery, a key, an ammeter, and a resistor of $30\ \Omega$ in series with a parallel combination of two resistors of $60\ \Omega$ each and a voltmeter across the $30\ \Omega$ resistor.
- (a) Find the equivalent resistance of the parallel combination of two $60\ \Omega$ resistors.
 - (b) Determine the total resistance of the circuit.
 - (c) (i) Calculate the current that will flow through the ammeter when the key is closed.

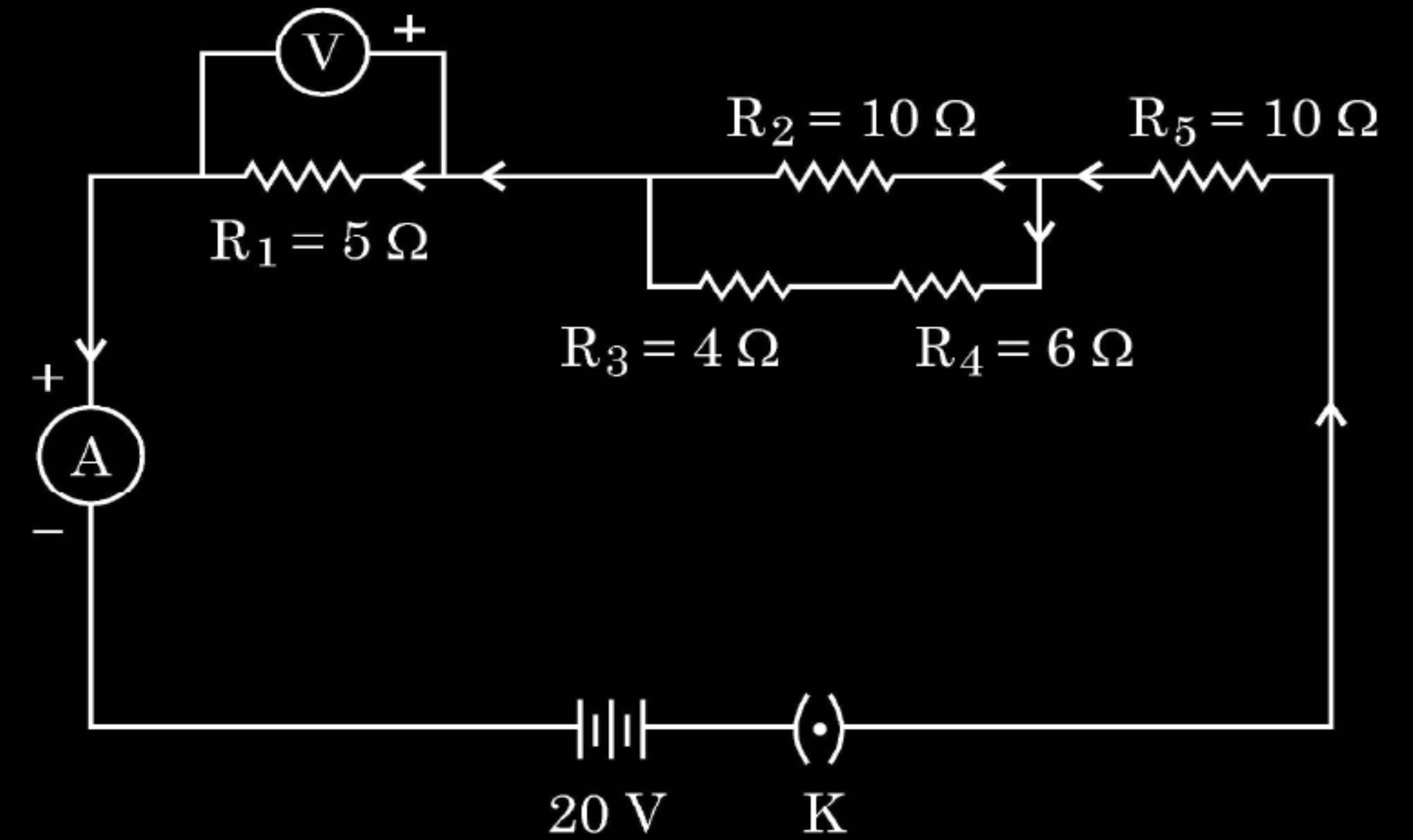
OR

- (ii) Will the potential difference across the $30\ \Omega$ resistor be the same as that across the parallel combination of two $60\ \Omega$ resistors ? Justify your answer.

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(a) $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$ $\frac{1}{R_p} = \frac{1}{60\ \Omega} + \frac{1}{60\ \Omega} = \frac{2}{60\ \Omega}$ $R_p = 30\ \Omega$	1/2
(b) $R_s = R_p + R_3$ $R_s = 30\ \Omega + 30\ \Omega = 60\ \Omega$	1/2
(c) (i) $I = \frac{V}{R}$ $= \frac{6V}{60\ \Omega} = 0.1A$	1/2
Deduct 1/2 mark for no / wrong unit. OR	
(c) (ii) Yes Justification should include following three points. - Both combinations in series so same current flow. - Resistance of the parallel combination of two 60 Ω resistors is also 30 Ω $V = I \times R$ Current is same and the resistance of the parallel combination of two 60 Ω resistors is also 30 Ω (or any other correct explanation)	1/2 1/2 1/2 1/2

Study the following circuit and find :



- (i) Effective resistance of the circuit
- (ii) Current drawn from the battery
- (iii) Potential difference across the $5\ \Omega$ resistor

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(i) Effective resistance of the circuit

$$R_s = R_3 + R_4 = 4\Omega + 6\Omega = 10\Omega$$

$$\frac{1}{R_p} = \frac{1}{R_s} + \frac{1}{R_2} = \frac{1}{10\Omega} + \frac{1}{10\Omega} = \frac{2}{10\Omega} = \frac{1}{5\Omega}$$

$$R_p = 5\Omega$$

$$\text{Total resistance of the circuit} = R_1 + R_p + R_5 = 5 + 5 + 10 = 20\Omega$$

(ii) Current drawn from the battery

$$V = 20V, R = 20\Omega$$

$$I = \frac{V}{R} = \frac{20V}{20\Omega}$$

$$I = 1 \text{ A}$$

(iii) Reading in voltmeter connected across 5Ω Resistance

$$V = I R$$

$$I = 1 \text{ A}$$

$$R = 5\Omega$$

$$V = 1 \text{ A} \times 5\Omega = 5 \text{ V}$$

$\frac{1}{2}$

$\frac{1}{2}$

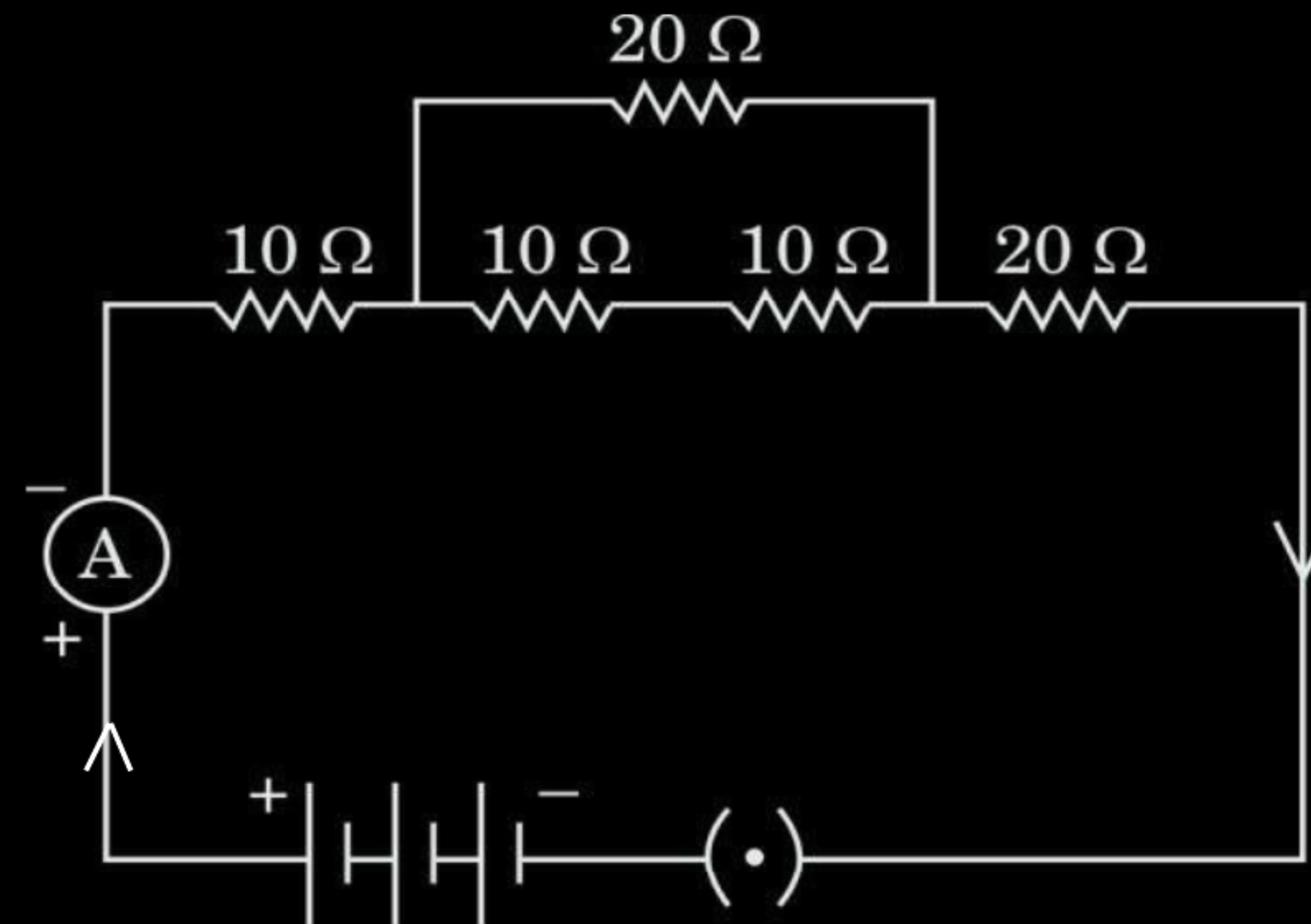
$\frac{1}{2}$

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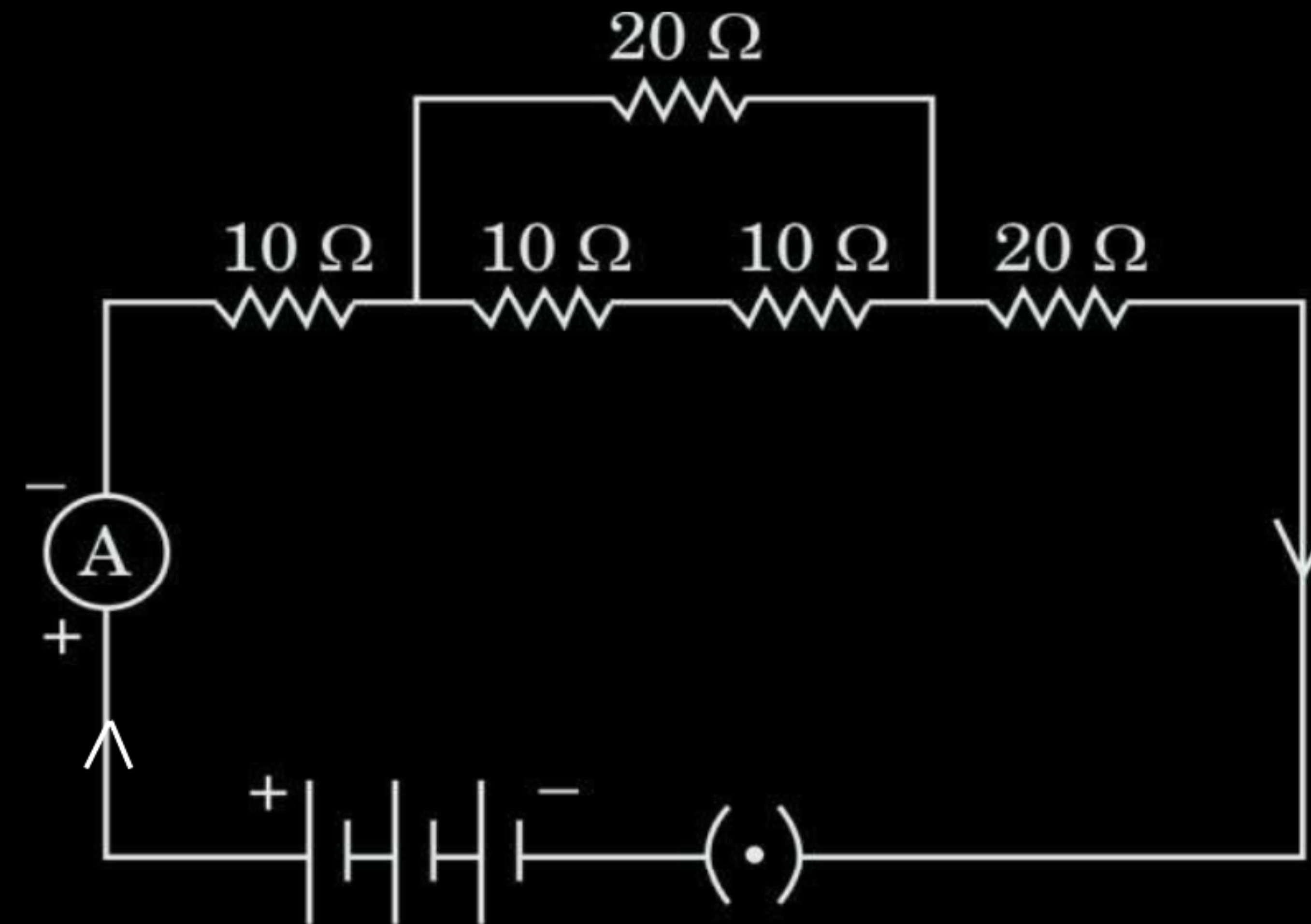
$\frac{1}{2}$

(b) Calculate the equivalent resistance of the following electric circuit : 2



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(b) Calculate the equivalent resistance of the following electric circuit : 2



CBSE 2022

(b) $R_S = R_3 + R_4 = 10 + 10 = 20 \, \Omega$

$$\frac{1}{R_P} = \frac{1}{R_2} + \frac{1}{R_S}$$

$$= \frac{1}{20} + \frac{1}{20} = \frac{1}{10} \, \Omega$$

$$R_P = 10 \, \Omega$$

Total equivalent resistance = $R = R_1 + R_P + R_5$

$$= R = 20 + 10 + 10 = 40 \, \Omega$$

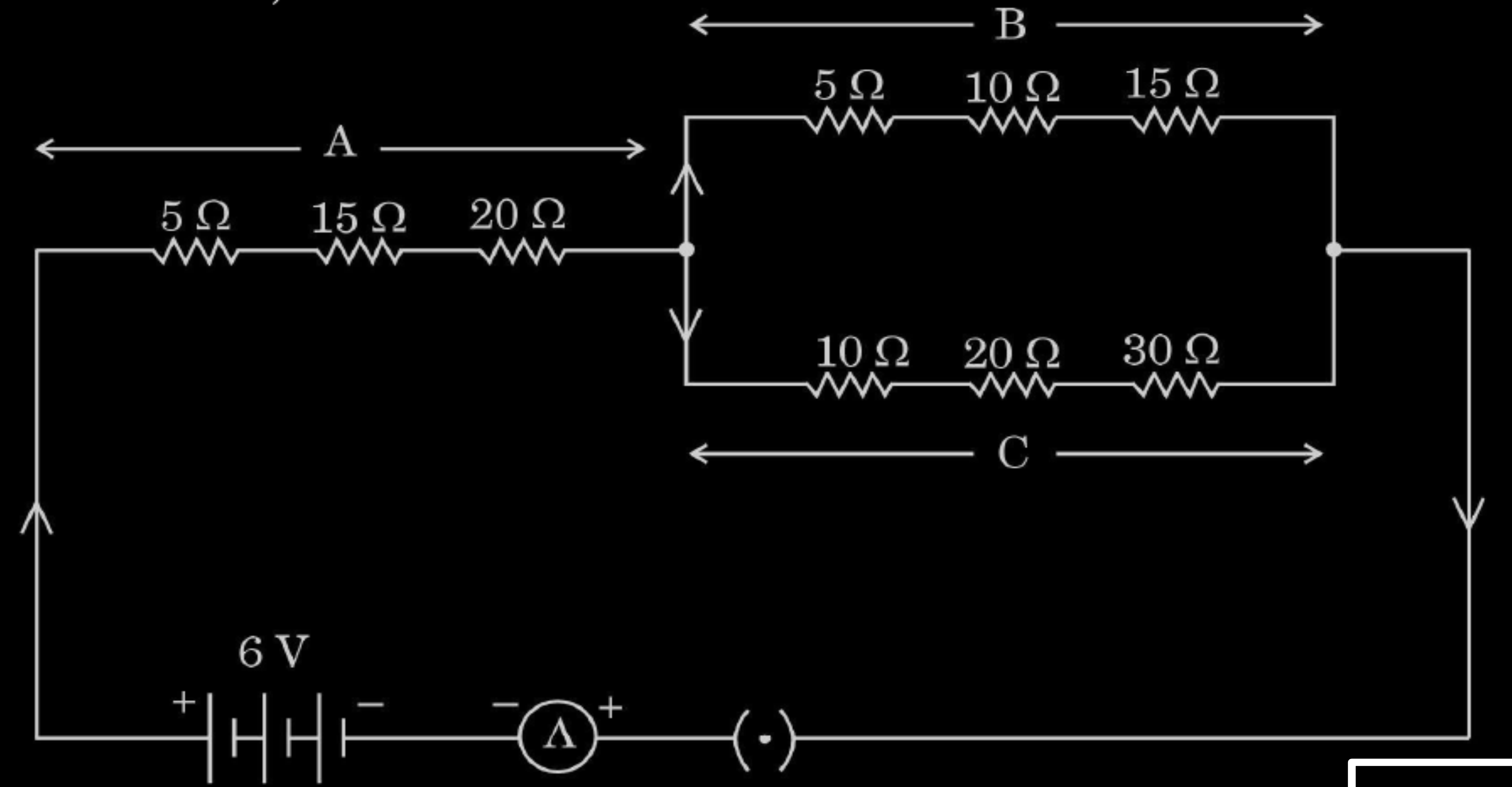
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

14. Study the following electric circuit in which the resistors are arranged in three arms A, B and C :



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- (a) Find the equivalent resistance of arm C. 1
- (b) Calculate the equivalent resistance of the parallel combination of the arms B and C. 1
- (c) (i) Determine the current that flows through the ammeter. 2
- OR**
- (ii) Determine the current that flows in the ammeter when the arm B is withdrawn from the circuit. 2

$$\begin{aligned}
 \text{(a)} \quad R &= R_1 + R_2 + R_3 \\
 &= 10\Omega + 20\Omega + 30\Omega \\
 &= 60\Omega
 \end{aligned}$$

1

$$\text{(b)} \quad \frac{1}{R_p} = \frac{1}{R_B} + \frac{1}{R_C}$$

$\frac{1}{2}$

$$\frac{1}{R_p} = \frac{1}{30\Omega} + \frac{1}{60\Omega}$$

$\frac{1}{2}$

$$R_p = 20\Omega$$

1

$$\begin{aligned}
 \text{(c)} \quad \text{(i)} \quad R &= R_s + R_p \\
 &= 40\Omega + 20\Omega = 60\Omega
 \end{aligned}$$

$\frac{1}{2} + \frac{1}{2}$

$$\therefore I = \frac{V}{R} = \frac{6V}{60\Omega} = \frac{1}{10} A = 0.1 A$$

OR

(c) (ii)

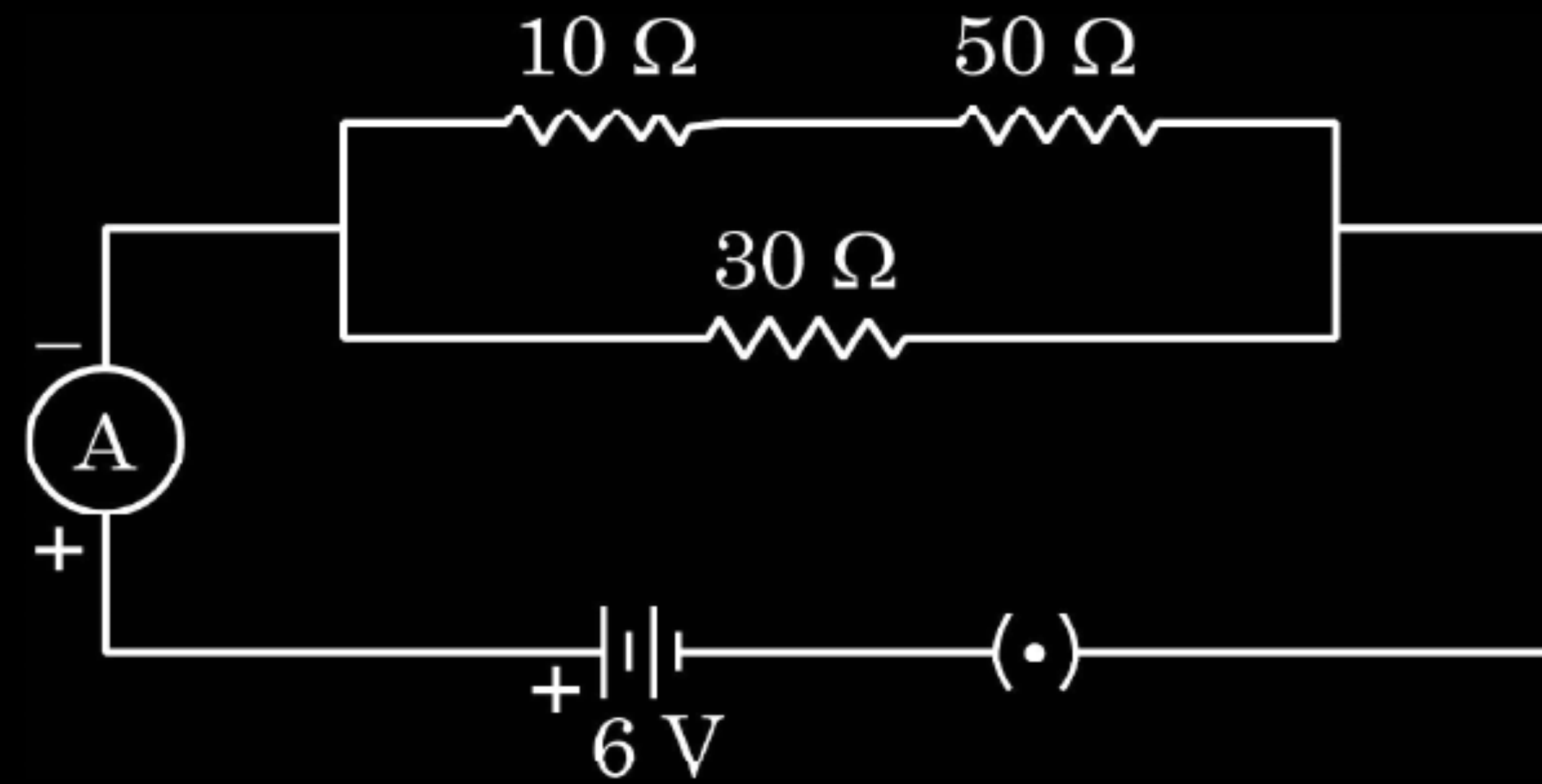
1

$$\text{Resistance, } R = 40\Omega + 60\Omega = 100\Omega$$

$\frac{1}{2} + \frac{1}{2}$

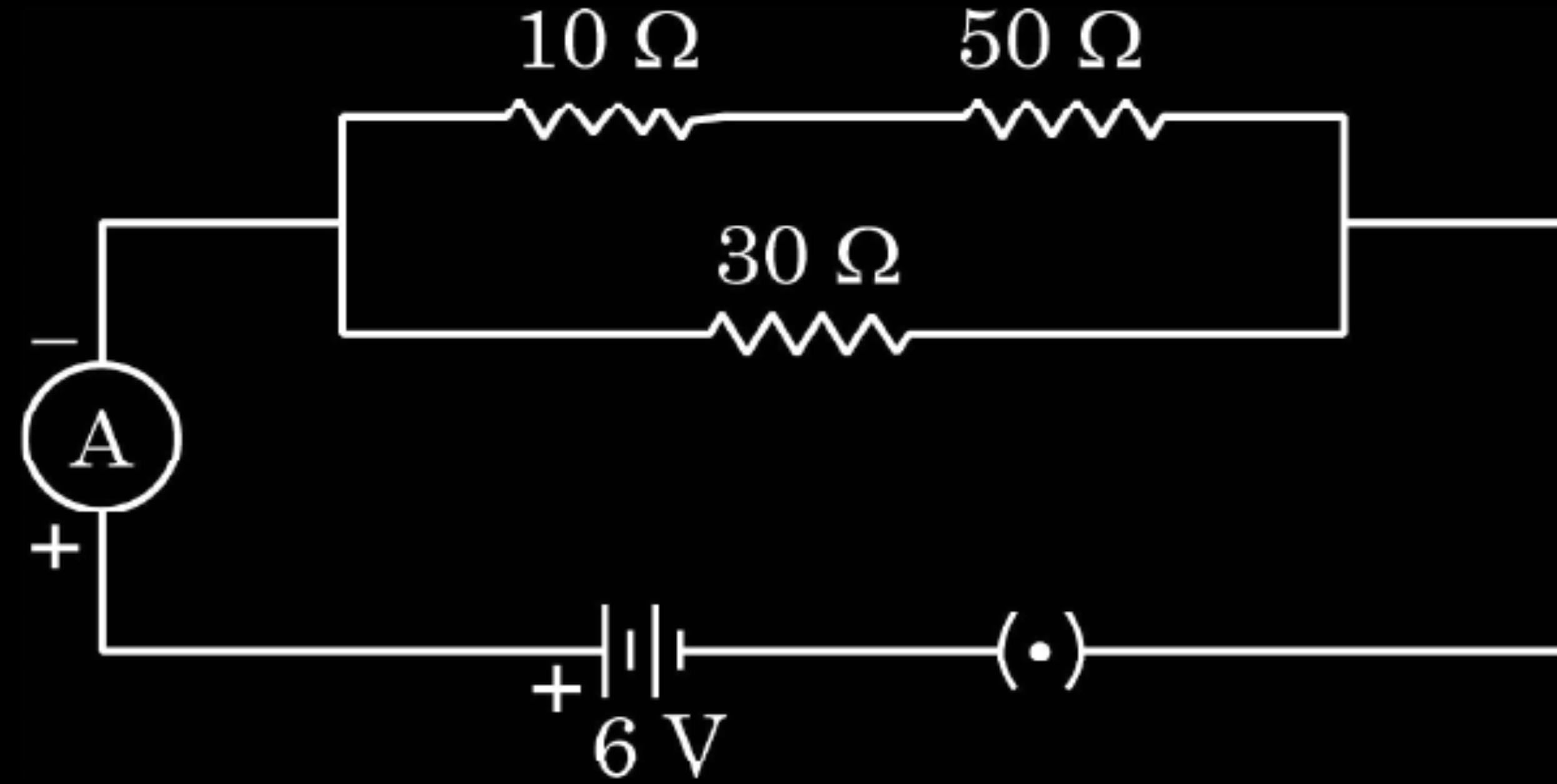
$$\therefore I = \frac{V}{R} = \frac{6V}{100\Omega} = 0.06 A$$

10. In the given circuit determine the value of :
- (i) total resistance of the circuit
 - (ii) current flowing through the ammeter.



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10. In the given circuit determine the value of :
- total resistance of the circuit
 - current flowing through the ammeter.



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$$(i) \quad R_S = R_1 + R_2$$

$$\text{Here } R_S = 10 \, \Omega + 50 \, \Omega = 60 \, \Omega$$

$$R_3 = 30 \, \Omega$$

$$\frac{1}{R_P} = \frac{1}{R_S} + \frac{1}{R_3}$$

$$\therefore R = \frac{R_S R_3}{R_S + R_3} = \frac{60 \, \Omega \times 30 \, \Omega}{(60 + 30) \, \Omega} = 20 \, \Omega$$

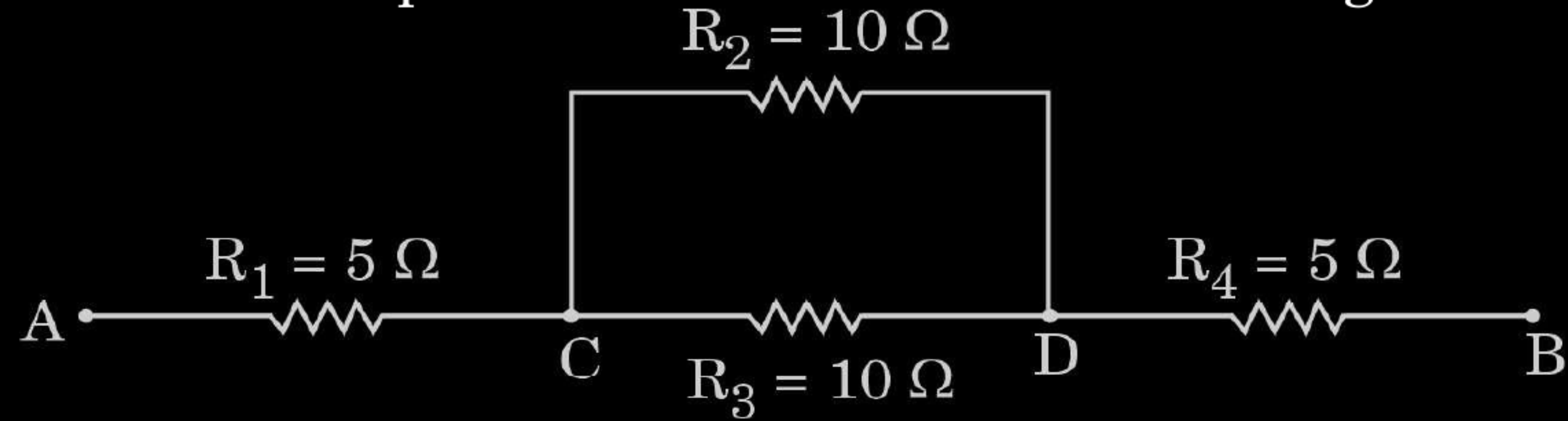
$$(ii) \quad I = \frac{V}{R}$$

$$= \frac{6 \, V}{20 \, \Omega} = 0.3 \, A$$

 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

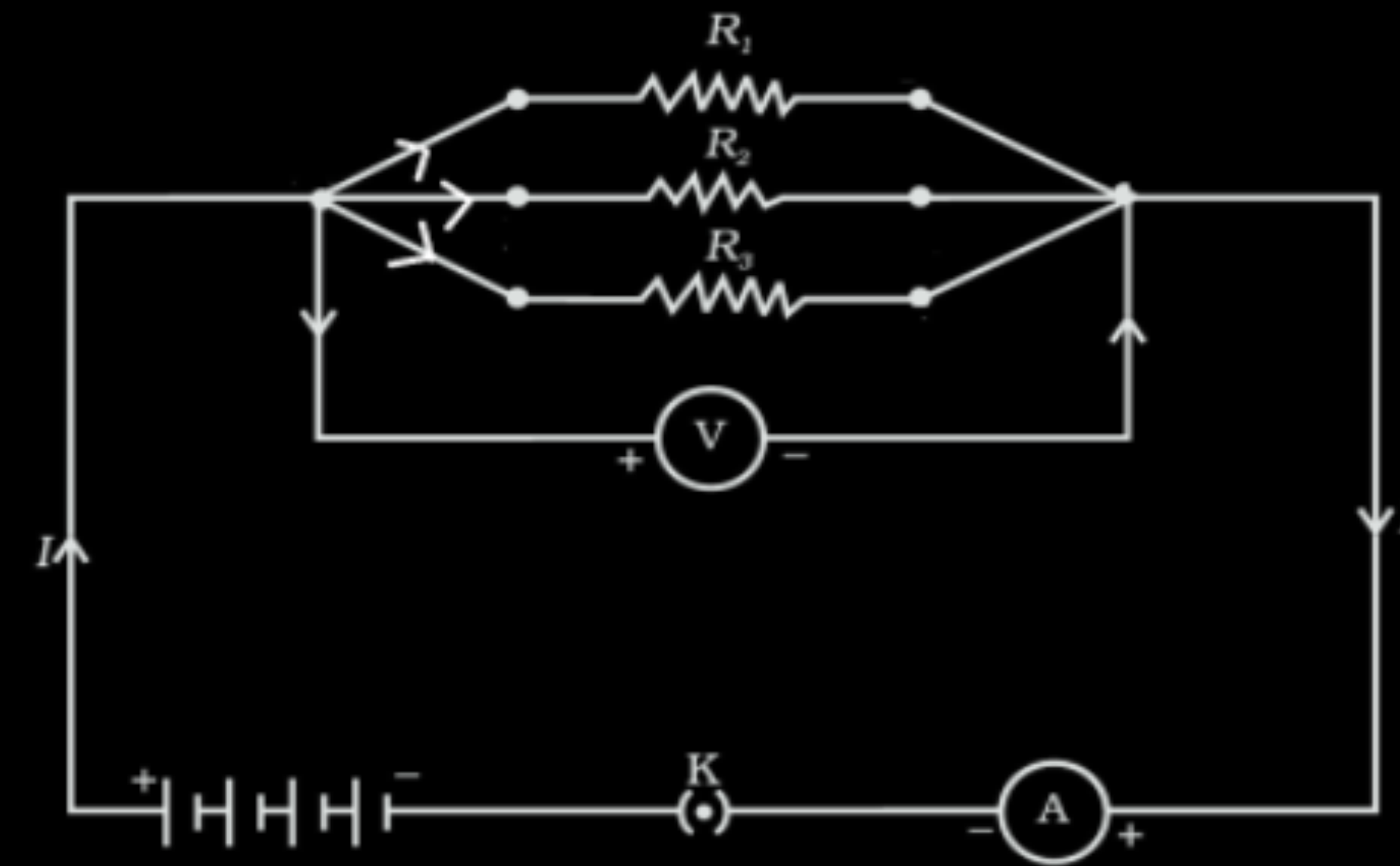
11. (a) Three resistors R_1 , R_2 and R_3 are connected in parallel and the combination is connected to a battery, an ammeter, a voltmeter and a key. Draw suitable circuit diagram to show the arrangement of these circuit components along with the direction of current flowing.
- (b) Calculate the equivalent resistance of the following network :

3



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(a)



Circuit diagram with given components

Direction

(b) Resistance between C and D is given by

$$\frac{1}{R_{CD}} = \frac{1}{10} + \frac{1}{10} = \frac{2}{10} = \frac{1}{5}$$

$$R_{CD} = 5 \Omega$$

$$D \text{ and } B = R_4 = 5 \Omega$$

$$\therefore \text{Total resistance is } R_S = R_{CD} + R_1 + R_4$$

$$\begin{aligned} R_{\text{total}} &= 5 \Omega + 5 \Omega + 5 \Omega \\ &= 15 \Omega \end{aligned}$$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

8. If a person has five resistors each of value $\frac{1}{5} \Omega$, then the maximum resistance he can obtain by connecting them is *1*

(A) 1Ω

(B) 5Ω

(C) 10Ω

(D) 25Ω

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8. If a person has five resistors each of value $\frac{1}{5} \Omega$, then the maximum resistance he can obtain by connecting them is 1

(A) 1Ω

(B) 5Ω

(C) 10Ω

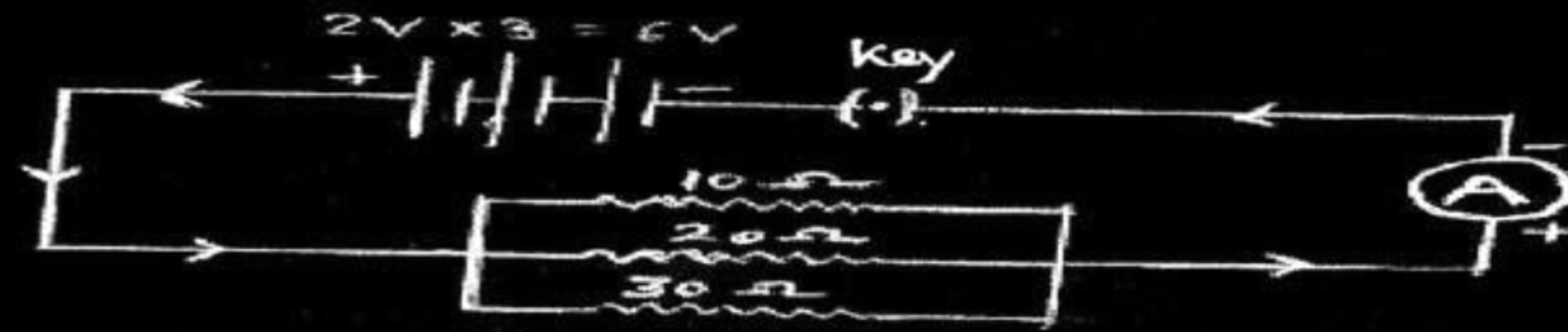
(D) 25Ω

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(A) 1Ω

29. Draw a schematic diagram of a circuit consisting of a battery of 3 cells of 2 V each, a combination of three resistors of $10\ \Omega$, $20\ \Omega$ and $30\ \Omega$ connected in parallel, a plug key and an ammeter, all connected in series. Use this circuit to find the value of the following :

- (a) Current through each resistor
- (b) Total current in the circuit
- (c) Total effective resistance of the circuit



2

(a) Three resistors are connected in parallel hence voltage across each is same i.e. 6V.

$$I_1 = \frac{V}{R_1} = \frac{6}{10} = 0.6 \text{ A}$$

$\frac{1}{2}$

$$I_2 = \frac{V}{R_2} = \frac{6}{20} = 0.3 \text{ A}$$

$\frac{1}{2}$

$$I_3 = \frac{V}{R_3} = \frac{6}{30} = 0.2 \text{ A}$$

$\frac{1}{2}$

$$\text{b) } I = I_1 + I_2 + I_3 = 1.1 \text{ A}$$

$\frac{1}{2}$

$$\text{c) } R_{\text{eff}} = \frac{V}{I}$$

$\frac{1}{2}$

$$= \frac{6}{1.1} = 5.4 \Omega$$

$\frac{1}{2}$

Two identical resistors, each of resistance $15\ \Omega$, are connected in (i) series, and (ii) parallel, in turn to a battery of $6\ \text{V}$. Calculate the ratio of the power consumed in the combination of resistors in each case.

5

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Two identical resistors, each of resistance $15\ \Omega$, are connected in (i) series, and (ii) parallel, in turn to a battery of 6 V . Calculate the ratio of the power consumed in the combination of resistors in each case.

5

$R_1 = R_2 = 15\Omega \quad V = 6V$	
i) In series :	
$R_s = R_1 + R_2 = 15\Omega + 15\Omega = 30\Omega$	$\frac{1}{2}$
$I = \frac{V}{R_s} = \frac{6V}{30\Omega} = 0.2\text{ A}$	$\frac{1}{2}$
$\therefore P_1 = VI = 6V \times 0.2\text{ A} = 1.2\text{ W}$	1
ii) In parallel	
$R_p = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{15 \times 15}{15 + 15} = \frac{225}{30} = 7.5\Omega$	$\frac{1}{2}$
$I = \frac{V}{R_p} = \frac{6V}{7.5\Omega} = 0.8\text{ A}$	$\frac{1}{2}$
$P_2 = VI = 6V \times 0.8\text{ A} = 4.8W$	1
Ratio of power = $\frac{P_1}{P_2} = \frac{1.2W}{4.8W} = \frac{1}{4}$	$\frac{1}{2}$
$\therefore P_1 : P_2 = 1 : 4$	$\frac{1}{2}$

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7. The maximum resistance which can be made using four resistors each of resistance $\frac{1}{2} \Omega$ is

- (a) 2Ω
- (b) 1Ω
- (c) 2.5Ω
- (d) 8Ω

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7. The maximum resistance which can be made using four resistors each of resistance $\frac{1}{2} \Omega$ is

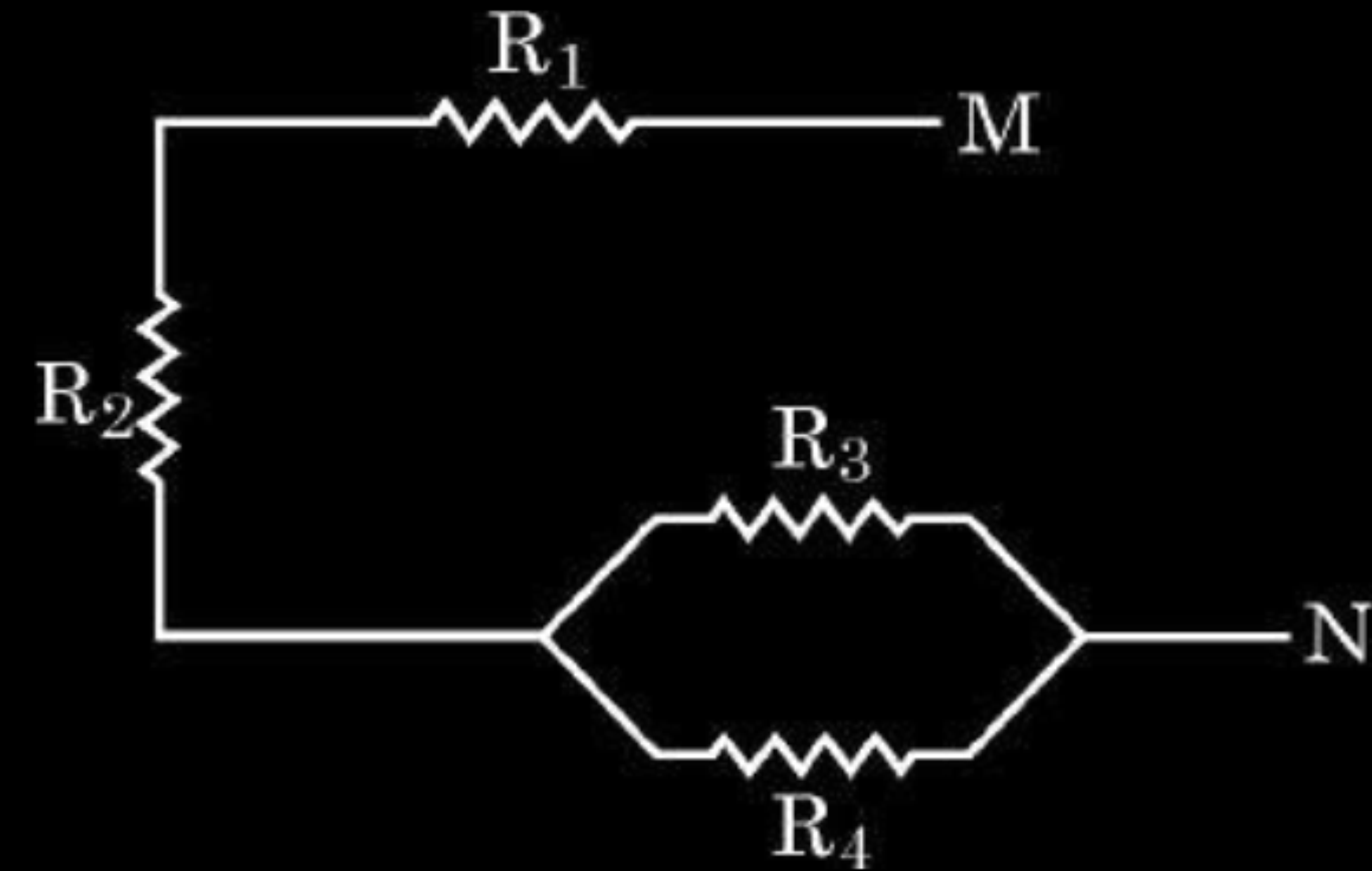
- (a) 2Ω
- (b) 1Ω
- (c) 2.5Ω
- (d) 8Ω

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1

(a) / 2Ω

29. (a) For the combination of resistors shown in the following figure, find the equivalent resistance between M & N.



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- (b) State Joule's law of heating.
(c) Why we need a 5 A fuse for an electric iron which consumes 1 kW power at 220 V ?
(d) Why is it impracticable to connect an electric bulb and an electric heater in series ?

(a) R_3 and R_4 are in parallel combination .

$\therefore R_{\text{parallel}}$ is given by

$$\frac{1}{R_p} = \frac{1}{R_3} + \frac{1}{R_4}$$

$$\frac{1}{R_p} = \frac{R_4 + R_3}{R_3 R_4}$$

$$R_p = \frac{R_3 R_4}{R_4 + R_3}$$

Now, R_1 R_2 and R_p are in series.

$$\therefore R_{\text{eq}} = R_1 + R_2 + R_p$$

$$= R_1 + R_2 + \frac{R_3 R_4}{R_4 + R_3}$$

(b) The heat produced in a resistor is directly proportional to

- square of current for a given resistance.
- the resistance for a given current and
- the time for which the current flows through the resistor.

(Note : if a candidate writes $H = I^2 R t$ give full credit).

$$(c) P = V I \quad \text{or} \quad I = \frac{P}{V}$$

$$I = \frac{1000 \text{ watt}}{220 \text{ volt}} = 4.54 \text{ A}$$

Since 4.54 ampere current flows in the circuit , a 5 A fuse must be used.

(d) Electric bulb & electric heater will not get currents and voltages as per their requirement.

1

1

1

1

1

6. The maximum resistance which can be made using four resistors each of $2\ \Omega$ is

(a) $2\ \Omega$

(b) $4\ \Omega$

(c) $8\ \Omega$

(d) $16\ \Omega$

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1

6. The maximum resistance which can be made using four resistors each of $2\ \Omega$ is

(a) $2\ \Omega$

(b) $4\ \Omega$

(c) $8\ \Omega$

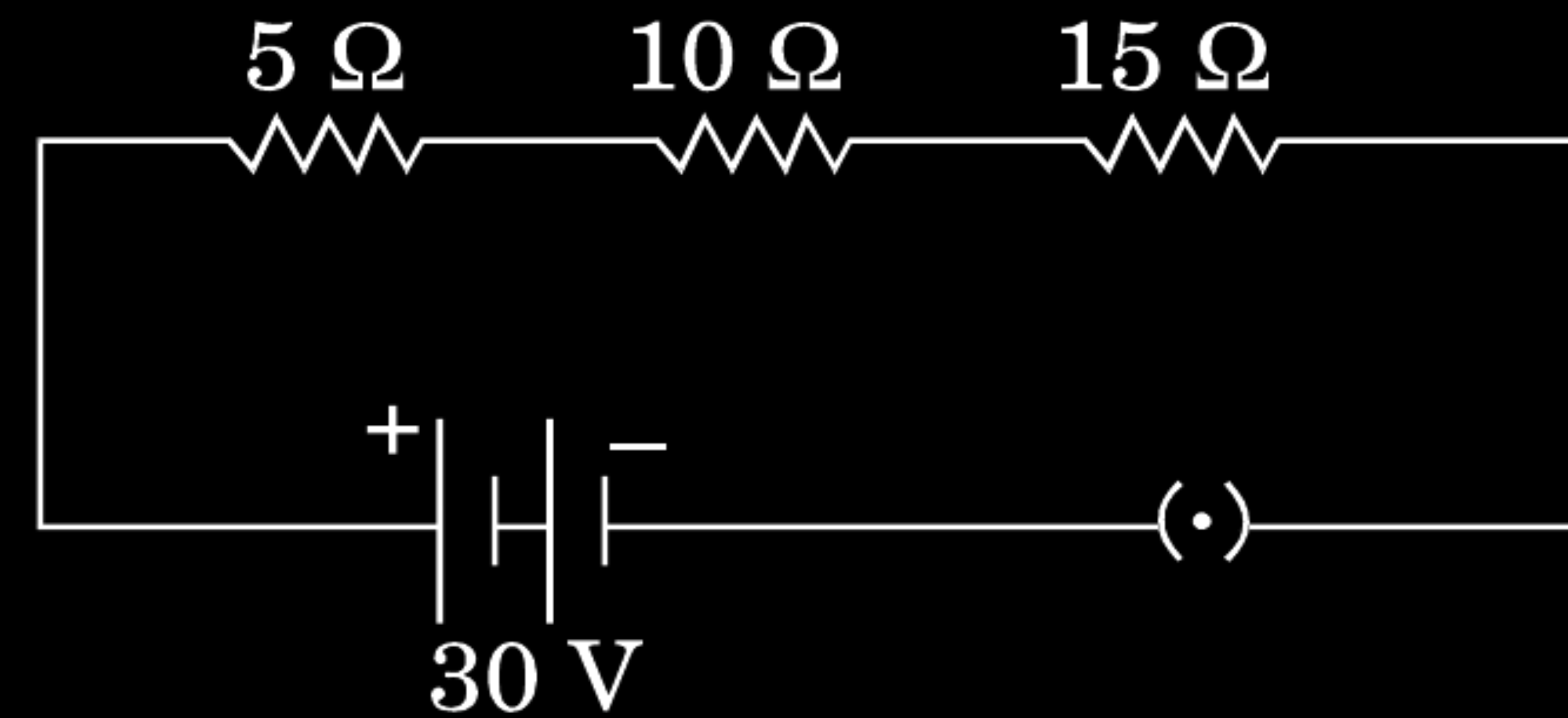
(d) $16\ \Omega$

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1

(c) / 8Ω

- (a) How will you infer with the help of an experiment that the same current flows through every part of a circuit containing three resistors in series connected to a battery ?
- (b) Consider the given circuit and find the current flowing in the circuit and potential difference across the $15\ \Omega$ resistor when the circuit is closed.



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- (a)
- (i) Join the three resistors of different values in series
 - (ii) Connect them with battery, an ammeter and plug key.
 - (iii) Plug the key and note the ammeter reading
 - (iv) Change the position of ammeter to anywhere in between the resistors and note the ammeter reading each time.
 - (v) The ammeter reading will remain same everytime. Therefore when resistors are connected in series same current flows through all resistors, when it is connected to a battery.

Note: If explained with the help of diagram give full credit

- (b) Total resistance of the circuit =

$$R = R_1 + R_2 + R_3 = 5 + 10 + 15 = 30 \text{ ohm}$$

Potential difference across the circuit / By ohm's law

$$V = IR \text{ or } I = \frac{V}{R} = \frac{30V}{30\text{ohm}} = 1A$$

$$\text{Potential difference across 15 ohm Resistor} = 1A \times 15 \text{ ohm} = 15 \text{ volt}$$

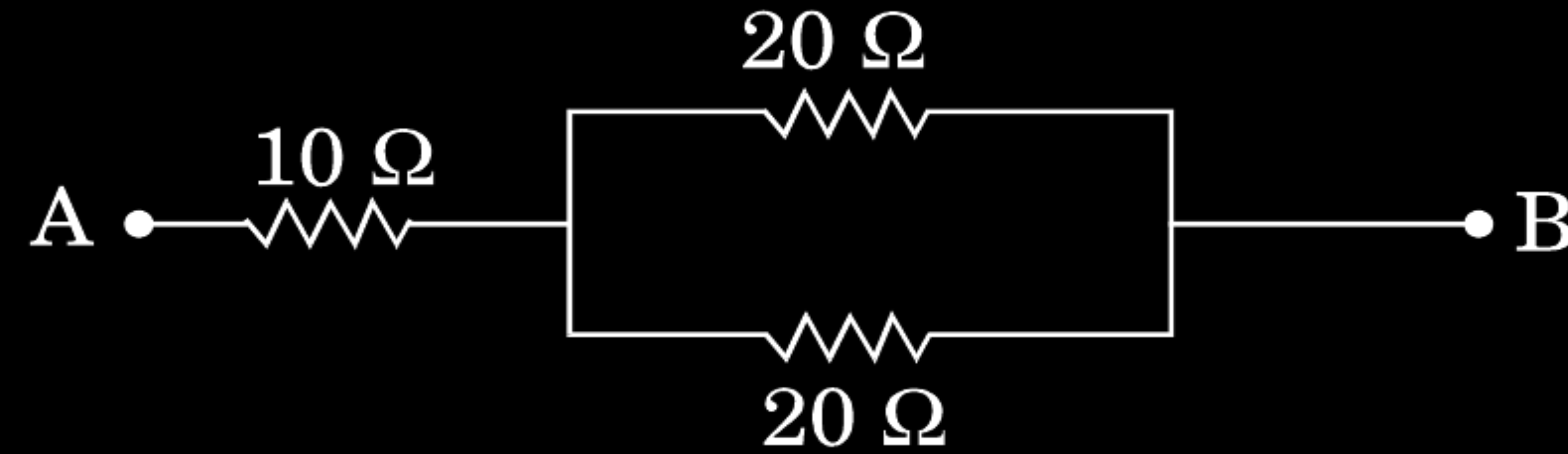
$\frac{1}{2} \times 5$

1

1

$\frac{1}{2}$

- (a) Three resistors R_1 , R_2 and R_3 are connected in parallel and the combination is connected to a battery, ammeter, voltmeter and key. Draw suitable circuit diagram and obtain an expression for the equivalent resistance of the combination of the resistors.
- (b) Calculate the equivalent resistance of the following network :



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(a)

Total current $I = I_1 + I_2 + I_3$

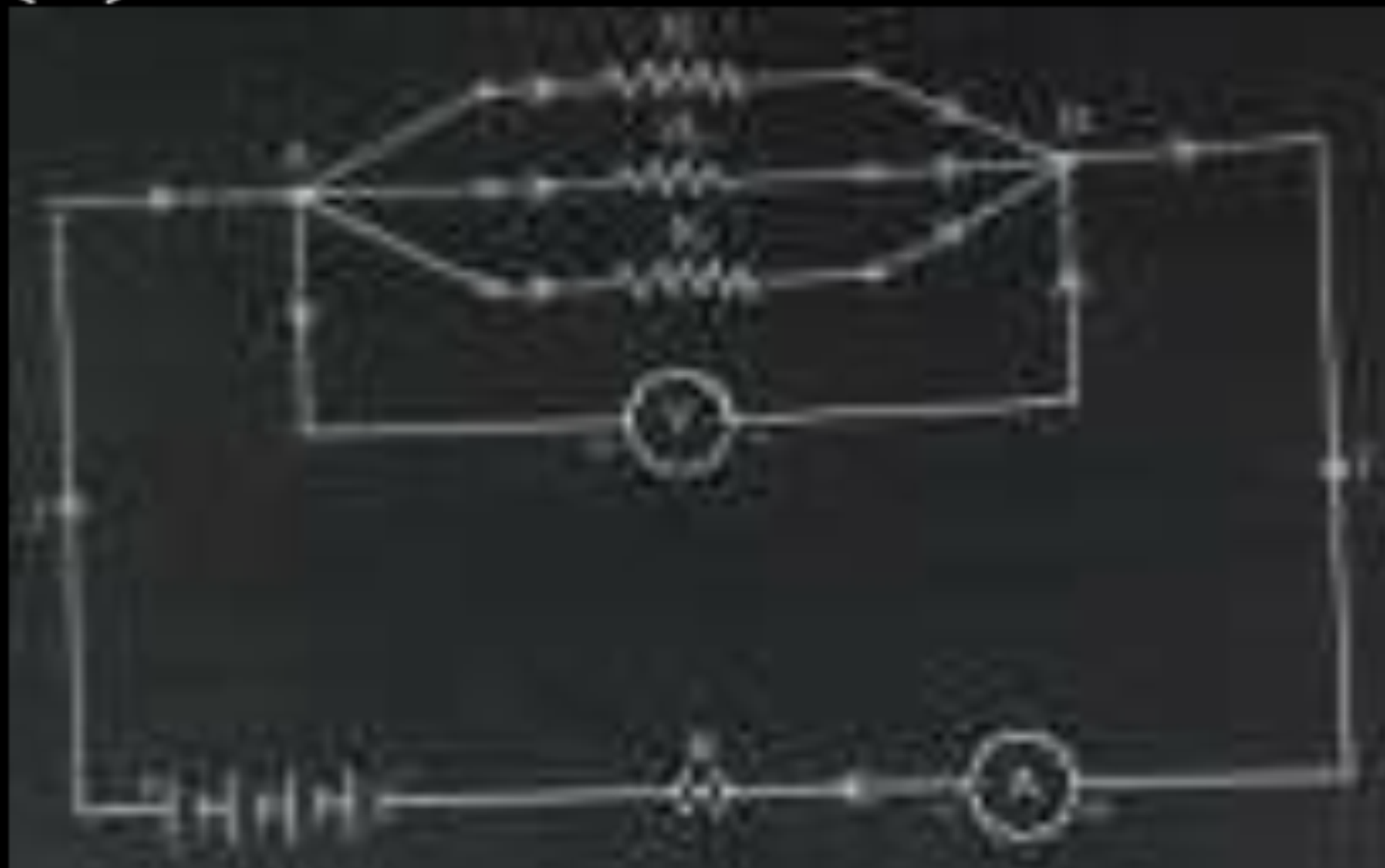
Let R_p be the equivalent resistance of R_1, R_2, R_3 . Then the total current $I = \frac{V}{R_p}$

$$I_1 = \frac{V}{R_1}, I_2 = \frac{V}{R_2}, I_3 = \frac{V}{R_3}$$

$$\therefore I = V \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) = \frac{V}{R_p}$$

$$\therefore \frac{1}{R_p} + \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

(b)



$$\frac{1}{R_p} = \frac{1}{20} + \frac{1}{20} = \frac{2}{20} = \frac{1}{10}$$

$$\Rightarrow R_p = 10 \text{ ohms}$$

$$\text{Equivalent resistance of the network} = R_{eq} = R_1 + R_p = 10 + 10 = 20 \text{ ohm}$$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

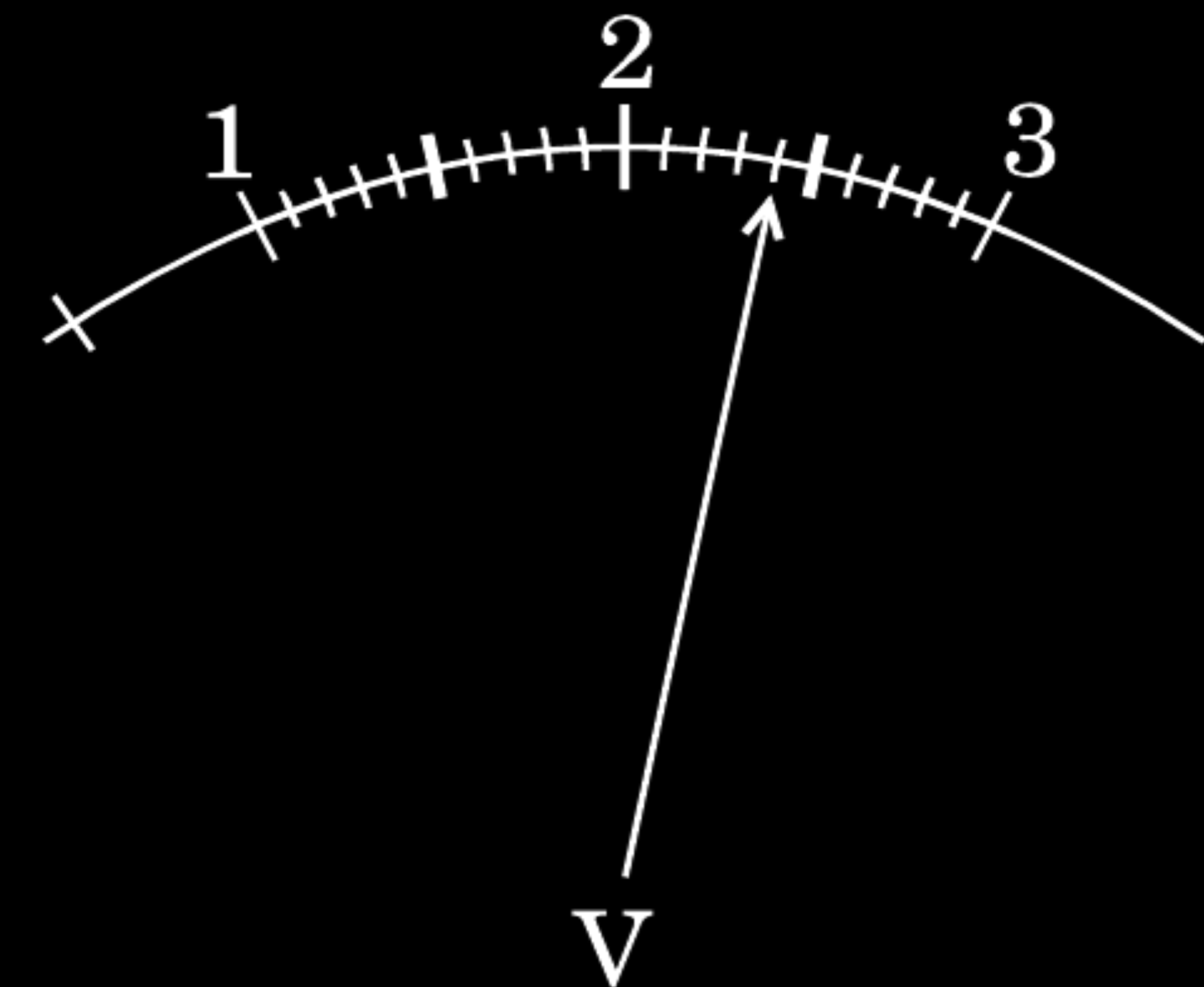
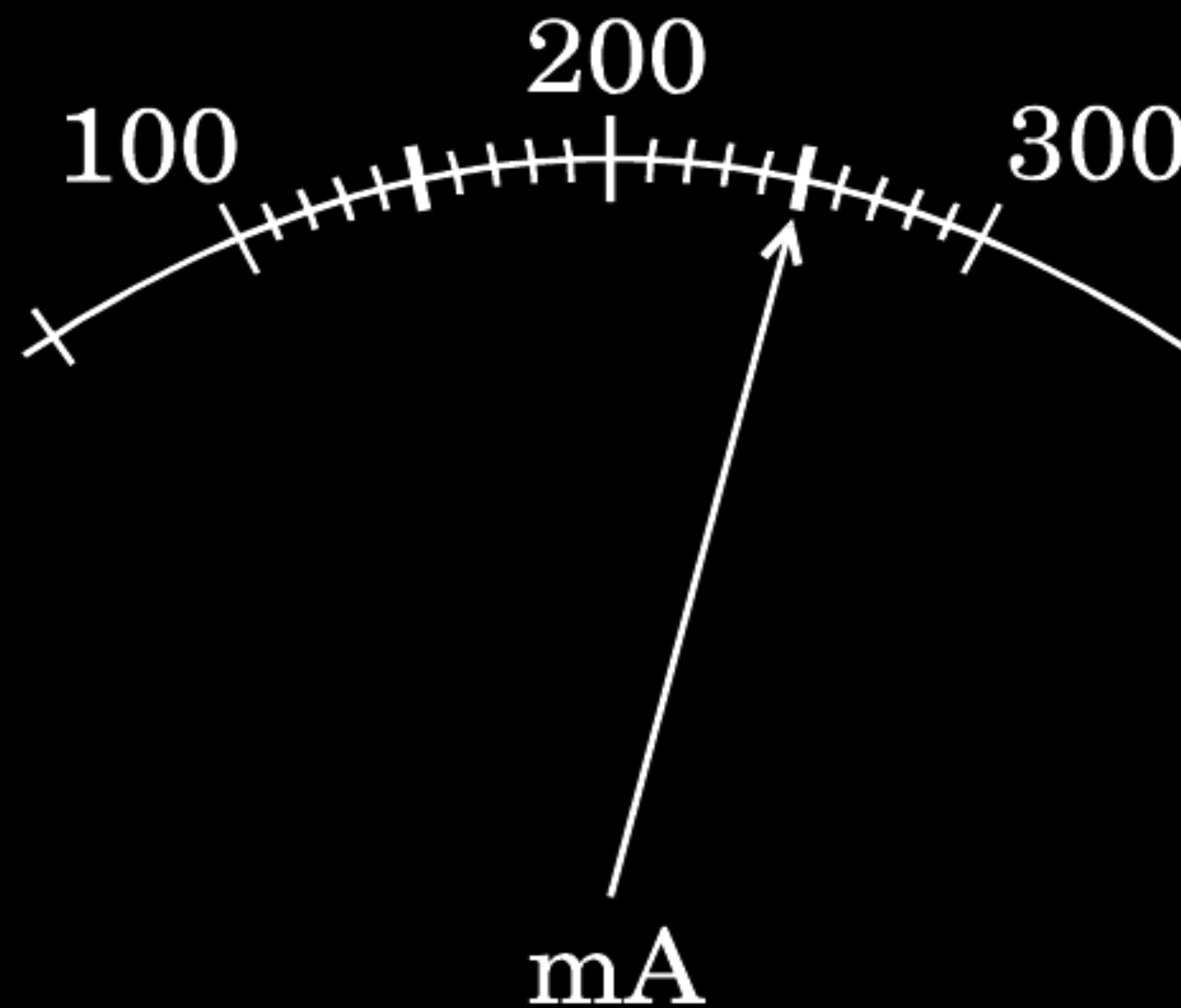
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1

The current flowing through a resistor connected in a circuit and the potential difference developed across its ends are as shown in the diagram by milliammeter and voltmeter readings respectively :

- (a) What are the least counts of these meters ?
- (b) What is the resistance of the resistor ?

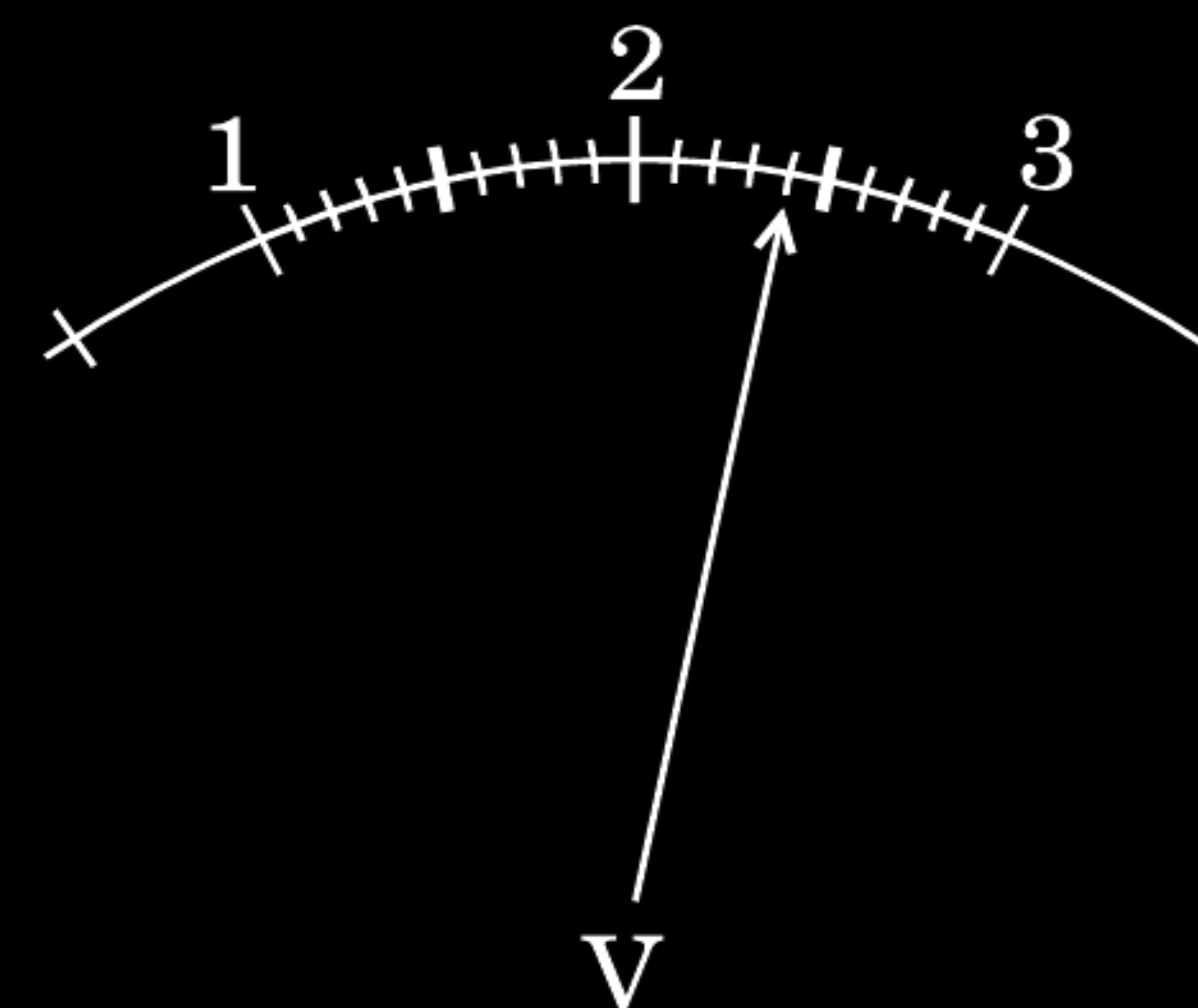
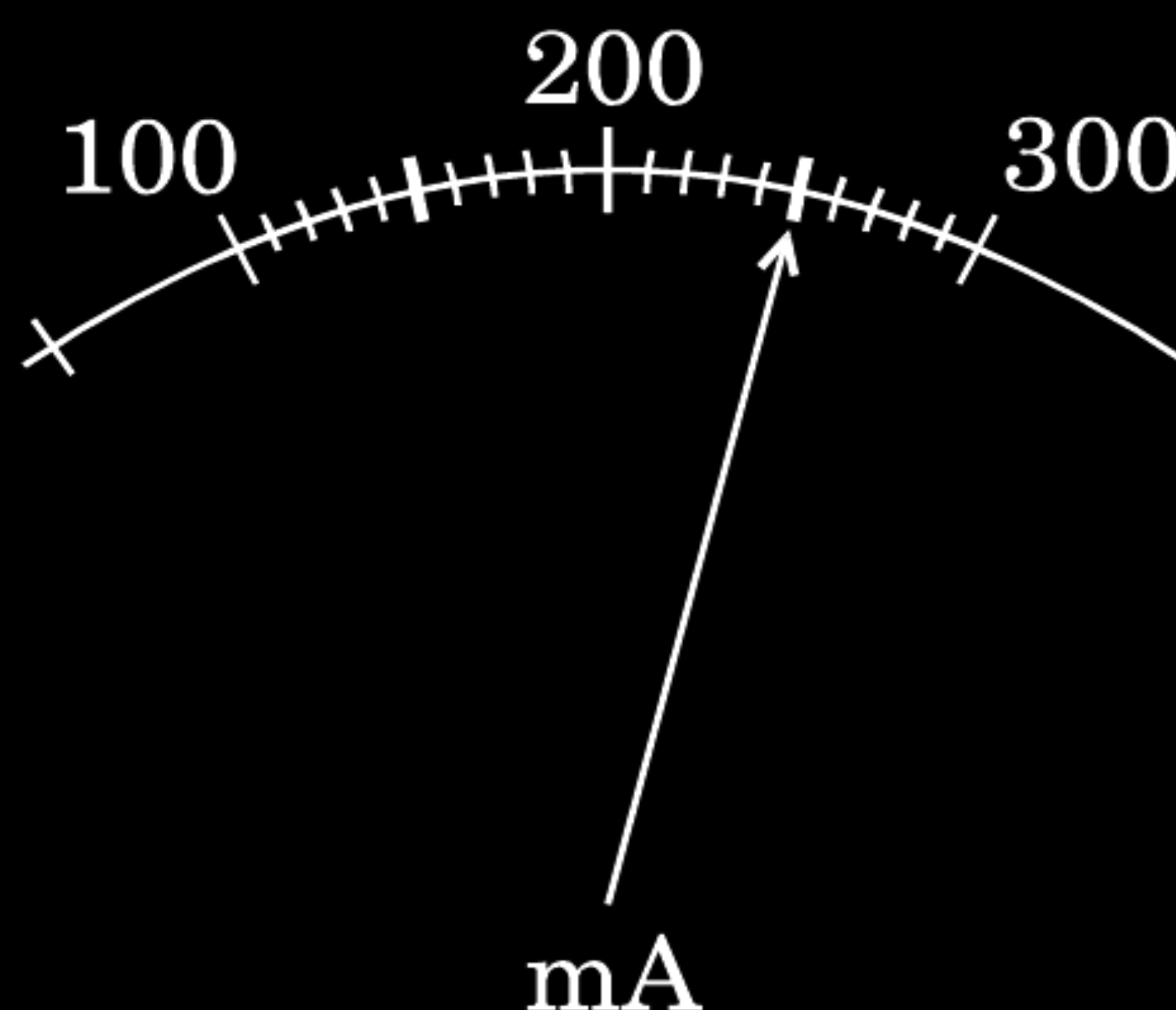
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The current flowing through a resistor connected in a circuit and the potential difference developed across its ends are as shown in the diagram by milliammeter and voltmeter readings respectively :

- (a) What are the least counts of these meters ?
- (b) What is the resistance of the resistor ?

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- (a) $\angle$ least count of ammeter = 10 mA
 $\angle$ least count of Voltmeter = 0.1 V

$\frac{1}{2} + \frac{1}{2}$

- (b) $\frac{2.4}{0.25} = 9.6 \text{ ohm}$ (250mA = 0.25A)

$\frac{1}{2} + \frac{1}{2}$

- (a) Three resistors of resistances R_1 , R_2 and R_3 are connected (i) in series, and (ii) in parallel. Write expressions for the equivalent resistance of the combination in each case.

(a)

• In series - $R_S = R_1 + R_2 + R_3$.

$\frac{1}{2}$

• In parallel - $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

$\frac{1}{2}$

Resistance is at minimum - $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$

1

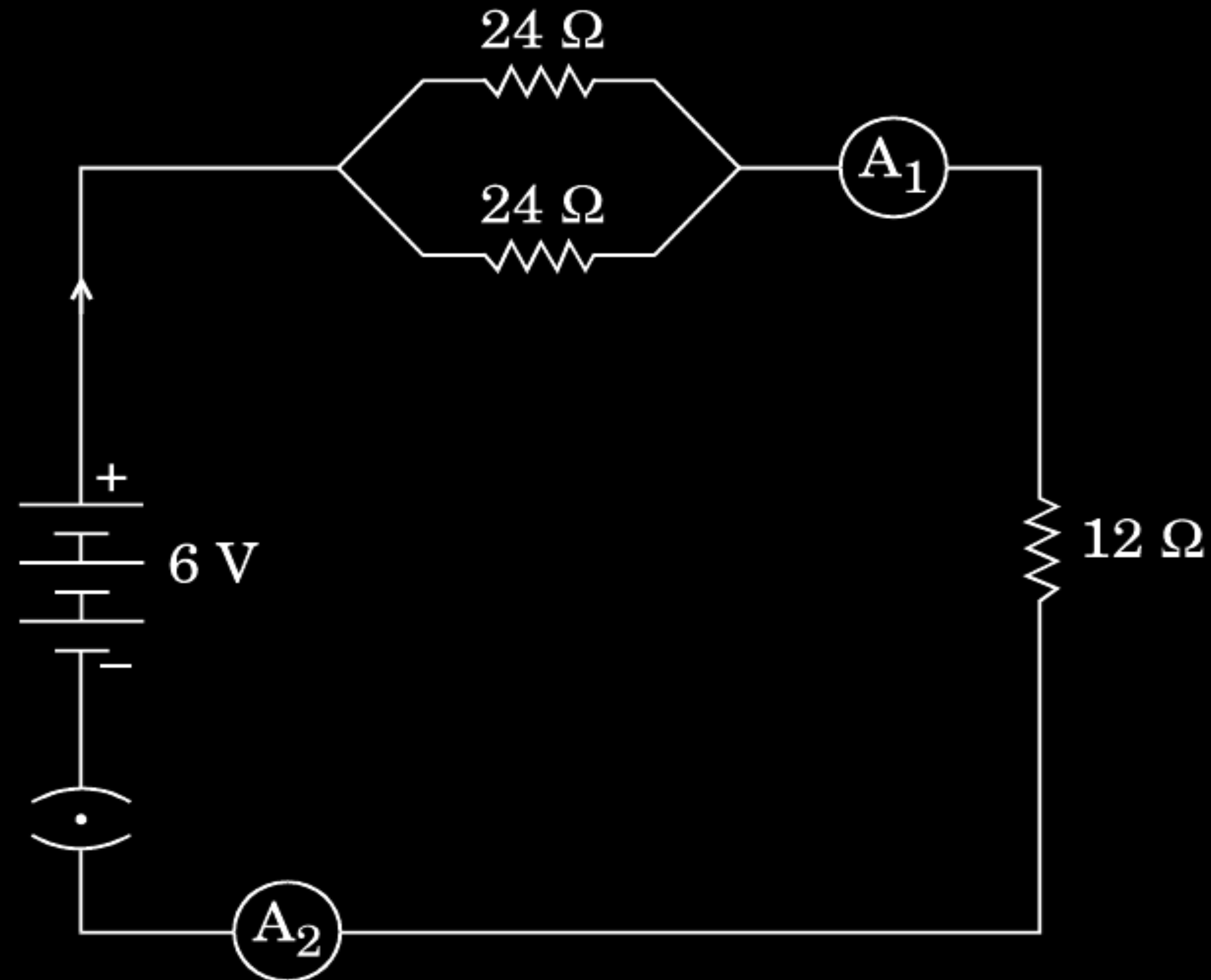
$$\frac{1}{12} + \frac{1}{12} = \frac{2}{12} = 6\Omega$$

Resistance is maximum - $R_S = R_1 + R_2$

1

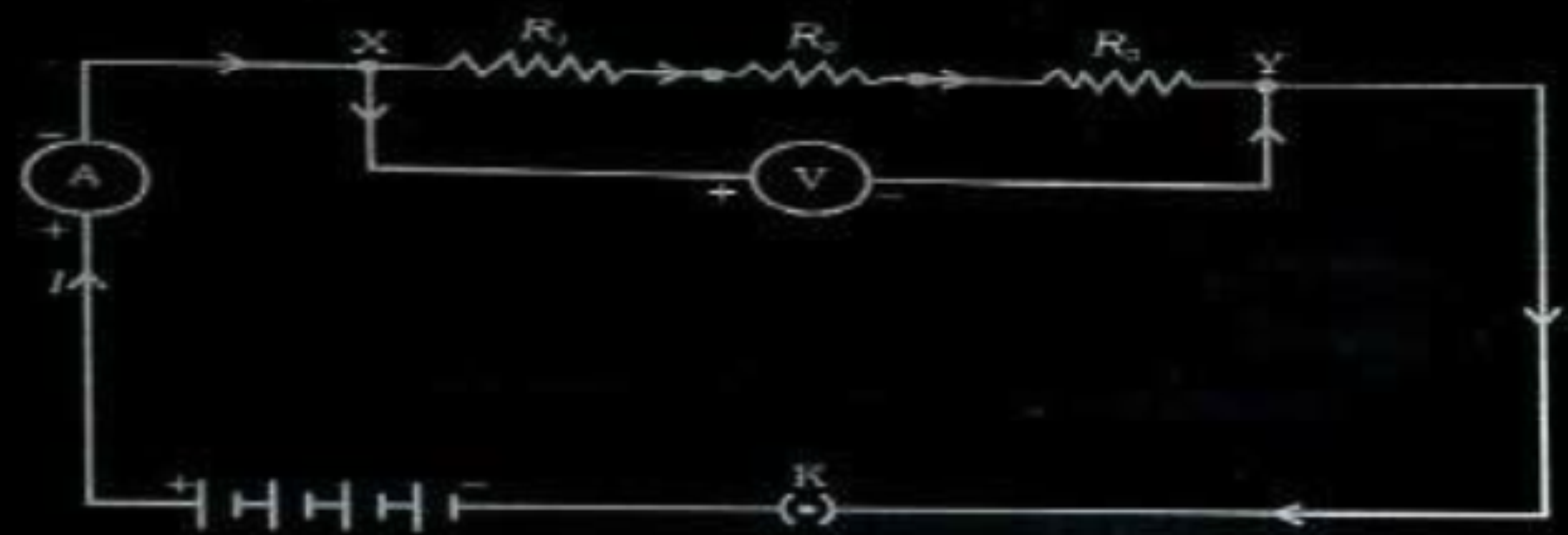
$$R_S = 12 + 12 = 24 \Omega$$

- (a) How will you infer with the help of an experiment that the same current flows through every part of the circuit containing three resistors R_1 , R_2 and R_3 in series connected to a battery of V volts ?
- (b) Study the following circuit and find out :
- (i) Current in $12\ \Omega$ resistor.
 - (ii) Difference in the readings of A_1 and A_2 , if any.



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(a)



- Three resistors R_1 , R_2 , R_3 are joined.

- They are connected with

2½

the battery and ammeter and a plug key.

- The ammeter reading is noted.
- Position of ammeter is changed to different position and readings taken each time.
- The reading remain same.

(If it is explained by diagram, give full credit)

(b)

$$(i) \quad \frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} \quad \frac{1}{R_1} = \frac{1}{R_1} + \frac{1}{R_1} \quad R_t = R_p + 12 \Omega$$

$$R_t = 24 \Omega$$

1½

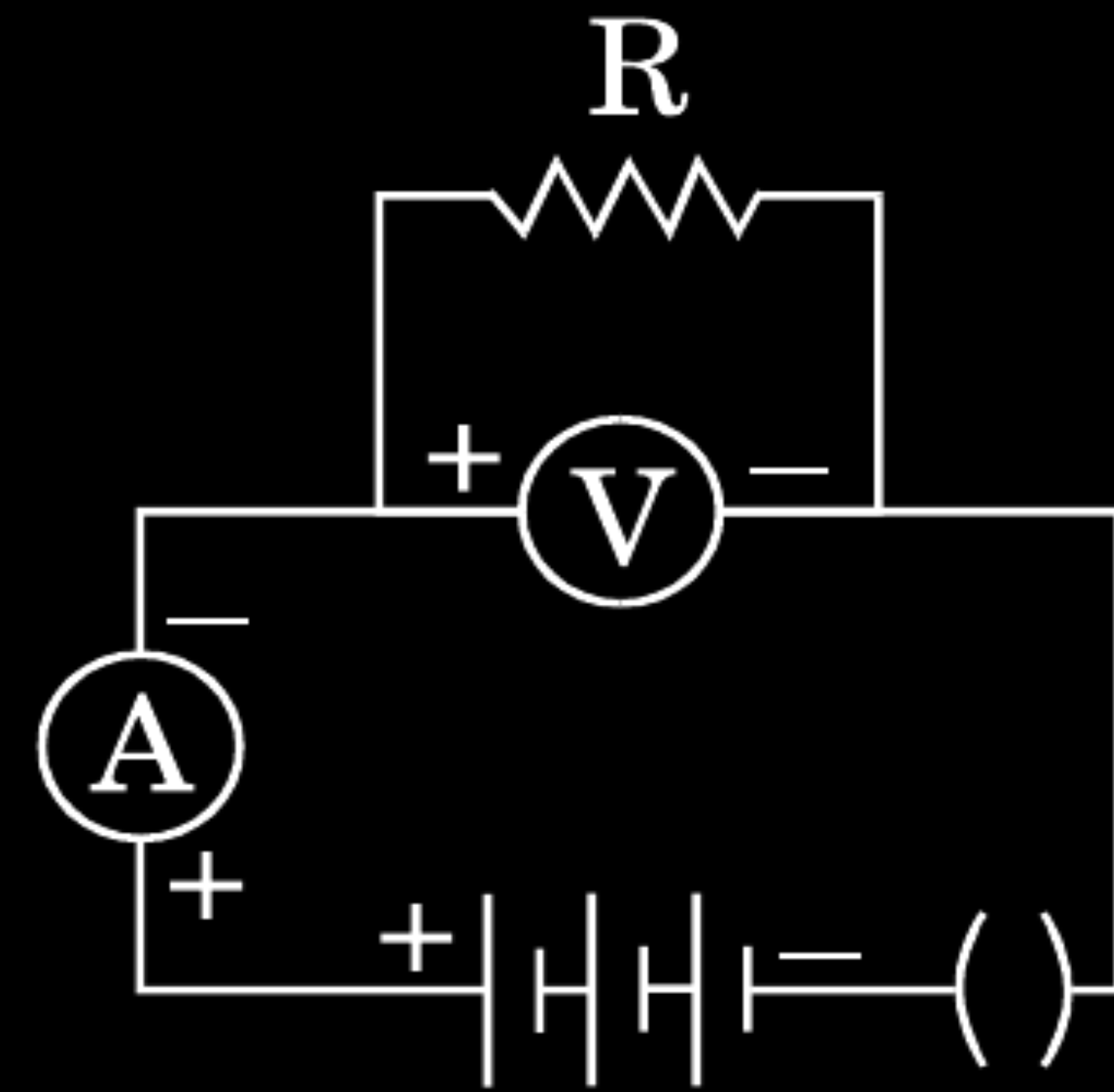
$$V = IR_T$$

$$I = 6/24 = 0.25 \text{ Ampere}$$

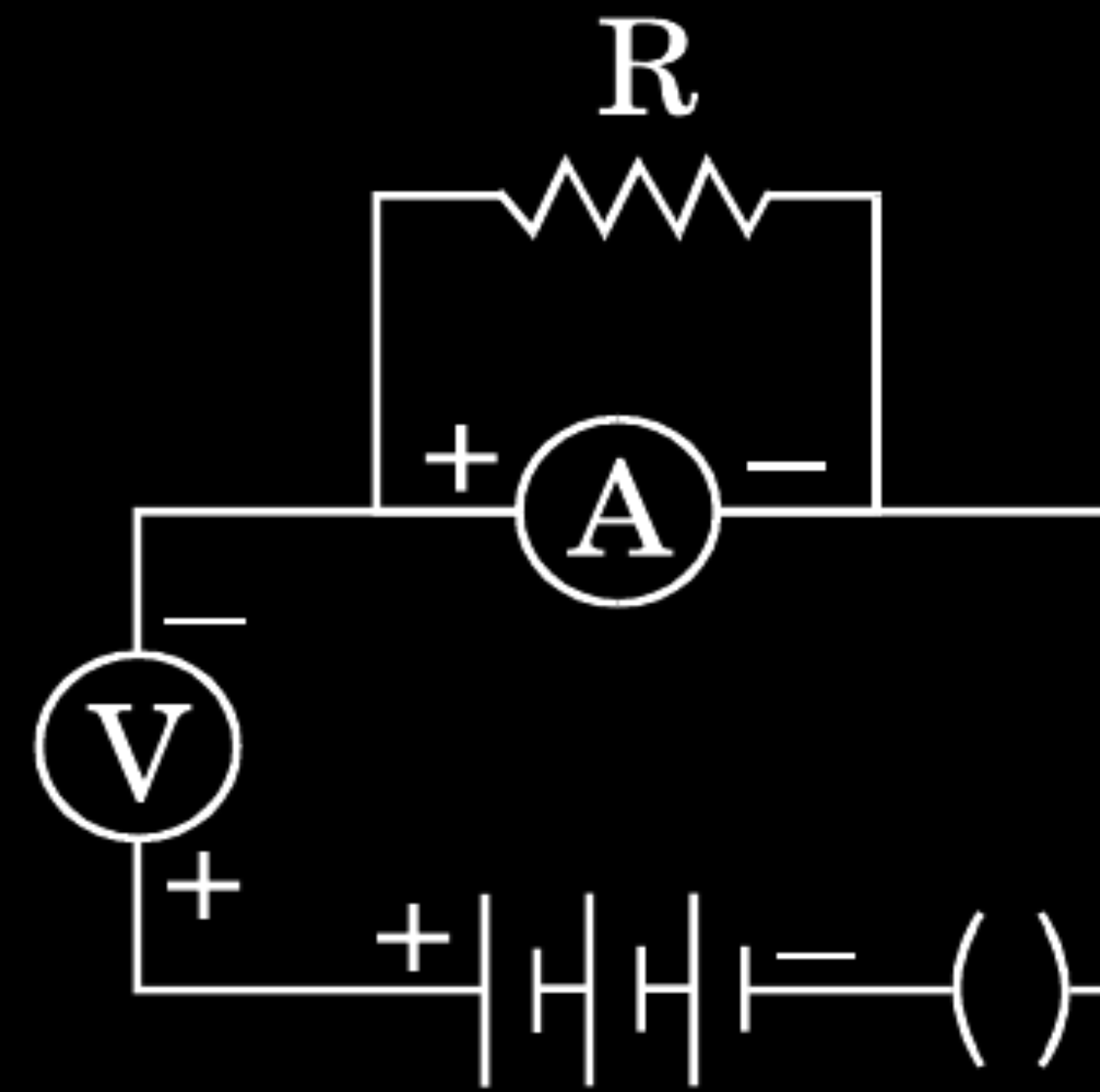
(ii) Same readings of A_1 and A_2

1

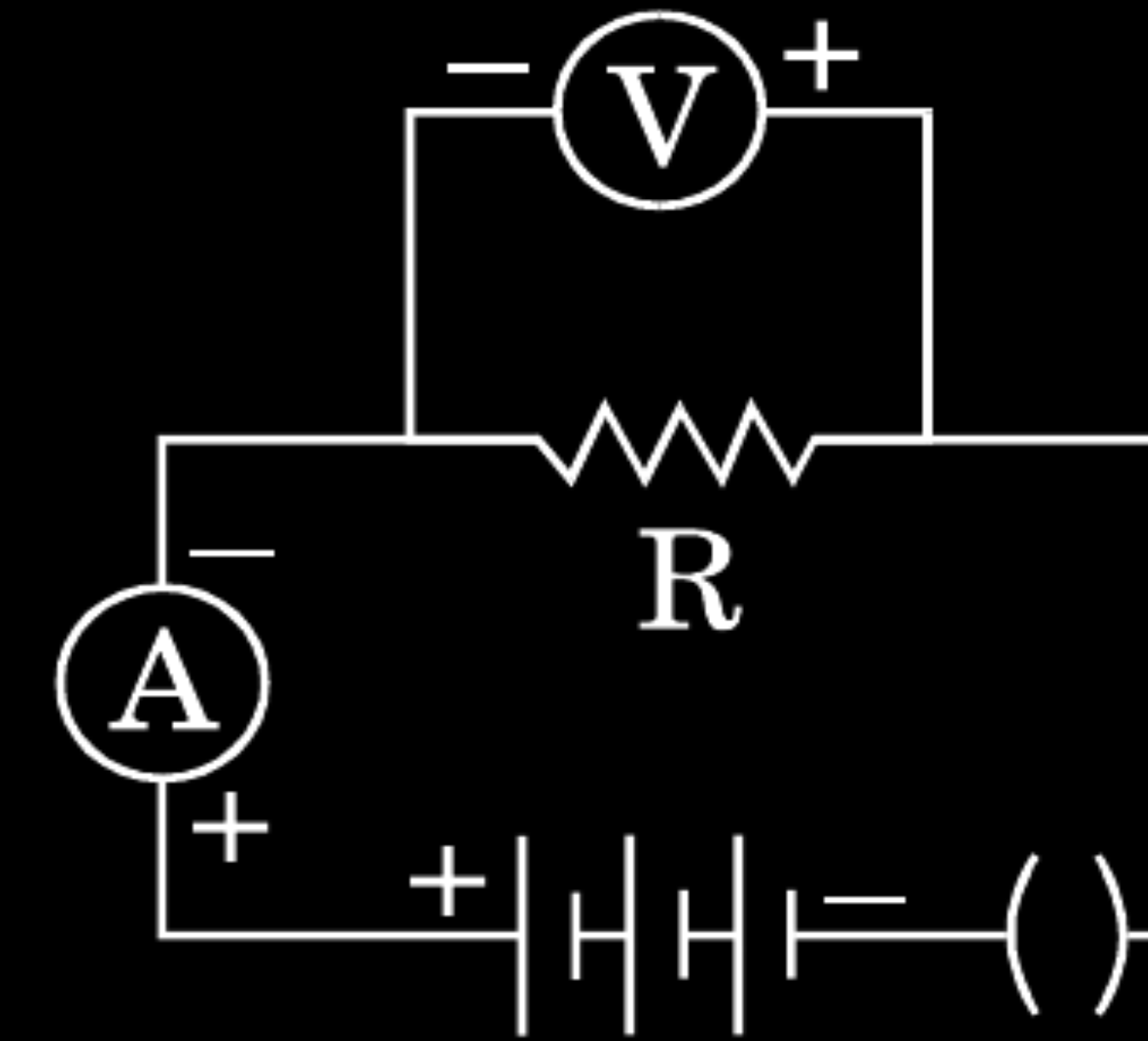
Which one of the following is the correct set-up for studying the dependence of the current on the potential difference across a resistor and why ?



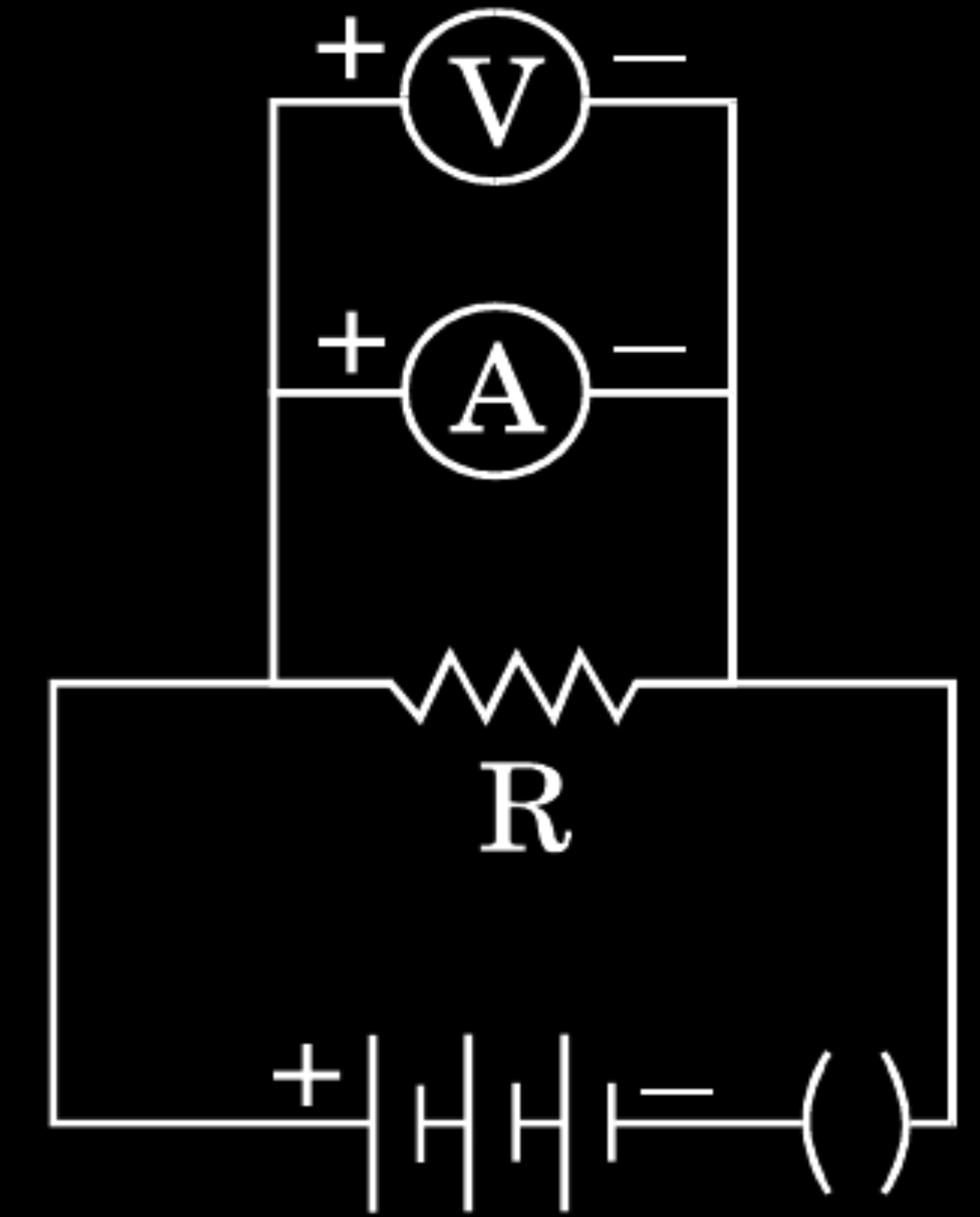
A



B

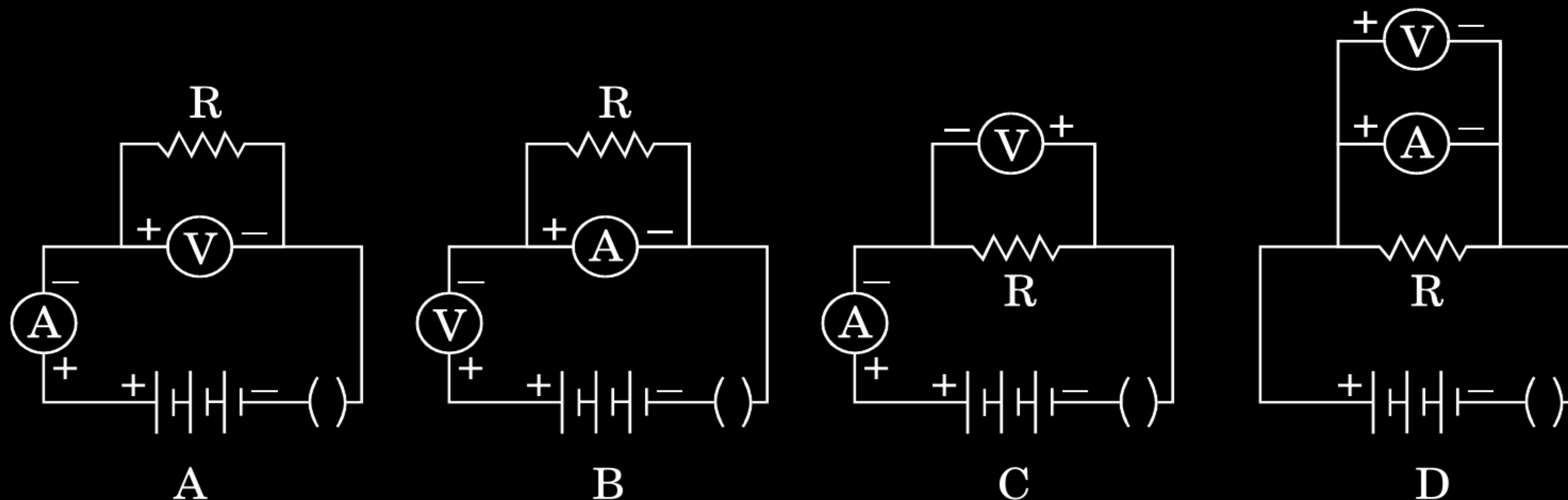


C



D

Which one of the following is the correct set-up for studying the dependence of the current on the potential difference across a resistor and why ?



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- Set up A is correct.
- Ammeter should be connected in series whereas voltmeter should be connected in parallel to the resistor across which potential difference is to be measured.
- Positive of voltmeter and ammeter should be connected to the positive of supply voltage.

$$\frac{1}{2}$$

$$\frac{1}{2} + \frac{1}{2}$$

$$\frac{1}{2}$$

The rest position of the needles in a milliammeter and voltmeter, not in use, are as shown in Figure A. When a student uses these instruments in his experiment, the readings of the needles are in the positions shown in Figure B. Determine the correct values of current and voltage the student should use in his calculations.

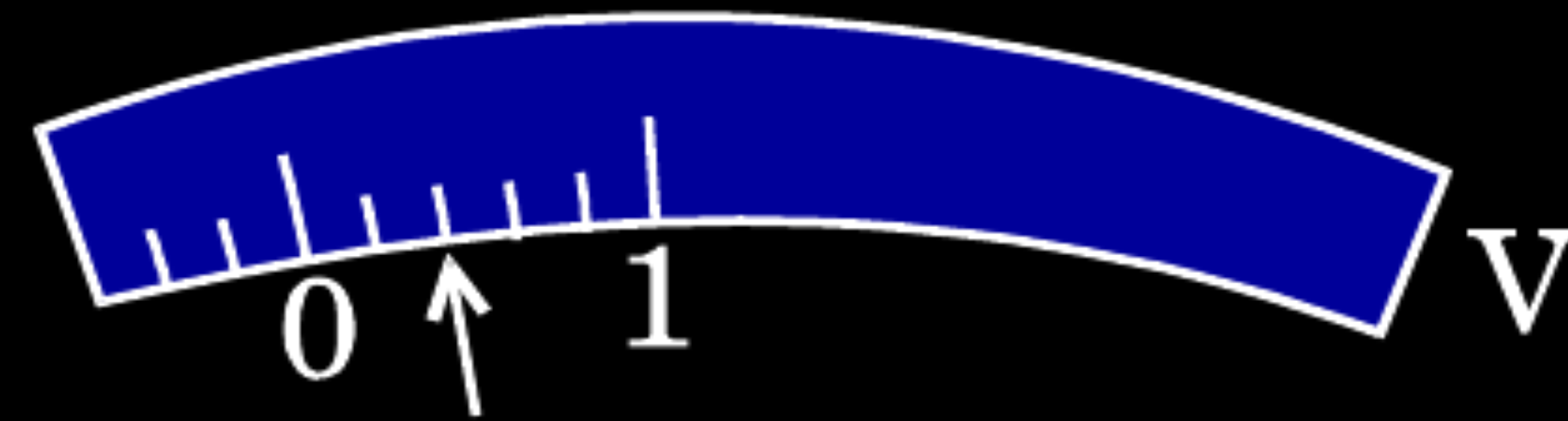


Figure A

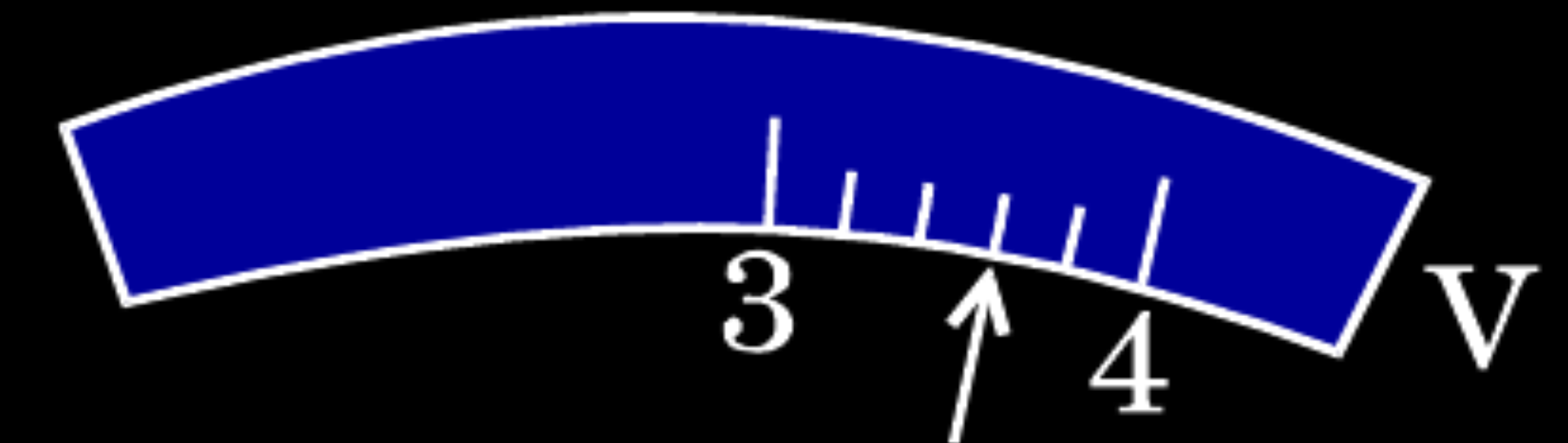
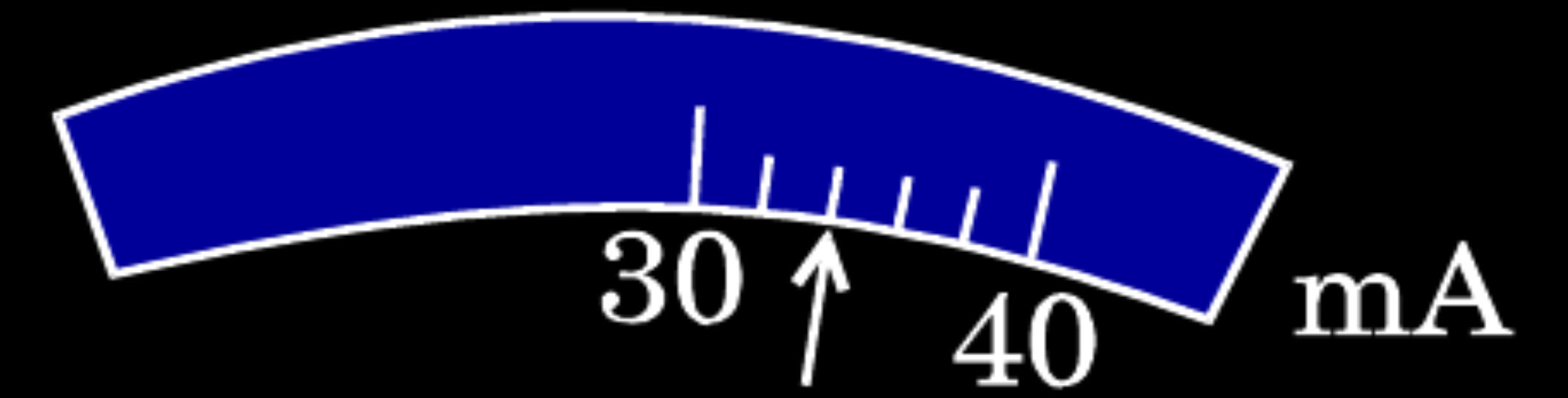


Figure B

The rest position of the needles in a milliammeter and voltmeter, not in use, are as shown in Figure A. When a student uses these instruments in his experiment, the readings of the needles are in the positions shown in Figure B. Determine the correct values of current and voltage the student should use in his calculations.

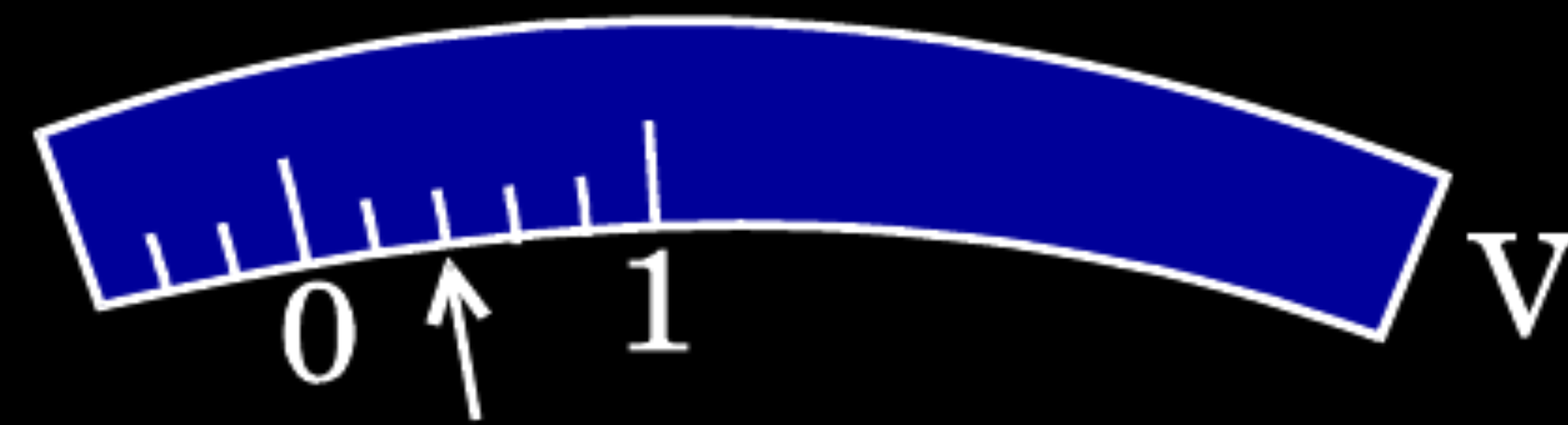
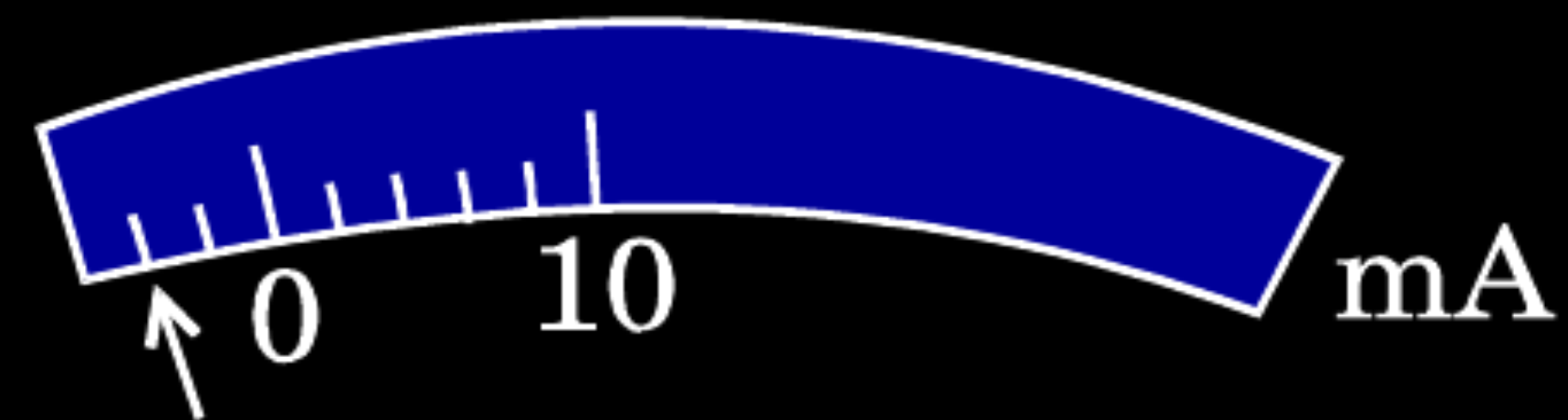


Figure A

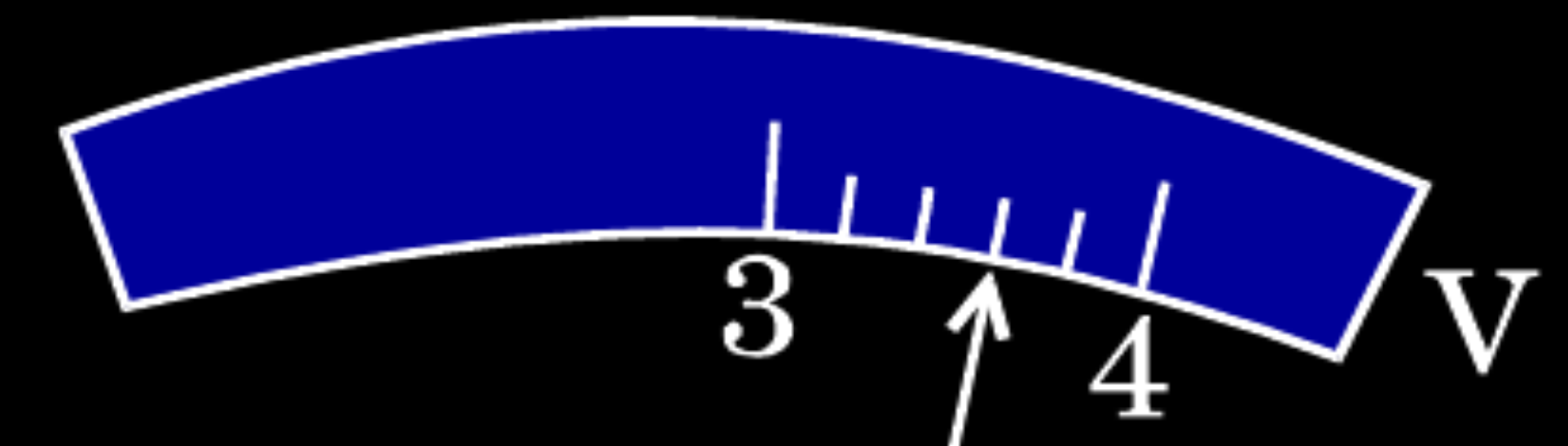
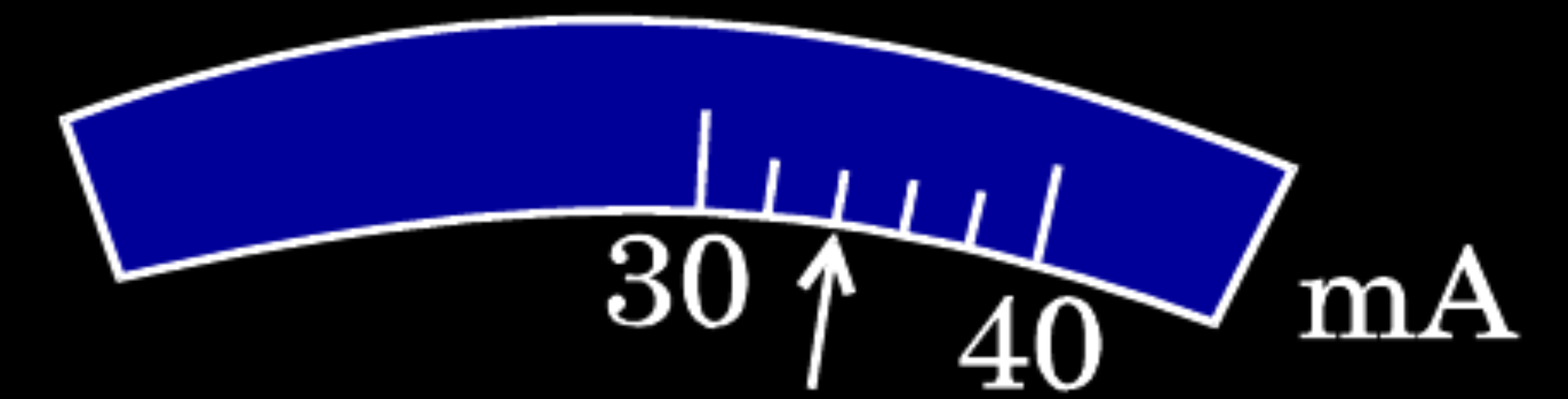


Figure B

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(i) 38mA, 3.2 V

| 1+1

- (a) With the help of a suitable circuit diagram prove that the reciprocal of the equivalent resistance of a group of resistances joined in parallel is equal to the sum of the reciprocals of the individual resistances.
- (b) In an electric circuit two resistors of $12\ \Omega$ each are joined in parallel to a $6\ \text{V}$ battery. Find the current drawn from the battery.

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From figure:

$$I = I_1 + I_2 + I_3$$

$$I_1 = \frac{V}{R_1}, \quad I_2 = \frac{V}{R_2}, \quad I_3 = \frac{V}{R_3}$$

$$\therefore \frac{V}{R_P} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

$$\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

b) $R_1 = R_2 = 12 \, \Omega$ $V = 6 \, V$

$$\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{12} + \frac{1}{12}$$

$$\therefore R_P = 6 \, \Omega$$

$$I = \frac{V}{R_P} = \frac{6V}{6\Omega} = 1A$$

1

1

1

1

$\frac{1}{2}$

$\frac{1}{2}$

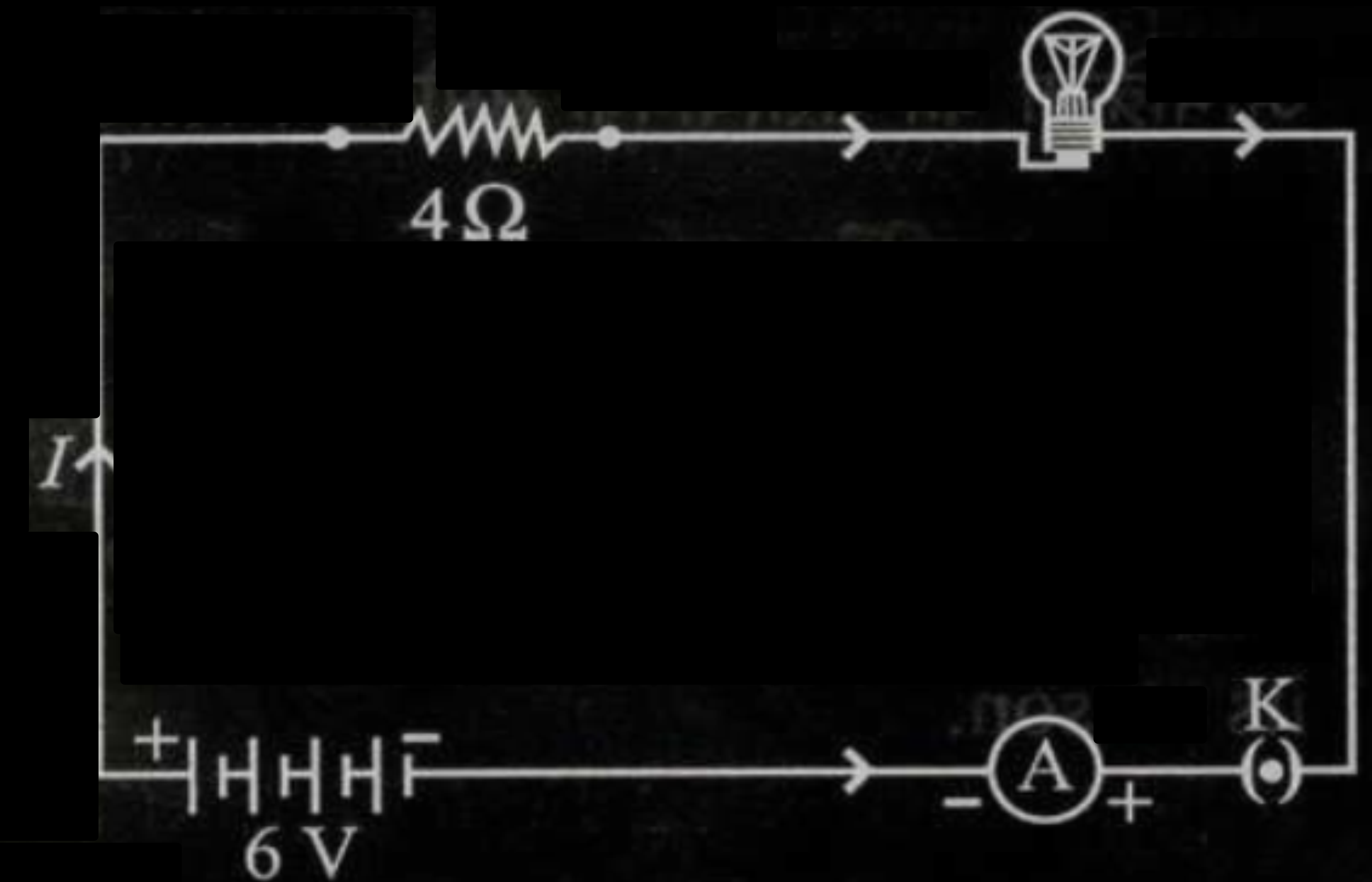
An electric lamp of resistance $20\ \Omega$ and a conductor of resistance $4\ \Omega$ are connected to a 6 V battery as shown in the circuit. Calculate

(a) the total resistance of the circuit,

(b) the current through the circuit.

(c) the potential difference across the (i) electric lamp and (ii) conductor, and

(d) power of the lamp.



$$\begin{aligned} \text{a) } R &= R_1 + R_2 \\ &= 20 \, \Omega + 4 \, \Omega = 24 \, \Omega \end{aligned}$$

$$\begin{aligned} \text{b) } I &= \frac{V}{R} \\ &= \frac{6V}{24 \, \Omega} = 0.25 \, A \end{aligned}$$

$$\begin{aligned} \text{c) (i) For electric lamp:} \\ V &= IR \\ &= \frac{6}{24} \times 20 = 5 \, V \end{aligned}$$

$$\begin{aligned} \text{(ii) For Conductor:} \\ V &= IR \\ &= \frac{6}{24} \times 4 = 1 \, V \end{aligned}$$

$$\begin{aligned} \text{d) } P &= VI \\ &= 5 \, V \times \frac{6}{24} A = 1.25 \, W \end{aligned}$$

1

1

1

1

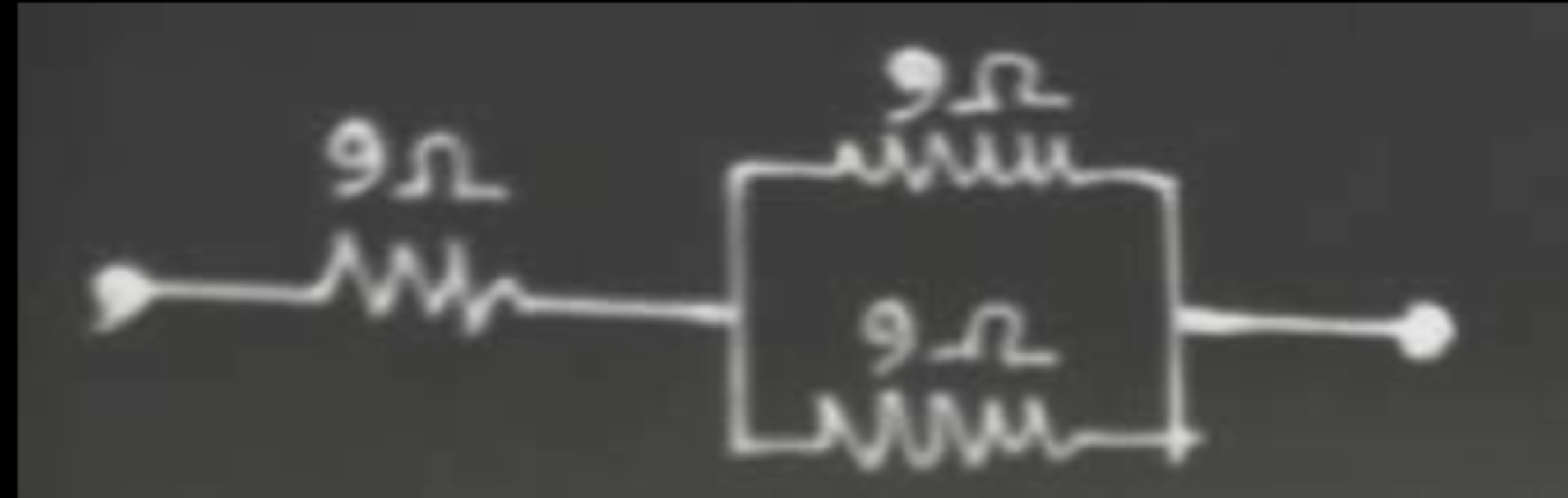
Show how would you join three resistors, each of resistance $9\ \Omega$ so that the equivalent resistance of the combination is (i) $13.5\ \Omega$, (ii) $6\ \Omega$?

CBSE 2018

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(i)



Two $9\ \text{ohm}$ resistors in parallel connected to one $9\ \text{ohms}$ in series

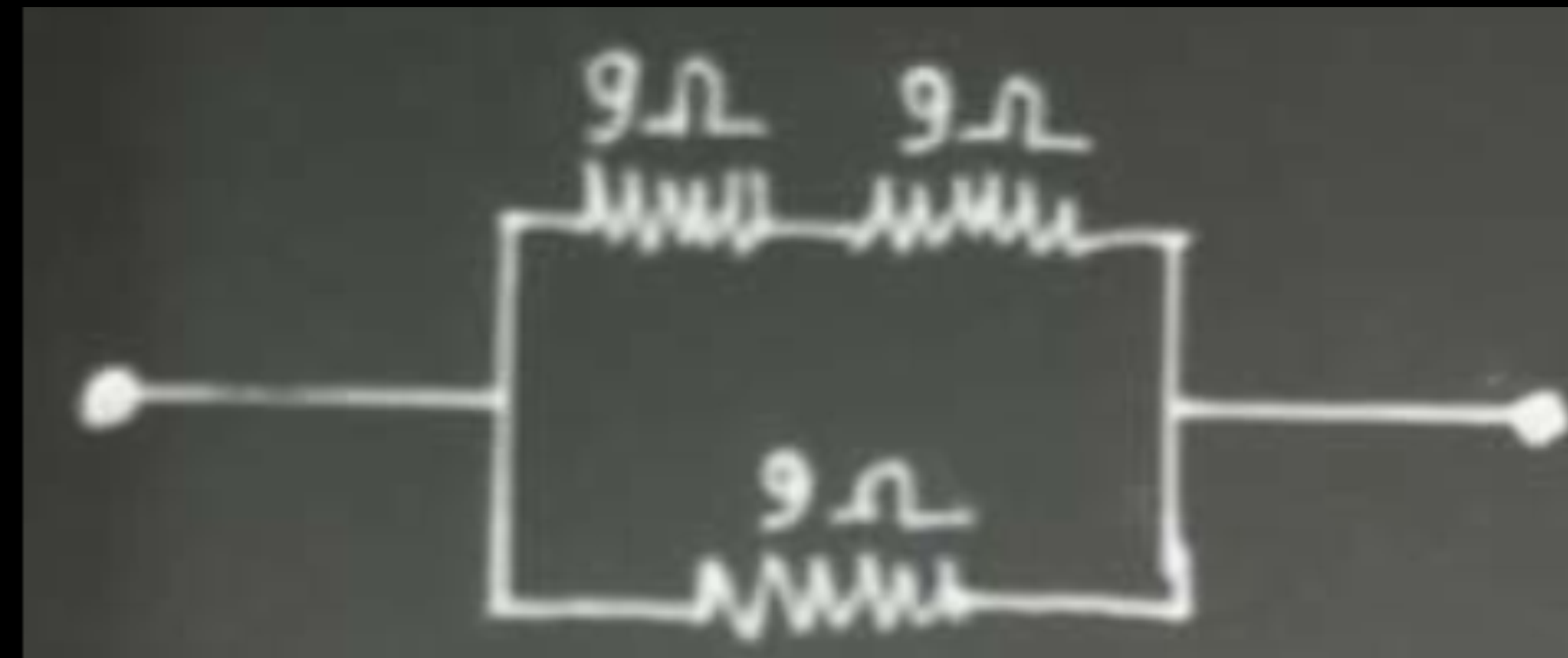
$$\frac{1}{R_p} = \frac{1}{9} + \frac{1}{9} = \frac{2}{9}$$

$$\therefore R_p = \frac{9}{2}\ \Omega$$

$$R = 9\ \Omega + \frac{9}{2}\ \Omega = 13.5\ \Omega$$

$1\ \frac{1}{2}$

(ii)



Two $9\ \text{ohm}$ resistors in series connected to one $9\ \text{ohms}$ in parallel

$$R_s = 9\ \Omega + 9\ \Omega = 18\ \Omega$$

$$\frac{1}{R} = \frac{1}{18} + \frac{1}{9} = \frac{3}{18}$$

$$\therefore R = 6\ \Omega$$

$1\ \frac{1}{2}$

Note: Deduct $\frac{1}{2}$ mark if calculations are not given.